Introduction to Cosmology

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1 Observational cosmology

1.1 Preliminaries

Cosmology: Science of the Universe as a whole, whose goal is to explain the current, past, and future state of the Universe by laws of Physics. Cosmology as an empirical science is a very special one, since **there is only one Universe we can observe:**

- rules (laws) deduced in our Universe cannot be tested in others;
- it is not always clear what are deep laws of Nature, and what are "cosmic accidents" that occur just in our Universe or to us (e.g., having an LMC 50 kpc away from us);
- we cannot even observe our Universe, but just our backward light-cone: only if simple can we hope to describe our Universe

The "basic" assumption in cosmology, is that the Universe is **simple**, and this way an extraordinary number of predictions can be made. Surprisingly, the Universe seems indeed to be simple (isotropic and homogeneous on large scales), and thus the predictions are in agreement with observations to a remarkable degree!

1.2 The Galaxy or Milky Way

- Spiral shape, with a bar
- $D \simeq 30$ kpc (note that $1 \text{ pc} \equiv 1 \text{ A.U.}/1 \text{ arcsec} \simeq 3.3$ ly)
- $L \simeq 10^{11} L_\odot$
- Flat rotation curve $\Rightarrow$ dark matter

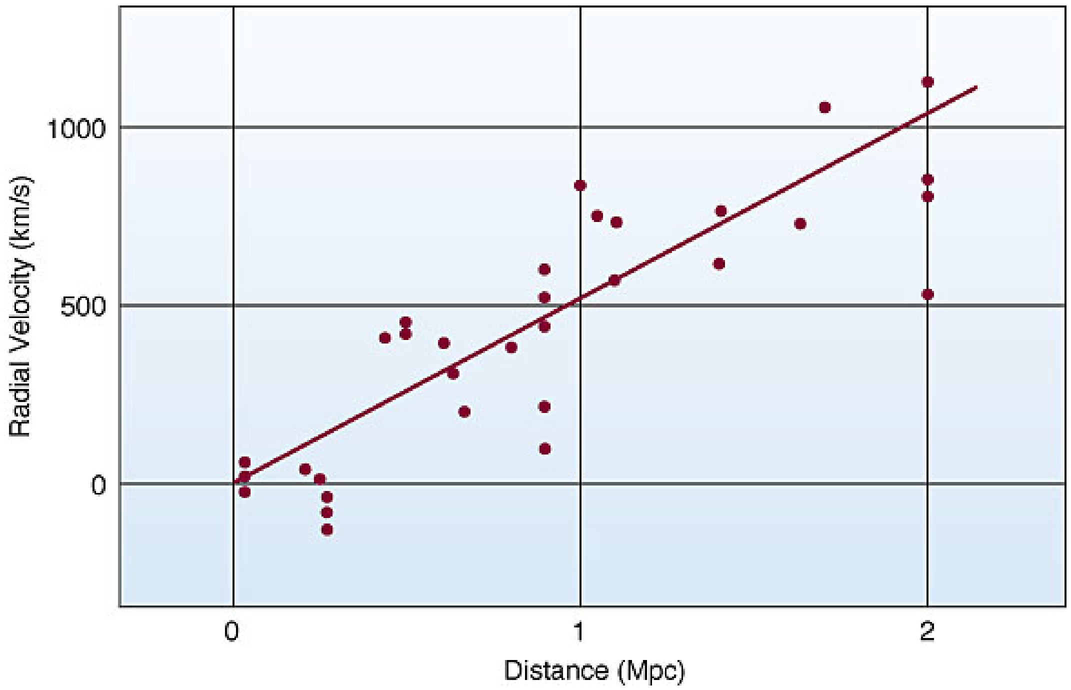

Figure 1: The original Hubble diagram.

1.3 Nearby galaxies

In the '20s Hubble realized that the *nebulae* are external objects, far away from out Galaxy.

- In these years *new telescopes* were built.
- Using these instruments, he managed to observe *Cepheids* in some nearby galaxies (such as Andromeda, M31, and Triangulum, M33).
- Since for a given period a Cepheid has a fixed absolute luminosity (*standard candle*), he could deduce from the apparent luminosity of the star its distance, and thus the distance of the galaxy.

⇒ This way everybody recognized that galaxies are *island universes*

1.4 Hubble's law

A further step was taken from the same astronomer, Hubble

- Using the Cepheids, Hubble measured the distance of several galaxies

- At the same time, he took *spectra* of their light

⇒ He estimated their velocity on the line of sight using the *Doppler* effect on known spectral lines

⇒ He showed that a linear relationship exists between the galaxy distance r and its velocity v:

$$\boxed{\boldsymbol{v} = H_0 \boldsymbol{r}} \tag{1.1}$$

- Hubble evaluated $H_0 \simeq 500\,\mathrm{km\ s^{-1}\ Mpc^{-1}}$, but the current estimated value is

$$\boxed{H_0 = 100\,h\ \mathrm{km\ s^{-1}\ Mpc^{-1}}\ ,} \tag{1.2}$$

with $h = 0.7 \pm 0.1$.

⇒ We are living in an expanding Universe.

1.5 Cosmological principles

- Observations show that the universe is isotropic around us (cf. galaxy distribution, QSO distribution, CMB)

- We are not special (Copernican principle)

⇒ The Universe is *isotropic around every point*

⇒ The Universe is *homogeneous*

Exercise 1.1 *Show the that isotropy around every point implies homogeneity.*

Exercise 1.2 *Show that Eq.* (1.1) *is the only expansion law compatible with the cosmological principles.*

1.6 Literature

- J.A. Peacock: Cosmological Physics, CUP, 1999
- J. Peebles: Principles of Physical Cosmology, Princeton, 1993
- T. Padmanabham: Structure formation in the Universe, CUP, 1993
- Kolb & Turner: The Early Universe, Addison Wesley, 1990

2 Friedmann cosmology

2.1 Comoving coordinates

It is useful to *label* points in the space using a coordinate system which is expanding together with the Universe:

$$\boxed{\boldsymbol{r}(t) = a(t)\boldsymbol{x}\ ,} \tag{2.1}$$

where $\boldsymbol{r}(t)$ is the *proper coordinate* (the physical one), $a(t)$ is the *scale factor* (used to factorize the expansion of the Universe), and $\boldsymbol{x}$ is the *comoving coordinate* (which thus does not change for objects moving with the Hubble flow). Note that $a(t)$ is chosen so that

$$a(t_0) = 1\ , \qquad\qquad \dot{a}(t_0) > 0\ . \tag{2.2}$$

2.2 Hubble's law

We have

$$\boldsymbol{v}(\boldsymbol{r},t) = \dot{\boldsymbol{r}} = \dot{a}\boldsymbol{x} = \frac{\dot{a}(t)}{a(t)}\boldsymbol{r} \equiv H(t)\boldsymbol{r}(t) \tag{2.3}$$

with

$$\boxed{H(t) = \frac{\dot{a}(t)}{a(t)}\ , \quad H_0 = \frac{\dot{a}(t_0)}{a(t_0)} = \dot{a}(t_0)\ .} \tag{2.4}$$

This yields the Hubble law:

$$\Delta\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{r} - \Delta\boldsymbol{r},t) - \boldsymbol{v}(\boldsymbol{r},t) = H(t)\Delta\boldsymbol{r}\ . \tag{2.5}$$

2.3 Cosmology and GR

General Relativity is the currently most complete theory of Gravitation, and should be used to describe physical systems each time one of the following assumptions does not hold:

- Gravitational fields are small, $\Phi \ll c^2$

- The scale of the system is much smaller than the curvature radius
- Velocities are small, $v \ll c$

A way to satisfy these hypotheses in cosmology is to consider the evolution of a small spherical portion of the Universe, where Newton's theory applies: then, the behaviour of this portion will reflect the evolution of the Universe (see below).
Two important notes:

- We can use the Newtonian Gravitation here only because we can neglect the matter outside the sphere does not affect the field gravitational field inside the sphere: this is a consequence of Gauss's theorem, which in GR is known as *Birchoff's theorem*.
- In principle, the same applies to an infinite distribution of charges. However, there is a fundamental difference between the electrical and the gravitational field: the former is measurable, while the second is not (because of the equivalence principle). In gravitation, only *tidal* fields can be measured! As a result, an infinite distribution of charges does not make sense (for a finite sphere of charges, the electrical field gives the center of the sphere, so the limit $R \to \infty$ is not allowed: where is the center then?). But an infinite distribution of matter is allowed in Newton's theory (the gravitational field increases linearly with R, which means that the tidal field is constant; also, the tidal fields is isotropic and no direction for a center is provided!).

2.4 Friedmann's equation

Consider a small expanding homogeneous sphere of density ρ and radius r. Energy conservation

$$E_{\text{kin}} + E_{\text{pot}} = \text{const} , \tag{2.6}$$

implies

$$\frac{v^2}{2}-\frac{GM}{r}=\frac{v^2}{2}-\frac{4\pi}{3}Gr^2\rho=\text{const}\,. \tag{2.7}$$

This equation holds in *proper* coordinates; using *comoving* coordinates we find $\boldsymbol{r}=a\boldsymbol{x}$ and $\boldsymbol{v}=\dot{a}\boldsymbol{x}$, so that

$$\dot{a}^2(t)=\frac{8\pi}{3}G\rho(t)a^2(t)-Kc^2\,, \tag{2.8}$$

where K is an integration constant to be determined later. This is **Friedmann's equation**.

2.5 Matter components

Friedmann's equation involves two unknowns, the scale factor $a(t)$ and the density $\rho(t)$, and thus needs to be supplemented with another equation. To this purpose, we use a version of the first law of thermodynamics:

$$\boxed{\mathrm{d}\left(\rho c^2 r^3\right)=-p\,\mathrm{d}\left(r^3\right)\,.} \tag{2.9}$$

We distinguish then three main matter components:

Dust (or matter): This is the "ordinary" cosmological matter, where the pressure vanishes $p=0$. Then Eq. (2.9) implies

$$\boxed{\rho_{\mathrm{m}}(t)=\rho_{\mathrm{m}0}a^{-3}(t)\,,} \tag{2.10}$$

i.e. the *mass conservation law*.

Radiation: In this case $p=\rho c^2/3$, so that

$$\frac{\mathrm{d}(\rho_{\mathrm{r}}a^3)}{\mathrm{d}a}=-\frac{\rho_{\mathrm{r}}}{3}\frac{\mathrm{d}(a^3)}{\mathrm{d}a}\,. \tag{2.11}$$

This equation can be solved by making the *Ansatz* $\rho_{\mathrm{r}}=\rho_{\mathrm{r}0}a^{\beta}$, which leads to $\beta=-4$. Hence

$$\boxed{\rho_{\mathrm{r}}(t)=\rho_{\mathrm{r}0}a^{-4}(t)\,.} \tag{2.12}$$

The dependence a^{-4} can be explained by noting that the *photon density* goes as a^{-3} (because of the photon conservation law, see above), but at the same time each photon is red-shifted by a factor a and its energy thus decrease by a^{-1}.

Vacuum energy: In this case it is postulated that $\rho_{\rm v} = \text{const}$, which implies

$$\boxed{p_{\rm v} = -\rho_{\rm v} c^2 \, .} \tag{2.13}$$

2.6 Density parameters

Consider the case $K = 0$ and specialize Friedmann's equation to today:

$$H_0^2 = \frac{8\pi}{3} G\rho_0 \, . \tag{2.14}$$

Hence, *if* $K = 0$, then the total current density of the Universe is

$$\boxed{\rho_0 = \frac{3H_0^2}{8\pi G} \equiv \rho_{\rm cr} \simeq 1.88 \times 10^{-29} h^2 \text{ g cm}^{-3} \, .} \tag{2.15}$$

This equation, generalized to other times, is simply

$$\rho_{\rm cr}(a) = \frac{3H^2(a)}{8\pi G} \, , \tag{2.16}$$

where

$$H^2(a) \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho(a) - \frac{Kc^2}{a^2} \, . \tag{2.17}$$

Now we have a "natural" density scale, and therefore we can define the dimensionless density parameters:

$$\Omega_{\rm m}(a) \equiv \frac{\rho_{\rm m}(a)}{\rho_{\rm cr}(a)} = \frac{8\pi G \rho_{\rm m}(a)}{3H^2(a)} \, , \tag{2.18}$$

$$\Omega_{\rm r}(a) \equiv \frac{\rho_{\rm r}(a)}{\rho_{\rm cr}(a)} = \frac{8\pi G \rho_{\rm r}(a)}{3H^2(a)} \, , \tag{2.19}$$

$$\Omega_{\rm v}(a) \equiv \frac{\rho_{\rm v}(a)}{\rho_{\rm cr}(a)} = \frac{8\pi G \rho_{\rm v}(a)}{3H^2(a)} \, . \tag{2.20}$$

Note that

- In most cases, these density parameters are considered "today" at $a = 1$.

- Many authors indicate Ω_v as Ω_Λ; we will follow this notation later.

- The total density parameter is defined as

$$\Omega_0 = \Omega_\mathrm{m} + \Omega_\mathrm{r} + \Omega_\mathrm{v} \ . \tag{2.21}$$

  Warning: in the literature Ω_0 is often used for Ω_m.

2.7 The curvature

Friedmann's equation today can be written as

$$Kc^2 = \frac{8\pi}{3} G\rho_0 - H_0^2 = H_0^2(\Omega_0 - 1) \ , \tag{2.22}$$

and hence K is determined by Ω_0.
$|K|^{-1/2}$ is the curvature radius, with $[K] = (\text{length})^{-2}$. The terminology "curvature" comes from GR, as will be explained shortly. We have

- $K > 0$ if $\Omega_0 > 1$,

- $K = 0$ if $\Omega_0 = 1$,

- $K < 0$ if $\Omega_0 < 1$.

2.8 Friedmann's equation in terms of density parameters

$$\left(\frac{\dot a}{a}\right)^2 = H^2 = \frac{8\pi}{3} G\rho(a) + \frac{H_0^2(1-\Omega_0)}{a^2} \ , \tag{2.23}$$

with

$$\rho(a) = \rho_{\mathrm{m}0} a^{-3} + \rho_{\mathrm{r}0} a^{-4} + \rho_\mathrm{v} \ . \tag{2.24}$$

Hence we find

$$\boxed{H^2(t) = H_0^2 \left[a^{-4}(t)\Omega_\mathrm{r} + a^{-3}(t)\Omega_\mathrm{m} + a^{-2}(t)(1-\Omega_0) + \Omega_\mathrm{v}\right] \ .} \tag{2.25}$$

2.9 Age of the Universe

$$H = \frac{\dot{a}}{a} = \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t} \Rightarrow \mathrm{d}t = \frac{\mathrm{d}a}{aH} \,, \tag{2.26}$$

which can be integrated

$$\boxed{t(a) = \int_0^a \frac{\mathrm{d}a'}{a'H(a')} = \frac{1}{H_0}\int_0^a \mathrm{d}a' \left[\frac{\Omega_\mathrm{r}}{a'^2} + \frac{\Omega_\mathrm{m}}{a'} + (1-\Omega_0) + \Omega_\mathrm{v} a'^2\right]^{-1/2} .} \tag{2.27}$$

- Age today: use $a = 1$;
- The age is proportional to $1/H_0 \simeq 14$ Gyr; the other "corrective factor" depends on the evolution of the Universe but is usually ~ 1.

2.10 Equation of motion

Let us differentiate Friedmann's equation

$$2\dot{a}\ddot{a} = \frac{8\pi G}{3}\left(\dot{\rho}a^2 + 2a\dot{a}\rho\right) \,, \tag{2.28}$$

and then eliminate $\dot{\rho}$ using the first law of thermodynamics:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left(\rho c^2 a^3\right) &= -p\frac{\mathrm{d}(a^3)}{\mathrm{d}t} \,, \\ \dot{\rho}a^3 + 3\rho a^2\dot{a} &= -\frac{p}{c^2}3a^2\dot{a} \,, \\ \dot{\rho}a^2 = -3a\dot{a}\left(\rho + \frac{p}{c^2}\right) &. \end{aligned}$$

It follows then

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) .} \tag{2.29}$$

Compare this with the naive Newtonian equation

$$\ddot{r} = -\frac{GM}{r^2} = -\frac{4\pi G}{3}\rho r \,. \tag{2.30}$$

That means that pressure acts like a source of gravity!

Exercise 2.1 *We “derived” that pressure is a source of gravity, but we did not use GR so far. Where is the swindle – or where was this equivalence of matter and energy put into our “derivation”?*

3 Matter and radiation dominated models

3.1 Einstein-de Sitter (EdS) model

Consider Friedmann's equation (2.23) in the case $\Omega_{\rm m} = 1$:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} G\rho_{0\rm m} a^{-3} = H_0^2 a^{-3} \, . \tag{3.1}$$

This differential equation can be easily integrated (using, for example, the technique of Sect. 2.9) or postulating a power-law behaviour

$$a = a_0 \left(\frac{t}{t_0}\right)^{\alpha} \, . \tag{3.2}$$

Then we obtain from the square root of Eq. (3.1)

$$\dot{a} = \frac{\alpha}{t_0}\left(\frac{t}{t_0}\right)^{\alpha-1} = H_0 \left(\frac{t}{t_0}\right)^{-\alpha/2} = H_0 a^{-1/2} \, . \tag{3.3}$$

- $\alpha = 2/3$, i.e. $a \propto t^{2/3}$: the Universe expands forever.
- $t_0 = 2/(3H_0)$ is the age of the Universe.

Note. In the past, when this model was favoured, there has been a controversial point regarding the age of the Universe. Taking our value of H_0, indeed, the age turned out to be $t_0 \simeq 9.3$ Gyr, which appears to be too small given the estimated age of stars in globular clusters. For this reason, many astronomers advocated a smaller Hubble constant, of the order of $h = 0.5$–0.6.

3.2 Other matter dominated models

For a matter dominated Universe Friedmann's equation takes the form

$$\dot{a}^2 = H_0^2 \left[\Omega_{\rm m} a^{-1} + 1 - \Omega_{\rm m}\right] \, . \tag{3.4}$$

Suppose initially $K > 0$, i.e. $\Omega_{\rm m} > 1$: then the solutions of Fridmann's equation can be written as

$$a(\eta) = A(1-\cos\eta) \, , \tag{3.5}$$

$$t(\eta) = B(\eta - \sin\eta) \, . \tag{3.6}$$

Exercise 3.1 *Show that Eqs.* (3.5) *and* (3.6) *satisfy Eq.* (2.23)*, find how the two constants* A *and* B *are related to* Ω_m *and* H_0*. Is there a connection of this result with the third law of Kepler?*

Exercise 3.2 *Show that at early times (*$a \ll 1$*) the Universe expands as* $a \propto t^{2/3}$*, similarly to the EdS Universe.*

- Eqs. (3.5) and (3.6) are the equations of a *cycloid*, i.e. of a point on the circumference of a wheel that rotates without slipping on a flat surface.
- After an expansion, for $t > \pi B$ a contraction of the Universe is observed.
- At $t = 2\pi B$ there is the *big-crunch*: the Universe collapses back to zero radius.

If, instead, $K > 0$, i.e. $\Omega_\mathrm{m} < 1$, then the solutions are given by

$$a(\eta) = A(\cosh\eta - 1)\,, \tag{3.7}$$

$$t(\eta) = B(\sinh\eta - \eta)\,. \tag{3.8}$$

- The Universe in this case expands forever.
- For $t \ll B$ the kinetic and potential energies are much larger, in modulus, than the total energy.
- For $t \gg B$ the potential energy becomes negligible and the solutions become freely expanding with $a \propto t$.

3.3 Radiation dominated models

At early times, when $a \ll 1$, the radiation density (because of its a^{-4} behaviour) dominates the matter, vacuum, and curvature densities. In these conditions we can neglect all densities in the Friedmann's equation but the ρ_r one:

$$\dot{a}^2 = \frac{8\pi}{3}\frac{G\rho_\mathrm{r0}}{a^2} = H_0^2\Omega_\mathrm{r}a^{-2}\,. \tag{3.9}$$

The solution of this equation can be found by putting

$$a(t) = \left(\frac{t}{t_0}\right)^{\alpha} , \tag{3.10}$$

By substitution in Eq. (3.9) we find $\alpha = 1/2$ and

$$t_0 = \frac{1}{2H_0\sqrt{\Omega_{\rm r}}} \, . \tag{3.11}$$

Hence, when radiation dominates the Universe expands “only” as $a \propto t^{1/2}$, instead of the faster Einstein-de Sitter expansion $a \propto t^{2/3}$; this difference is due to the different scaling laws of the radiation and matter densities.

4 Special Relativity

4.1 Principle of Special Relativity

Special Relativity is built upon a general principle:

> Laws of nature are invariant under transformation of the Poincaré group, which includes translations, Lorentz-boosts, and spatial rotations:
>
> $$x^{\mu'} = \Lambda^{\mu'}{}_{\mu} x^{\mu} + a^{\mu} \ , \quad \mu, \mu' \in \{0,1,2,3\} \ .$$
>
> In this equations both a^{μ} and $\Lambda^{\mu'}{}_{\mu}$ are constants.

Note that we make use of Einstein's summation rule: a sum over *repeated* indices (up and down) *on the same side of an equation* is implicit:

$$\Lambda^{\mu'}{}_{\mu} x^{\mu} \equiv \sum_{\mu=0}^{3} \Lambda^{\mu'}{}_{\mu} x^{\mu} \ .$$

The tensor $\Lambda^{\mu'}{}_{\mu}$ satisfies

$$\Lambda^{\mu'}{}_{\mu} \Lambda^{\nu'}{}_{\nu} \eta_{\mu'\nu'} = \eta_{\mu\nu} \ , \tag{4.1}$$

where $\eta_{\mu\nu}$ is *Minkowski's metric*:

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \ . \tag{4.2}$$

Hence, Poincaré (or Lorentz) transformations are those which leave the *proper time* invariant:

$$c^2 \mathrm{d}\tau^2 = c^2 \mathrm{d}t^2 - \mathrm{d}\boldsymbol{x}^2 = \eta_{\alpha\beta} \mathrm{d}x^{\alpha} \mathrm{d}x^{\beta} \ , \tag{4.3}$$

where $x^0 = ct$:

$$\begin{aligned} c^2 \mathrm{d}\tau'^2 &= \eta_{\alpha'\beta'} \mathrm{d}x^{\alpha'} \mathrm{d}x^{\beta'} = \eta_{\alpha'\beta'} \Lambda^{\alpha'}{}_{\alpha} \mathrm{d}x^{\alpha} \Lambda^{\beta'}{}_{\beta} \mathrm{d}x^{\beta} \\ &= \eta_{\alpha\beta} \mathrm{d}x^{\alpha} \mathrm{d}x^{\beta} = c^2 \mathrm{d}\tau^2 \,. \end{aligned} \tag{4.4}$$

We restrict our attention to the *proper Lorentz group*:

- $\Lambda^0{}_0 \geq 0$, so that the direction of time is preserved (in fact, $\Lambda^0{}_0 \geq 1$);
- $\det\Lambda = +1$, so that the parity does not change.

In this case, Λ can be decomposed into spatial rotations:

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix} , \tag{4.5}$$

where $R^{\mathrm{T}} R = \mathrm{Id}$ and $\det R = 1$, and Lorentz boosts

$$\Lambda = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} , \tag{4.6}$$

with $\beta = v/c$ and $g\gamma = \left(1-\beta^2\right)^{-1/2}$.

4.2 Four vectors

A four vector is a quantity that transforms in the same way as coordinates:

$$V^{\mu} \to V^{\mu'} = \Lambda^{\mu'}{}_{\mu} V^{\mu} \,. \tag{4.7}$$

This is a **controvariant 4-vector**. One defines a **covariant 4-vector** as a quantity V_μ that transforms as

$$V_{\mu} \to V_{\mu'} = \Lambda_{\mu'}{}^{\mu} V_{\mu} \,. \tag{4.8}$$

In this equation $\Lambda_{\mu'}{}^{\nu'} = \eta_{\mu'\mu}\eta^{\nu'\nu}\Lambda^{\mu}{}_{\nu}$ and $\eta_{\mu\nu} = \eta^{\mu\nu}$ (the metric tensor is symmetric). Note that we have

$$\Lambda_{\alpha}{}^{\gamma}\Lambda^{\alpha}{}_{\beta} = \eta_{\alpha\alpha'}\eta^{\gamma\gamma'}\Lambda^{\alpha'}{}_{\gamma'}\Lambda^{\alpha}{}_{\beta} = \delta^{\gamma}_{\beta}\,, \tag{4.9}$$

since $\eta_{\alpha\alpha'}\Lambda^{\alpha'}{}_{\gamma}\Lambda^{\alpha}{}_{\beta} = \eta_{\gamma\beta}$.
Note: The relation (4.9) can be better understood by noting that

$$\Lambda_{\alpha'}{}^{\alpha} = \frac{\partial x^{\alpha}}{\partial x^{\alpha'}}\,, \qquad\qquad \Lambda^{\alpha'}{}_{\alpha} = \frac{\partial x^{\alpha'}}{\partial x^{\alpha}}\,, \tag{4.10}$$

and by applying the Theorem of the inverse function.

4.3 Examples of four vectors

The coordinates of a particle are a 4-vector, as is the difference of the coordinates at two points along its world-line. For two neighboring points or "events," we can divide by the proper-time $\mathrm{d}\tau$ between the events (which is invariant), i.e. the interval between the events as measured by an observer moving with the particle and which is a scalar, to obtain the 4-velocity

$$U^{\mu} = \frac{\mathrm{d}x^{\mu}}{\mathrm{d}\tau}\,, \tag{4.11}$$

which is a contravariant 4-vector. If the particle has 3-velocity $\boldsymbol{u}$ relative to our inertial frame, then the two events in our frame have temporal coordinate separation $\mathrm{d}t = \gamma\mathrm{d}\tau$, with $\gamma \equiv (1 - \boldsymbol{u}\cdot\boldsymbol{u}/c^2)^{-1/2}$ as usual, and hence the particle's 4-velocity is related to its coordinate velocity by $U^0 = \mathrm{d}x^0/\mathrm{d}\tau = c\mathrm{d}t/\mathrm{d}\tau = c\gamma$ and $U^i = \mathrm{d}x^i/\mathrm{d}\tau = \gamma\mathrm{d}x^i/\mathrm{d}t$ or

$$U^{\mu} = \gamma\begin{pmatrix} c \\ \boldsymbol{u}\end{pmatrix}\,. \tag{4.12}$$

The product of the four velocity with the *rest mass* of a particle (an invariant) is a four vector, and is denoted by $P^{\mu} = mU^{\mu}$.
Another example of a four (contravariant) vector is the four acceleration, defined as $A^{\mu} = \mathrm{d}U^{\mu}/\mathrm{d}\tau$.

Exercise 4.1 *Write down the components of the four acceleration.*

4.4 Why are there two types of vectors?

Let $\boldsymbol{e}_i$ be a set of basis vectors in $\mathbb{R}^n$, and let $\boldsymbol{a}$ be a vector. The components of $\boldsymbol{a}$ can be defined as

$$\boldsymbol{a} = \sum_{i=1}^{n} a^i \boldsymbol{e}_i \,, \qquad \text{or as} \qquad a_i = \boldsymbol{a} \cdot \boldsymbol{e}_i \,. \tag{4.13}$$

If the $\boldsymbol{e}_i$ are Cartesian vectors there is no difference, since

$$a_i = \sum_{j=1}^{n} a^j \boldsymbol{e}_j \cdot \boldsymbol{e}_i = a^i \quad \text{if} \quad \boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij} \,. \tag{4.14}$$

In Special Relativity, one always works with Cartesian coordinates, so this distinction is not so important. For example, in the Minkoski space

$$\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \eta_{ij} \,, \tag{4.15}$$

so that

$$V^\mu = (V^0, V^1, V^2, V^3) \Rightarrow V_\mu = (V^0, -V^1, -V^2, -V^3) \,. \tag{4.16}$$

In General Relativity, however, it is not possible to choose Cartesian coordinates everywhere, and thus one needs to distinguish between the two types of vectors.
Covariant and contravariant vectors are related through the metric:

$$V_\mu = \eta_{\mu\nu} V^\nu \,, \qquad V^\mu = \eta^{\mu\nu} V_\nu \,, \tag{4.17}$$

The product (contraction) of a covariant and contravariant vector is a scalar invariant:

$$U_\alpha V^\alpha = \eta_{\alpha\beta} U^\alpha V^\beta = \eta_{\alpha_\beta} \Lambda^\alpha{}_{\alpha'} \Lambda^\beta{}_{\beta'} U^{\alpha'} V^{\beta'} = \eta_{\alpha'\beta'} U^{\alpha'} V^{\beta'} \,. \tag{4.18}$$

The gradient of a scalar is a covariant quantity

$$\frac{\partial}{\partial x^\alpha} \to \frac{\partial}{\partial x^{\alpha'}} = \frac{\partial x^\alpha}{\partial x^{\alpha'}} \frac{\partial}{\partial x^{\alpha'}} = \Lambda_\alpha{}^{\alpha'} \frac{\partial}{\partial x^{\alpha'}} \,. \tag{4.19}$$

4.5 Four tensors

A four tensor is a quantity with several indices which transforms like

$$T_{\mu'\nu'} = \Lambda_{\mu'}{}^{\mu}\Lambda_{\nu'}{}^{\nu}T_{\mu\nu} \, , \tag{4.20}$$

$$S^{\mu'\nu'} = \Lambda^{\mu'}{}_{\mu}\Lambda^{\nu'}{}_{\nu}S^{\mu\nu} \, , \tag{4.21}$$

$$Q^{\lambda'}_{\mu'\nu'} = \Lambda_{\mu'}{}^{\mu}\Lambda_{\nu'}{}^{\nu}\Lambda^{\lambda'}{}_{\lambda}Q^{\lambda}_{\mu\nu} \, , \tag{4.22}$$

and so on. Vectors are tensors of rank 1.
A tensor can be constructed in several ways using other quantities:

Linear combination: if $R^{\mu\nu}$ and $S^{\mu\nu}$ are tensors of the same rank, so is $T^{\mu\nu} = aR^{\mu\nu} + bS^{\mu\nu}$.

Direct product: if $R^{\mu\nu}$ and S_{λ} are tensors, then $T^{\mu\nu}{}_{\lambda} = R^{\mu\nu}S_{\lambda}$ is also a tensor.

Contraction: if $R^{\mu\nu}{}_{\alpha\beta}$ is a tensor, then $S^{\mu}{}_{\alpha} = R^{\mu\beta}{}_{\alpha\beta}$ is a tensor.

Derivative: if $R^{\mu\nu}$ is a tensor, then $S_{\alpha}{}^{\mu\nu} = \partial R^{\mu\nu}/\partial x^{\alpha}$ is a tensor.

Physical laws in Special Relativity take the form of tensor equations. This ensures that the laws are manifestly **covariant**, i.e. do not depend on the coordinate system used (i.e. on the motion of the inertial observer).

Example: the force equation is written as

$$f^{\mu} = m\frac{\mathrm{d}^2 x^{\mu}}{\mathrm{d}\tau^2} = \frac{\mathrm{d}p^{\mu}}{\mathrm{d}\tau} \, , \tag{4.23}$$

where f^{μ} is the relativistic force, $p^{\mu} = m\gamma(c^2, \boldsymbol{v})$ is the four-momentum. Note that $p^{\mu}p_{\mu} = m^2c^2$ is a scalar, as expected.

4.6 Energy-momentum (stress-energy) tensor

In Relativity it is convenient to define an energy momentum tensor $T^{\mu\nu}$ to represent the energy and momentum densities. Consider a collection of particles with momenta $\{p_i^\mu\}$ and trajectories $x_i^\mu(t) = \big(ct, \boldsymbol{x}_i(t)\big)$, and define

$$T^{\mu\nu} = \sum_i p_i^\mu \frac{\mathrm{d}x_i^\nu}{\mathrm{d}t} \delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{x}_i(t)\big) = \sum_i \frac{p_i^\mu p_i^\nu}{E_i} \delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{x}_i(t)\big) . \tag{4.24}$$

This quantity is covariant (i.e. a tensor) because:

- the space-time volume element $\mathrm{d}t\,\mathrm{d}V = \mathrm{d}x^0\,\mathrm{d}x^1\,\mathrm{d}x^2\,\mathrm{d}x^3$ is a Lorentz invariant;
- the combination of $\delta^{(3)}/\mathrm{d}t$ in Eq. (4.24) transforms like $1/\mathrm{d}t\,\mathrm{d}V$ and is thus invariant;
- the direct product of $p_i^\mu\,\mathrm{d}x_i^\nu$ makes a 2-rank tensor.

Note that from the last expression of Eq. (4.24) $T^{\mu\nu} = T^{\nu\mu}$.
The components of $T^{\mu\nu}$ are

T^{00}: the energy density of particles;

$T^{0\mu}$: the momentum density;

T^{kl}: the momentum current.

For a *perfect fluid* at rest we have

$$T = \begin{pmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} . \tag{4.25}$$

This expression, when the fluid has bulk velocity U^μ, transforms into

$$\boxed{T^{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) U^\mu U^\nu - p\eta^{\mu\nu} .} \tag{4.26}$$

The energy-momentum conservation law takes the form

$$\frac{\partial}{\partial x^\mu} T^{\mu\nu} = T^{\mu\nu}_{,\mu} = 0 \,. \tag{4.27}$$

Exercise 4.2 *Show that the spatial components of this equation becomes*

$$\boxed{\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v} = -\frac{1-\boldsymbol{v}/c^2}{\rho + p/c^2}\left(\nabla p + \frac{\boldsymbol{v}^2}{c^2}\frac{\partial p}{\partial t}\right) \,.} \tag{4.28}$$

Exercise 4.3 *Show that from the contraction* $U_\nu T^{\mu\nu}_{,\mu}$ *one has*

$$\boxed{U^\mu \frac{\partial \rho}{\partial x^\mu} + \left(\rho + \frac{p}{c^2}\right)\frac{\partial U^\mu}{\partial x^\mu} = 0 \,.} \tag{4.29}$$

Exercise 4.4 *Using the results of the last two exercises, show that the energy-momentum conservation in the non-relativistic limit* $v \ll c$ *takes the form*

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\boldsymbol{v}) = 0 \,, \tag{4.30}$$

$$\rho\left[\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v})\right] = -\nabla p \,, \tag{4.31}$$

i.e. the continuity and Euler equations.

5 General Relativity

The best theory for gravity is Einstein's general theory of relativity. This is a classical theory of *geometrodynamics*, and has not yet been accommodated within a proper quantum mechanical framework.

> Spacetime is a mainfold M on which there is defined a Lorentz metric $g_{\mu\nu}$; the curvature of g is related to the matter distribution in spacetime by Einstein's equation.

Hence, in general relativity matter causes curvature of space-time, which in turn influences the orbits of particles etc. As Wheeler has colorfully put it

> space tells matter how to move – matter tells space how to curve.

5.1 Principle of Mach

In the original Newtonian formulation, the physical laws can only be applied to a special class of reference, the *inertial frame*, while for non-inertial reference systems a set of "unphysical" forces is present (for example the centrifugal and Coriolis' force). The limited applicability of his laws, induced Newton to assume that the space itself had some physical properties, and hence he formulated the idea of an *absolute space*. Early on, however, it was recognized that the hypothesis of an absolute space was not at all an explanation of the existence of the inertial frames, but rather an even stronger reiteration of this observation.

One of the first philosophers to address the problem of inertial frames was the Irish bishop Berkeley (1685–1753): he claimed that the origin of inertial forces was to be found in the distant stars. This idea was developed from the observation that the inertial forces measured on the Earth (for example the Coriolis force in a Foucault's pendulum) showed beautifully that the angular speed of our non-inertial frames is identical to the angular speed at which we see the night sky in rotation. Moreover, he added, speaking of an “absolute space” makes no sense, since it is not observable: for example, if the Universe were composed by a sphere only, it would be impossible to identify any movement of such a sphere. These idea were then further developed by Ernst Mach (1872). His point of view is basically identical to the one of Berkeley, but his theory is particularly important because it deeply impressed and inspired Einstein for his General Relativity.

5.2 Inertial induction

It is worth to try to estimate the influence of distant massive bodies on accelerated objects. To the purpose, we note that there is a well-known parallel between the gravitational and the electrostatic forces, both inversely proportional to the square distance: in practice, in Gravitation the mass plays the role of the electric charge. In addition, from the electromagnetism theory we know that an accelerated charge e_1 close to a second, static charge e_2 experiences a “force” proportional to

$$F \propto \frac{e_1 e_2}{r} a \,, \tag{5.1}$$

where r is the distance between the charges and a is the acceleration of e_1. Note that, in contrast with electrostatic forces, the expression (5.1) is proportional to $1/r$, i.e. it is a (very) long range force. Following the ideas of Mach, we might assumed that a similar force is experiences by a mass m_1 accelerated with respect to a second mass m_2:

$$F \propto \frac{m_1 m_2}{r} a \,. \tag{5.2}$$

An informal justification of the distance dependence of F can be easily given. Laws of the form $F \propto r^{\beta}$ with exponents $\beta \leq -2$ can be excluded, because in these cases inertial forces would be dominated by nearby masses. For example, in case of $\beta = -2$ we would have a force with the same dependence of gravity, and in such case the Earth would be the dominating other terms; hence, for example, Foucault’s pendulum in this case would not show any rotation. Hence, if we confine ourselves to integer exponents, the choice $\beta = -1$ seems a natural one.

We now try to push the parallel with electromagnetism further, by trying to evaluate the numerical coefficient appearing in the relation (5.2). We first note that for electrical charges we are considering the two forces

$$F_1 = k\frac{e_1 e_2}{r^2}\,, \qquad\qquad F_2 = k\frac{e_1 e_2}{c^2 r}\Theta a\,, \tag{5.3}$$

where k is the dielectric constant of vacuum and Θ is a factor (of order unity) that takes into account the angular dependence of the second force. Note that in the case of F_2 we inserted a factor c^2 (which, among other things, makes the expression for F_2 dimensionally correct).

We postulate now that something similar happens for gravitational forces:

$$F_1 = G\frac{m_1 m_2}{r^2}\,, \qquad\qquad F_2 = G\frac{m_1 m_2}{c^2 r}\Theta a\,. \tag{5.4}$$

We now try to evaluate the effects of F_2 on a body of mass m. To this purpose, we need to sum all contributions in F_2 from the various bodies in the Universe. Since $F_2 \propto 1/r$, it is clear that distant objects are more important than nearby ones. In particular, if we assume a constant density ρ of objects and neglect the angular dependence Θ, we find the following expression for F_2 if we consider objects up to a distance R:

$$F_2 \simeq \frac{2\pi G\rho}{c^2} R^2 m a\,. \tag{5.5}$$

The problem of this equation is clearly that, in principle, for an *infinite* Universe we would have a divergent expression (cf. Olbert's paradox). In reality, it is reasonable to assume that, because of the expansion of the Universe, objects at distances larger than $R \sim c/H_0$ will not contribute significantly to F_2. Using this limit, we obtain

$$F_2 \simeq \frac{2\pi G\rho}{H_0^2} m a\,. \tag{5.6}$$

Interestingly, if we identify ρ with $\rho_0 = \rho_\mathrm{m} + \rho_\mathrm{r} + \rho_\mathrm{v}$, we find

$$F_2 \simeq \frac{3}{4}\Omega_0 m a\,. \tag{5.7}$$

In other words, since $\Omega_0 \simeq 1$, our approximate calculation was able to provide us with the second law of dynamics: the inertial force F_2 is within a numerical factor of order of unity equal to ma.

5.3 Principle of Equivalence

> At every spacetime point one can choose a locally inertial frame such that in a small region around this point, laws of nature are the same as in an unaccelerated Cartesian coordinate system in absence of gravity.

This principle is the consequence of two observations

- All things fall equally fast, since the "inertial mass" is equivalent to the "gravitational mass":

$$m_{\rm i}\ddot{\boldsymbol{x}} = m_{\rm g}\boldsymbol{g} \ , \qquad\qquad m_{\rm i} = m_{\rm g} \equiv m \ . \tag{5.8}$$

- In an accelerated system, gravity can be transformed away locally (e.g., elevator in free fall, or satellite in Earth orbit).

Note that gravity can be transformed away only *locally in small regions*: tidal gravitational forces cannot be transformed away!

5.4 The metric

Central to general relativity is the *metric tensor* $g_{\mu\nu}$ which is the generalization to curved space-times of the Minkowski metric $\eta_{\mu\nu}$ from special relativity. The basic equation of general relativity is an identity between the curvature tensor $G^{\mu\nu}$, which is constructed from second derivatives of the metric, and the stress-energy tensor $T^{\mu\nu}$ describing the matter.
The metric tensor $g_{\mu\nu}$ (which in GR replaces the tensor $\eta_{\mu\nu}$) is used to evaluated *distances* between events:

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu \ . \tag{5.9}$$

Example 1: in 2D polar coordinates (r,ϕ), the distance between two neighbouring points is

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\,\mathrm{d}\phi^2\ , \tag{5.10}$$

and thus the metric tensor takes the simple form

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}\ . \tag{5.11}$$

Example 2: The metric

$$\mathrm{d}s^2 = \mathrm{d}r^2 + \sin^2 r\,\mathrm{d}\phi^2\ , \qquad g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 r \end{pmatrix} \tag{5.12}$$

describes the surface of a unit sphere.

Exercise 5.1 *Write the metric for the 3D polar coordinates* (r,θ,ϕ).

Exercise 5.2 *Try to generalize the metric of Eq.* (5.12) *to the surface of a 3D sphere (i.e. of a sphere in a 4D space).*

5.5 Geodesics

The concept of "straight lines" in a curved space takes the form of geodesics, curves $y^i(\lambda)$ connecting two points A and B where the curve length

$$\ell = \int_A^B \sqrt{\mathrm{d}s^2} \tag{5.13}$$

is *stationary*. In Relativity, one talks about three kind of geodesics

Timelike: characterized by $\mathrm{d}s^2 > 0$, these describe the motion of particles under the effect of gravity only (hence, in GR the motion of planets around the Sun is along "straight lines"; the fact that the actual orbit is not a line ios only due to the curved spacetime around the Sun).

Null: characterized by $\mathrm{d}s^2 = 0$, described the path of light rays (or other massless particles).

Spacelike: characterized by $\mathrm{d}s^2 < 0$, do not have relevance for us.

Exercise 5.3 *Use Euler-Lagrange equations to find the geodesics for the metric* (5.10)*, and show that they correspond to straight lines in polar coordinates.*

Exercise 5.4 *Similarly, use Euler-Lagrange equations to find the geodesics on the sphere* (5.12)*: these correspond to maximal circles on the unit sphere.*

5.6 Coordinate transformation and covariance

Consider a coordinate transformation $x^\mu \to x^{\mu'}$.

- A *scalar* is invariant under this transformation.
- A *vector* transforms according to

$$\boxed{V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} V^\mu \,, \qquad V_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} V_\mu \,.} \tag{5.14}$$

  For example, the gradient of a scalar field Φ

$$\frac{\partial \Phi}{\partial x^{\mu'}} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial \Phi}{\partial x^\mu} \tag{5.15}$$

  is a covariant vector.

- A *tensor* transforms like

$$\boxed{T^{\mu'\nu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} \frac{\partial x^{\nu'}}{\partial x^\nu} T^{\mu\nu}} \tag{5.16}$$

  and like

$$\boxed{g_{\mu'\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} g_{\mu\nu} \,.} \tag{5.17}$$

5.7 Equation of motion

Consider a particle that is freely falling under the influence of gravity. Because of the principle of equivalence, there is a *local* inertial coordinate system where the metric is Minkowskian:

$$d\tau^2 = \eta_{\mu\nu}\, d\xi^\mu\, d\xi^\nu\,. \tag{5.18}$$

This coordinate system corresponds to a the freely falling system or to a system that is moving with uniform velocity with respect to that one. Hence, the equation of motion of the particle there is simply

$$\frac{d^2\xi^\mu}{d\tau^2} = 0\,. \tag{5.19}$$

We now rewrite Eq. (5.19) in a general coordinate system x^μ. Since by assumption $\xi^\mu(x^\mu)$ is *locally* invertible, we have

$$0 = \frac{d}{d\tau}\left(\frac{d\xi^\mu}{d\tau}\right) = \frac{d}{d\tau}\left(\frac{\partial\xi^\mu}{\partial x^\nu}\frac{dx^\nu}{d\tau}\right) = \frac{\partial\xi^\mu}{\partial x^\nu}\frac{d^2x^\nu}{d\tau^2} + \frac{\partial^2\xi^\mu}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\,. \tag{5.20}$$

By multiplying this relation by $\partial x^\alpha/\partial\xi^\mu$, and by using the inverse function theorem, we obtain

$$\boxed{\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0\,,} \tag{5.21}$$

where

$$\Gamma^\mu_{\nu\lambda} = \frac{\partial x^\mu}{\partial\xi^\alpha}\frac{\partial^2\xi^\alpha}{\partial x^\nu\partial x^\lambda}\,. \tag{5.22}$$

The quantity $\Gamma^\mu_{\nu\lambda}$ is called the *affine connection* (or Christoffel symbol). Correspondingly, in the new coordinate system x^μ the metric is

$$d^2\tau = \eta_{\mu\nu}\frac{\partial\xi^\mu}{\partial x^\alpha}dx^\alpha\frac{\partial\xi^\nu}{\partial x^\beta}dx^\beta = g_{\alpha\beta}\, dx^\alpha\, dx^\beta\,, \tag{5.23}$$

with the metric tensor

$$\boxed{g_{\alpha\beta} = \frac{\partial\xi^\mu}{\partial x^\alpha}\frac{\partial\xi^\nu}{\partial x^\beta}\eta_{\mu\nu}\,.} \tag{5.24}$$

If we differentiate g with respect to x^ν and substitute the result in the affine connection, we obtain

$$\boxed{\Gamma^{\alpha}_{\lambda\mu} = \frac{1}{2} g^{\alpha\nu} \left(\frac{\partial g_{\mu\nu}}{\partial x^{\lambda}} + \frac{\partial g_{\lambda\nu}}{\partial x^{\mu}} - \frac{\partial g_{\mu\lambda}}{\partial x^{\nu}} \right) .} \tag{5.25}$$

Exercise 5.5 *Show that the equation of motion can also be obtained using the variational principle*

$$\delta \int_{\gamma} \mathscr{L}\, \mathrm{d}p = 0 \,, \qquad \textit{with} \qquad \mathscr{L} = g_{\mu\nu} U^{\mu} U^{\nu} \,. \tag{5.26}$$

In these equation p is any parameter along the curve γ, and U^μ is the 4-velocity. For example, by using the proper time as parameter p we have

$$\delta \int_{\gamma} g_{\mu\nu} U^{\mu} U^{\nu} \, \mathrm{d}\tau = 0 \,, \tag{5.27}$$

which is manifestly covariant.

5.8 Covariant differentiation

Exercise 5.6 *Let V^μ be a vector, and consider its derivative*

$$A^{\mu}_{\nu} \equiv \frac{\partial V^{\mu}}{\partial x^{\nu}} \tag{5.28}$$

in two different coordinate systems x^μ and $x^{\mu'}$. Show that A^μ_ν does not transform as a tensor, and that in particular a term involving the second derivatives of $x^{\mu'}$ with respect to x^μ appears.

Exercise 5.7 *Show that the quantity*

$$A^{\mu}_{\nu} \equiv \frac{\partial V^{\mu}}{\partial x^{\nu}} + \Gamma^{\mu}_{\nu\lambda} V^{\lambda} \,, \tag{5.29}$$

does transform as a tensor.

The quantity of Eq. (5.29) is the *covariant derivative* of a vector and is usually indicated as $A^{\mu}_{;\nu}$q. Similarly, one can define covariant derivatives of higher order tensors:

$$T^{\mu\nu}_{;\lambda} = T^{\mu\nu}_{,\lambda} + \Gamma^{\mu}_{\lambda\rho} T^{\rho\nu} + \Gamma^{\nu}_{\lambda\rho} T^{\mu\rho} \,, \tag{5.30}$$

$$V_{\mu;\lambda} = V_{\mu,\lambda} - \Gamma^{\rho}_{\mu\lambda} V_{\rho} \,, \tag{5.31}$$

$$T^{\mu}_{\nu;\lambda} = T^{\mu}_{\nu,\lambda} + \Gamma^{\mu}_{\lambda\rho} T^{\rho}_{\nu} - \Gamma^{\rho}_{\nu\lambda} T^{\mu}_{\rho} \,. \tag{5.32}$$

In general, one needs a $+\Gamma$ term for each controvariant index, and a $-\Gamma$ term for each covariant index.
The chain rule applies:

$$\left(A^{\mu}{}_{\nu} B^{\lambda}\right)_{;\rho} = A^{\mu}{}_{\nu;\rho} B^{\lambda} + A^{\mu}{}_{\nu} B^{\lambda}{}_{;\rho} \,. \tag{5.33}$$

Moreover, the covariant derivative of the metric tensor vanishes:

$$\boxed{g_{\mu\nu;\lambda} = 0 \,,} \tag{5.34}$$

which implies that covariant derivative and contraction commute:

$$\left(g^{\mu\nu} V_{\nu}\right)_{;\lambda} = g^{\mu\nu} V_{\nu;\lambda} = V^{\mu}_{;\lambda} \,. \tag{5.35}$$

A covariant derivaties thus transforms tensors into tensors, and reduces to normal derivatives in inertial systems where $\Gamma^{\mu}_{\nu\lambda} = 0$. This suggests a simple way to obtain laws in General Relativity: use the laws of Special Relativity, and replace derivatives with covariant derivatives. This works because one can always find a local inertial frame where the laws of Special Relativity apply (equivalence principle).

5.9 Towards a theory of gravity

Since the metric $g_{\mu\nu}$ determines the trajectories of particles (geodesics), it must contain information on the gravitational field. The gravitational field is determined by the matter contents, i.e. the source of gravity.
Out of $g_{\mu\nu}$ and its first derivative, no new tensor can be constructed: indeed, one can always find a coordinate systems where $g_{\mu\nu,\lambda}$ vanishes,

and this implies that such a vector would vanish in any coordinate system (this property is reminiscent of the equivalence principle).
Hence, one needs second derivatives of g. Further, a natural *source* for the gravity is the stress-tensor $T^{\mu\nu}$. We then look for a two-rank tensor with the following properties:

1. Contains only up to second derivatives of the metric, ande is linear in $\partial^2 g$ and quadratic in ∂g (since gravity is scale-free).

2. Has vanishing divergence, similarly to $T^{\mu\nu}_{;\nu}$.

3. Is symmetric, like the stress-energy tensor.

4. For weak stationary fields, its $(0,0)$ components reduces to $-\nabla^2 g_{00}$ (this comes from the equivalence principle applied to the metric).

There is a unique tensor that has these properties and is called the *Einstein tensor* $G_{\mu\nu}$: with this tensor Einstein's field equation is

$$\boxed{G_{\mu\nu} = -\frac{8\pi G}{c^4} T_{\mu\nu} \,.} \tag{5.36}$$

6 Cosmological models

6.1 Robertson-Walker metric

Consider a *isotropic* and *homogeneous* Universe, with a set of *fundamental observers* which are at rest with respect to the Hubble flow. In these conditions the metric for the Universe must assume a simple form:

- First, note that there exists an **absolute time**: the observers can synchronize their clocks by agreeing that, for example, time 0 is when the Universe reaches a given density.

- Then, use as time coordinate for the metric the **proper time** of the fundamental observers, i.e. the absolute time defined above. This way, the metric assumes the form

  $$c^2 \, \mathrm{d}\tau^2 = c^2 \, \mathrm{d}t^2 - \mathrm{d}S^2 \,, \tag{6.1}$$

  where $\mathrm{d}S^2$ indicates the spatial part of the metric.

- The spatial part $\mathrm{d}S^2$ must be "as symmetric as possible" which in practice forces $\mathrm{d}S^2$ to be like the surface element on a sphere (see below).

The metric then takes the form (Robertson 1935; Walker 1936)

$$\boxed{c^2 \, \mathrm{d}\tau^2 = c^2 \, \mathrm{d}t^2 - a^2(t) \left[\mathrm{d}w^2 + f_K^2(w) \, \mathrm{d}\Omega^2 \right] \,,} \tag{6.2}$$

where $\mathrm{d}\Omega$ is the soling angle element, $\mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta \, \mathrm{d}\phi^2$ and

$$f_K(w) = \begin{cases} 1/\sqrt{K} \sin(\sqrt{K} w) & \text{if } K > 0 \,, \\ w & \text{if } K = 0 \,, \\ 1/\sqrt{-K} \sinh(\sqrt{-K} w) & \text{if } K < 0 \,. \end{cases} \tag{6.3}$$

Note that $1/\sqrt{|K|}$ represents here a sort of curvature radius.

6.2 Friedmann-Lemaître cosmological models

If we substitute the RW metric into Einstein's equations we find two relations for the scale factor $A(t)$:

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}G\rho - \frac{Kc^2}{a^2} + \frac{\Lambda}{3}\,,} \tag{6.4}$$

and

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda}{3}\,.} \tag{6.5}$$

These equations, combined together, give

$$\frac{\mathrm{d}\left(a^3\rho c^2\right)}{\mathrm{d}t} + p\frac{\mathrm{d}\left(a^3\right)}{\mathrm{d}t} = 0\,. \tag{6.6}$$

Note: coordinates used by Peacock

Peacock writes the metric as

$$c^2\,\mathrm{d}\tau^2 = c^2\,\mathrm{d}t^2 - R^2(t)\left[\mathrm{d}r^2 + S_k^2(w)\,\mathrm{d}\Omega^2\right]\,,$$

with the same t (cosmic time) and angular coordinates. To relate them to ours, one must have:

$$a(t)\,\mathrm{d}w = R(t)\,\mathrm{d}r = R_0 a(t)\,\mathrm{d}r$$

which implies that $w = R(t_0)r$. Also

$$a(t)f_K(w) = R(t)S_k(r) = a(t)R(t_0)S_k(r)\,.$$

If we now choose $R_0 \equiv R(t_0) = 1/\sqrt{|K|}$, then we have

$$S_k(r) = \begin{cases} \sin r & \text{if } k = +1\,, \\ r & \text{if } k = 0\,, \\ \sinh r & \text{if } k = -1\,. \end{cases}$$

Hence, Peacock uses a discrete variable $k \in \{-1, 0, +1\}$ in addition to the scale R; in contrast, we use a single continuous variable, K.

6.3 Comoving distance

Because of the symmetry of the metric, radial curves with θ and ϕ constant are spatial projections of geodesics. For light rays, one has $\mathrm{d}\tau^2 = 0$, and therefore a geodetic must satisfy the differential equation

$$c^2\,\mathrm{d}t^2 = a^2(t)\,\mathrm{d}w^2\,. \tag{6.7}$$

This implies that the comoving distance of a source that emits light at the epoch t is given by

$$w = \int_t^{t_0} \frac{c\,\mathrm{d}t}{a(t)}\,. \tag{6.8}$$

6.4 Cosmological redshift

Consider a source located at comoving distance w and an observer, and assume that both the source and the observer move with the Hubble flow. In these conditions, the comoving distance is invariant, and thus the integral that appears in Eq (6.8) does not change with time:

$$0 = \int_{t+\Delta t_\mathrm{e}}^{t_0+\Delta t_0} \frac{c\,\mathrm{d}t}{a(t)} - \int_t^{t_0} \frac{c\,\mathrm{d}t}{a(t)} \simeq \frac{c\Delta t_0}{a(t_0)} - \frac{c\Delta t_\mathrm{e}}{a(t)}\,, \tag{6.9}$$

which implies

$$\boxed{\Delta t_\mathrm{e} = a(t)\Delta t_0\,.} \tag{6.10}$$

In other words, the time interval Δt_e is stretched to $\Delta t_0 = \Delta t_\mathrm{e}/a(t) > \Delta t_\mathrm{e}$ due to the cosmic expansion. For example, for the frequencies of fotons we have $\nu_\mathrm{e} = \nu_\mathrm{obs}/a(t) > \nu_\mathrm{obs}$: photons are redshifted due to time dilation:

$$\boxed{(1+z) \equiv \frac{\nu_\mathrm{e}}{\nu_\mathrm{obs}} = \frac{1}{a(t)}\,.} \tag{6.11}$$

Note: The variables a, t, w, and z can all be used to characterize distance of comoving sources, as long as they are monotonic; z however is the only quantity which is directly observable.

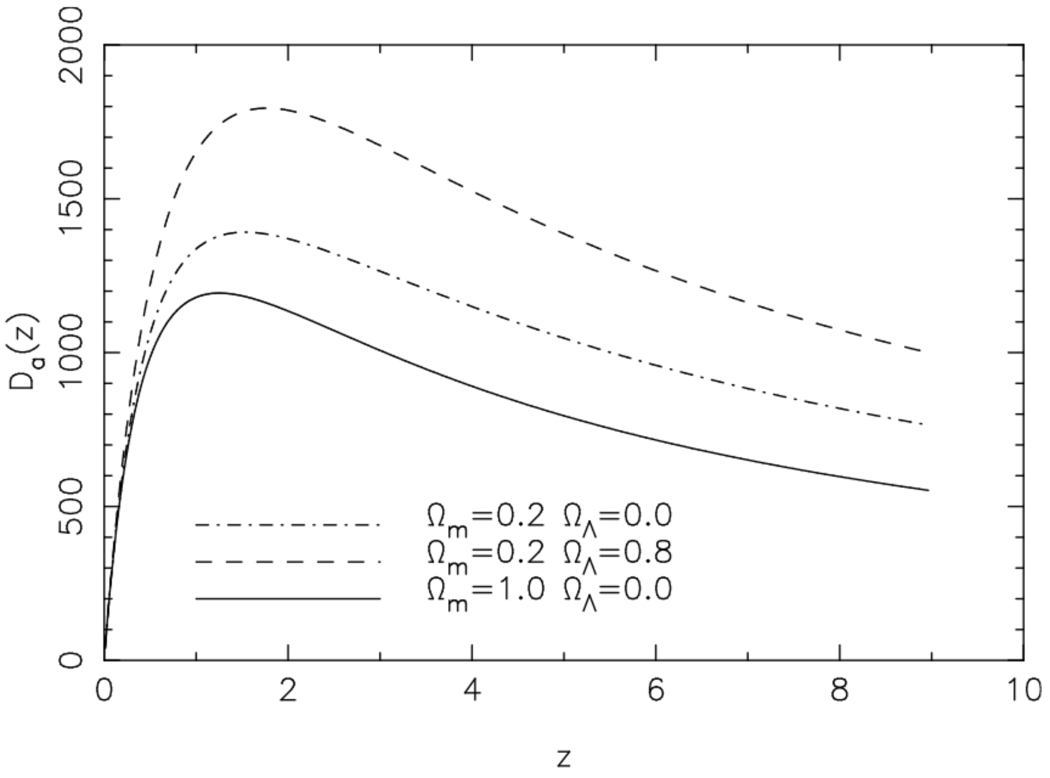

Figure 2: Angular diameter distance in various cosmological models.

6.5 Angular diameter distance

Consider two light rays with angular separation $\mathrm{d}\theta$ at the observer, located at the "pole" of the coordinate systems (i.e., at $w = 0$). The physical separation of the rays, $\mathrm{d}L$, is related to $\mathrm{d}\theta$ through

$$\mathrm{d}L = a(t) f_K(w)\, \mathrm{d}\theta \;, \tag{6.12}$$

as shown by the RW metric with $\mathrm{d}t^2 = \mathrm{d}w^2 = 0$. We can then define an angular diameter distance as

$$\boxed{D_A \equiv \frac{\mathrm{d}L}{\mathrm{d}\theta} = a(t) f_K(w) = \frac{f_K(w)}{1+z}\;.} \tag{6.13}$$

Sometimes it is also useful to define a *comoving* angular diameter distance as $D_A/a(t)$ (this distance does not change as the Universe expands, as does the "normal" angular diameter distance).

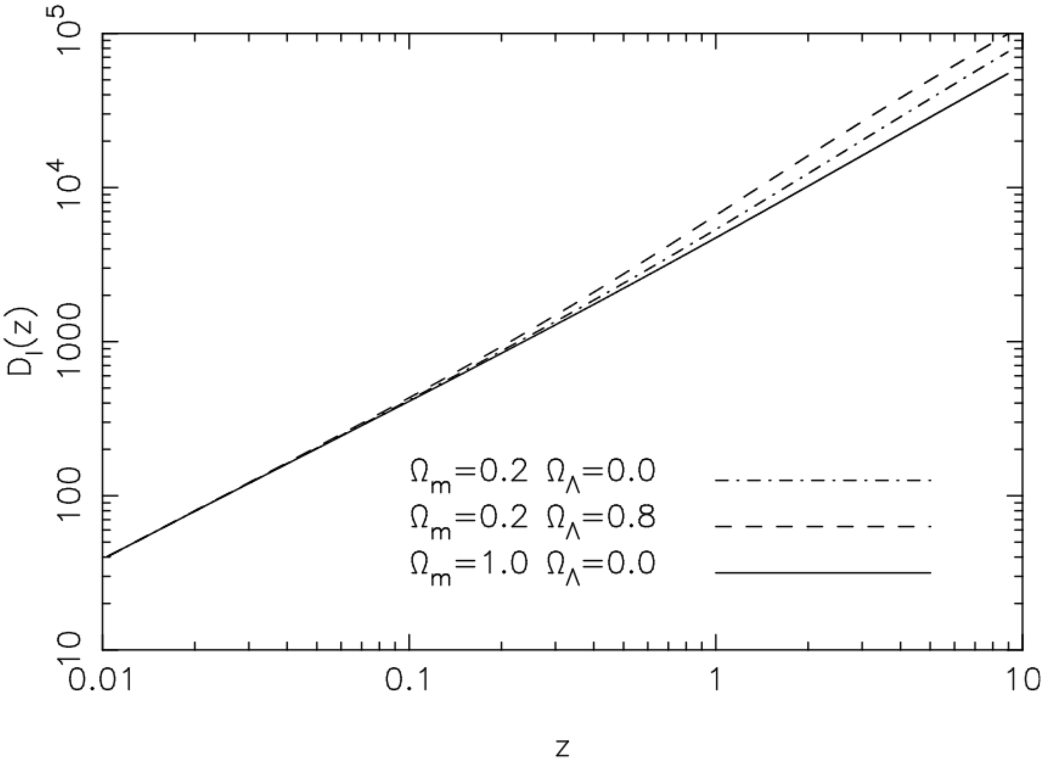

Figure 3: Luminosity distance in various cosmological models.

6.6 Luminosity distance

Consider a light source with total luminosity L located at the "pole" $w = 0$ that emits at a time t. The surface of the wave front at $z = 0$ is given by

$$\mathscr{A} = 4\pi f_k^2(w) a^2(t_0) = 4\pi f_k^2(w) \ . \tag{6.14}$$

In addition, each emitted photon is redshifted by $(1+z)^{-1}$ and the arrival frequency is reduced by another factor $(1+z)^{-1}$. In summary, the observed flux is given by

$$S = \frac{L}{4\pi f_k^2(w)}(1+z)^{-2} \ . \tag{6.15}$$

This justify the definition of luminosity distance as

$$\boxed{D_L \equiv \sqrt{\frac{L}{4\pi S}} = (1+z) f_K(w) \ .} \tag{6.16}$$

6.7 Surface brightness

Consider the ratio $\mathscr{B}$ between the observed flux of a source and its angular surface on the sky: the former is given by S of Eq. (6.15), while the latter is proportional to $1/D_A^2$. Hence, $\mathscr{B}$ is proportional to

$$\boxed{\mathscr{B} \propto S D_A^2 = (1+z)^{-4}\,.} \tag{6.17}$$

In summary, in the local Universe ($z \ll 1$) the surface brightness is *conserved*; at high-redshift, instead, there is a *cosmological dimming* factor $(1+z)^{-4}$. This is a very general property (which holds also in difference cosmological models).

6.8 Distances in the EdS model

As shown in Sect. 3.1, in the EdS cosmological model we have $a(t) = (t/t_0)^{2/3}$ and $t_0 = 2/(3H_0)$. Moreover, since $\Omega_0 = \Omega_\mathrm{m} = 1$, the Universe is spatially flat ($K = 0$) and $f(w) = w$.
From Eq. (6.8) we find

$$w = \int_t^{t_0} \frac{c\,\mathrm{d}t}{(t/t_0)^{2/3}} = \frac{2c}{H_0}\left(1 - \sqrt[3]{t/t_0}\right)\,. \tag{6.18}$$

Hence, the angular diameter distance is

$$D_A = \frac{w}{1+z} = \frac{2c}{H_0(1+z)}\left(1 - \frac{1}{\sqrt{1+z}}\right)\,. \tag{6.19}$$

Note: In the EdS cosmological model (and in many others as well) the angular diameter distance D_A is not monotonic with z; the luminosity distance $D_L(z)$, in contrast, is in general monotonic. Among other things, this implies that the angular size of distant galaxies does not necessarily decrease with z, while their luminosity does.

6.9 Expansion law

Friedmann's equation can be written as

$$\boxed{H^2(t) = \left(\frac{\dot{a}(t)}{a(t)}\right)^2 = H_0^2\left[\frac{\Omega_\mathrm{r}}{a^4(t)} + \frac{\Omega_\mathrm{m}}{a^3(t)} + \Omega_\Lambda + \frac{\Omega_\mathrm{c}}{a^2(t)}\right]\,,} \tag{6.20}$$

where $\Omega_\Lambda \equiv \Omega_\mathrm{v}$ (this notation is more commonly used) and $\Omega_\mathrm{c} = 1 - \Omega_\mathrm{m} - \Omega_\mathrm{r} - \Omega_\Lambda$.

Various terms are important at different times, owing the different a-dependence:

- At early times ($a \ll 1$) the radiation dominates, and thus the Universe expands as $a(t) \propto t^{1/2}$ (cf. Sect. 3.3;
- Later on, matter dominates, and the Universe expands as $a(t) \propto t^{2/3}$ (see Sect. 3.2);
- Finally, later the vacuum energy dominates, provided $\Lambda \neq 0$.

6.10 Radiation density

The current Universe is filled with a black-body radiation at $T \simeq 2.7$ K. For thermal radiation, most of the energy is in photons with $E \sim kT$. Imagine now the Universe to be a periodic box of side L, then the spacing of wave numbers is $\delta k = 2\pi/L$. Since the energy of a photon is given by $E = c\hbar k$, the number of photons in the volume L^3 is given by

$$N \sim \left(\frac{k}{\delta k}\right)^3 = \left(\frac{E}{hc}\right)^3 L^3 \tag{6.21}$$

Hence, the density of photons is $c^2\rho \sim NE/L^3$ or $\rho \sim E^4/(h^3c^5)$. The exact result is provided by the Stefan-Bolzmann law and is

$$\rho_\mathrm{CMB} = \frac{1}{c^2}\frac{\pi^2}{15}\frac{(kT_\mathrm{CMB})^4}{(\hbar c)^3} \simeq 4.5 \times 10^{-34}\ \mathrm{g\ cm^{-3}}\,. \tag{6.22}$$

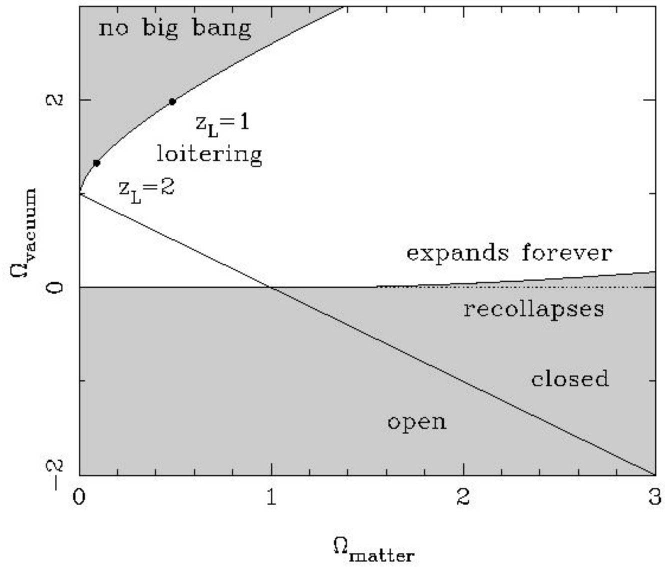

Figure 4: Qualitative behaviour of the Universe expansion.

This implies a density of

$$\boxed{\Omega_{\rm CMB} = \frac{\rho_{\rm CMB}}{\rho_{\rm cr}} \simeq 2.4 \times 10^{-5}\,h^{-2}\,.} \tag{6.23}$$

If neutrinos are also considered as “massless” particles and included in the radiation budget, then

$$\Omega_{\rm r} \simeq 4.2 \times 10^{-5}\,h^{-2}\,. \tag{6.24}$$

Since $\rho_{\rm m} \propto a^{-3}$ and $\rho_{\rm r} \propto a^{-4}$, radiation will equal matter when

$$a = a_{\rm eq} = \frac{\Omega_{\rm r}}{\Omega_{\rm m}} = 4.2 \times 10^{-5}\,\Omega_{\rm m}^{-1}\,h^{-2}\,, \tag{6.25}$$

and thus at redshift

$$\boxed{1 + z_{\rm eq} = 23\,800\,\Omega_{\rm m}\,h^2\,.} \tag{6.26}$$

Hence, in all “geometrical” cosmological calculations radiation can be neglected unless for very early times.

6.11 Past and future expansion

Consider again Friedmann's equation for the case $\Omega_\mathrm{r} = 0$ (which as explained above applies up to very high redshift):

$$\frac{H^2(a)}{H_0^2} = \Omega_\Lambda \left(1 - \frac{1}{a^2}\right) + \Omega_\mathrm{m} \left(\frac{1}{a^3} - \frac{1}{a^2}\right) + \frac{1}{a^2} \,. \tag{6.27}$$

We know that today $H = H_) > 0$, i.e. the Universe is expanding; can H be negative at some time? This can happen only if the r.h.s. of Eq. (6.27) vanishes for some a. Let $x = 1/a$ and consider the function

$$f(x) = \Omega_\Lambda (1 - x^2) + \Omega_\mathrm{m} (x^3 - x^2) + x^2 = 0 \,. \tag{6.28}$$

We distinguish various cases:

$\Omega_\Lambda = 0$: The equation simplifies to $\Omega_\mathrm{m}(x^3 - x^2) + x^2 = 0$, which has the solution $x = 1 - 1/\Omega_\mathrm{m} < 1$. There is a positive solution only for $\Omega_\mathrm{m} > 1$, in which case the Universe recollapese at

$$a_\mathrm{coll} = \frac{\Omega_\mathrm{m}}{\Omega_\mathrm{m} - 1} > 1 \,. \tag{6.29}$$

This solution corresponds to the case considered in Sect. 3.2. For $\Omega_\mathrm{m} < 1$ the Universe expands forever.

$\Omega_\Lambda > 0$ and $\Omega_\mathrm{m} < 1$: Since in this case $f(x) > 0$ for $x < 1$, such a Universe will expand forever in the future.

$\Omega_\mathrm{m} > 1$: In this case a future recollapse is avoided provided that

$$\Omega_\Lambda > 4\Omega_\mathrm{m} \cos^3 \left(\frac{\arccos\left(\Omega_\mathrm{m}^{-1} - 1\right)}{3} + \frac{4\pi}{3} \right) \,. \tag{6.30}$$

Finally, note that for Ω_Λ sufficiently large, $f(x)$ vanishes for $x > 1$, i.e. in the past! This is called a *loitering Universe*: a has a minimum in the past. However, this model is excluded since for $\Omega > 0.1$ (as provided by observations, e.g. in clusters of galaxies) the loitering redshift would be $z_\mathrm{L} \sim 2$; this is however in contrast with the observation of high-redshift galaxies. In summary:

A lointering Universe is excluded, and thus a Big Bang ($a = 0$) is unavoidable in the framework of this theory. Since $T_{\text{CMB}} \propto a^{-1} = 1 + z$, the Universe was hot in the past and cooled during the expansion.

6.12 Expansion for early times

In Sect. 2.10 we considered separately the matter and radiation dominated models; we now consider a Universe where both terms are present. This limit is very accurate and important at early times, when $a \ll 1$, since then the curvature and vacuum terms in Fridmann's equation can be neglected.
We have

$$H^2(a) = H_0^2 \left(\frac{\Omega_\text{r}}{a^4} + \frac{\Omega_\text{m}}{a^3} \right) = H_0^2 \Omega_\text{m} a^{-3} \left(1 + \frac{a_\text{eq}}{a} \right) . \tag{6.31}$$

Using $\mathrm{d}t = \mathrm{d}a/\dot{a} = \mathrm{d}a/\left[aH(a)\right]$ we find

$$\begin{aligned} t = \int_0^t \mathrm{d}t' &= \int_0^a \frac{\sqrt{a}\,\mathrm{d}a}{H_0\sqrt{\Omega_\text{m}}\sqrt{1+a_\text{eq}/a}} \\ &= \frac{2}{3H_0\sqrt{\Omega_\text{m}}} \left[a^{3/2} \left(1 - \frac{2a_\text{eq}}{a} \right) \sqrt{1 + \frac{a_\text{eq}}{a}} + 2a_\text{eq}^{2/3} \right] . \end{aligned} \tag{6.32}$$

In the limit of matter- and radiation-dominated phases we find the known results

$$t(a) = \frac{1}{H_0\sqrt{\Omega_\text{m}}} \begin{cases} \frac{a^2}{2\sqrt{a_\text{eq}}} & \text{for } a \ll a_\text{eq} \,, \\ \frac{2}{3} a^{3/2} & \text{for } a \gg a_\text{eq} \,, \end{cases} \tag{6.33}$$

i.e. $a(t) \propto t^{1/2}$ and $a(t) \propto t^{2/3}$, respectively.

6.13 Flatness and horizon problems

Although the success of the standard cosmology is astonishing, a few problems with the Fridmann-Lemaître cosmological models have been noted.

Let us consider the evolution of the matter density parameter:

$$\Omega_\mathrm{m} \equiv \frac{\rho_\mathrm{m}(a)}{\rho_\mathrm{cr}(a)} = \frac{8\pi G\rho_\mathrm{m}(a)}{3H^2(a)} = \left(\frac{H_0}{H(a)}\right)^2 \Omega_{\mathrm{m},0} a^{-3}\,. \tag{6.34}$$

Since similar relations are found for the other matter components, the evolution of the (total) matter parameter Ω_0 is

$$\begin{aligned}\Omega_0(a) &= \left(\frac{H_0}{H}\right)^2 \left(\frac{\Omega_{\mathrm{m},0}}{a^3} + \frac{\Omega_{\mathrm{r},0}}{a^4} + \Omega_\Lambda\right) \\ &= \frac{\Omega_{\mathrm{m},0}a^{-3} + \Omega_{\mathrm{r},0}a^{-4} + \Omega_\Lambda}{\Omega_{\mathrm{m},0}a^{-3} + \Omega_{\mathrm{r},0}a^{-4} + \Omega_\Lambda + (1-\Omega_0)a^{-2}}\,.\end{aligned} \tag{6.35}$$

This equation can be rewritten as

$$\boxed{(1-\Omega_0)(a) = \frac{1-\Omega_0}{1-\Omega_0 + \Omega_{\mathrm{r},0}a^{-2} + \Omega_{\mathrm{m},0}a^{-1} + \Omega_\Lambda a^2}\,.} \tag{6.36}$$

First, note that since the denominator only vanishes for a loitering Universe, an open (respectively, closed) Universe remains open (closed) forever. Second, $\Omega_0 \to 1$ as $a \to 0$: in particular one has

$$(1-\Omega_0)(a) = a^2 \frac{1-\Omega_0}{\Omega_\mathrm{r}} \qquad \text{for } a \ll 1\,. \tag{6.37}$$

Hence, at early times the Universe always look as an EdS. Moreover, since today $\Omega_0 \simeq 1$, the fine tuning at early times must have been extremely good (this is even more surprising if $\Omega_0 \neq 1$).
Another problem is related to the *horizon*: since $c\,\mathrm{d}t = -a\,\mathrm{d}w$, the light can only propagate over a finite distance since the Big Bang. The maximum (comoving) distance that can be covered is given by

$$w_\mathrm{h} = \int_0^t \frac{c\,\mathrm{d}t'}{a(t')} = \int_0^a \frac{c\,\mathrm{d}a}{a^2 H(a)}\,. \tag{6.38}$$

At early times, but for a matter dominated Universe, the integral is

$$w_\mathrm{h} = \frac{c}{H_0}\int_0^a \frac{a^{3/2}\,\mathrm{d}a}{a^2\sqrt{\Omega_\mathrm{m}}} = \frac{2c\sqrt{a}}{H_0\sqrt{\Omega_\mathrm{m}}}\,. \tag{6.39}$$

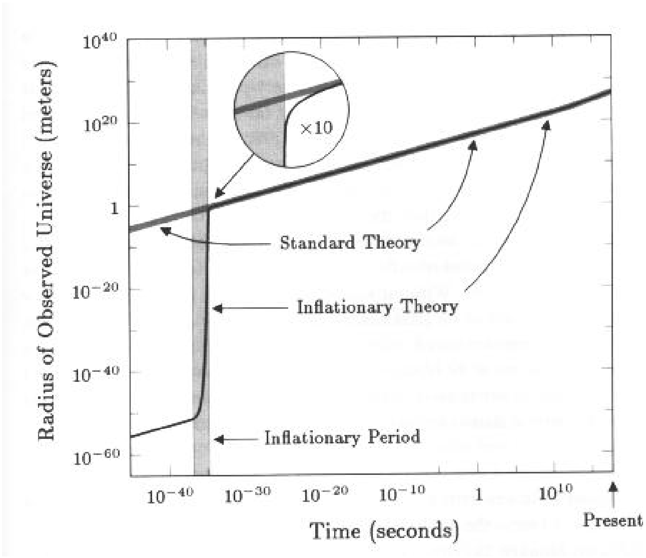

Figure 5: Inflationary models predict an accelerated expansion at early times.

The physical horizon scale is $r_{\rm h} = aw_{\rm h}$, and this scale correspond to an angular diameter $r_{\rm h}/D_A(a)$ when observed at redshift $z = 0$. In particular, at the epoch of recombination

$$\theta_{\rm h} = \frac{r_{\rm h}(a_{\rm rec})}{D_A(a_{\rm rec})} = \sqrt{\frac{\Omega_{\rm m}}{z_{\rm rec}}} \simeq \sqrt{\Omega_{\rm m}}\, 2^\circ \, . \tag{6.40}$$

Hence, points in the CMB at angular distances larger than 2° have never been in causal contact!
A possible solution to both problems is to allow the Universe to have a different expansion law at early time. Suppose in general that $a(t) \propto t^\nu$, so that we have

$$r_{\rm h}(a) = aw_{\rm h} = a(t) \int_0^t \frac{c\,{\rm d}t'}{a(t')} = \frac{ct}{1-\nu} \, . \tag{6.41}$$

We now note that a perturbation of size λ_0 today had a size approximately equal to $\lambda(t) = a(t)\lambda_0 \propto t^\nu$ at early times. Hence, the ratio between the perturbation size and the horizon goes as $t^{\nu-1}$, i.e. increases as we go back in time if $\nu < 1$: this is at the origin of the horizon problem. In order to reverse this, we need $\nu > 1$, i.e. $\ddot{a} > 0$ at some early time. However, from Friedmann's equation we can have $\ddot{a} > 0$ only if $p < -\rho c^2/3$, i.e. a negative pressure (similar to the one induced by a cosmological constant). A period with these properties is called *inflation*. Finally, we note that a sufficiently long inflationary period implies a flat Universe.

7 Cosmological tests

In this chapter we will consider some important classical tests of cosmological models.

7.1 Sunyaev-Zel'dovich effect

The hot gas found in galaxy clusters in principle gives us an opportunity to measure distances within the universe through purely physical processes. The reason is that we see the effects of the hot gas in two way: through the emission of X-rays from a cluster, and the modification of the microwave background by a cluster. This gives use two measurements to disentangle two unknowns: the density of the hot gas, and the physical size of the system. By measuring the physical size of the system and comparing it to the angle the cluster subtends on the sky, we can derive a distance. The X-ray emission seen from clusters of galaxies come primarily through the bremsstrahlung process, which is the creation of X-rays when free electrons scatter with ions. The rate at which this process produces X-rays depends on two parameters: the temperature and density of the gas. The temperature can be determined from the shape of the continuum part of the X-ray spectrum. The second quantity we can measure, the flux produced by the cluster, gives us a measure of the density-squared times the diameter of the hot gas sphere. The trick is to find a measure that will disentangle the density from the diameter. Fortunately, the hot gas in a cluster does more than just emit X-rays: it also scatters the CMB. When this radiation Compton scatters with the hot electrons, it acquires energy from the electrons: low energy CMB photons get scatter to much higher energies. This scattering distorts the shape of the microwave background in the direction of a galaxy cluster, an effect that is called the Sunyaev-Zel'dovich effect, after the two Russian scientists who pointed it out. The amount of distortion depends on three properties: the temperature of the electrons, which we know from X-ray observations, the density of electrons, and the diameter of the hot gas sphere. By measuring the distortion of the microwave background in the direction of a galaxy cluster, we have a measure of the density times the diameter. The X-ray observations provide us with a measurement of $\eta^2 D$, where η is the electron density and D is the diameter of the gas sphere, and microwave measurements provide a measurement of ηD. These observations therefore give us a direct measure of the diameter of the hot gas sphere. Combining this with the angle that the sphere subtends on the sky, and we have a measurement of the distance to the galaxy cluster. In reality, this simple picture is complicated by many factors, including a non-constant density of the X-ray emitting gas, and instrumental difficulties. As a result, measurements of the Sunyaev-Zel'dovich effects are far from trivial.

7.2 Supernovæ Type Ia

A supernova (SN) is a stellar explosion that creates an extremely luminous object. During the explosion of a supernova, as much energy as a normal star would emit during its lifetime is released. At the end of the explosion, most of the mass of the star is in form of ejecta, moving at a speed of on tenth of c. Our interest is focused mainly in SN of type Ia (SNIa), which are made by a white dwarf star that accumulates material from a companion star. When the white dwarf reaches (or approaches) the Chandrasekhar limit, $M \simeq 1.44\ \mathrm{M}_\odot$, it undergoes a violent nuclear fusion of its interior, which completely disrupt the star.

A very important property of SNIa is the fact that they can be used as *standard candles*. In particular, the absolute luminosity of a SNIa is approximately constant (especially after the correction by an effect called Δm_{15}–M_{max} relationship). Hence, SNIa can be used to obtain a measurement of the luminosity distance-redshift relation, from which we can then infer the cosmological parameters.

8 Thermal history

We know that today the Universe is filled with a Cosmic Microwave Background (CMB) with temperature $T_0 \simeq 2.73$. Hence, in the past the Universe was very hot, with CMB temperatures scaling as

$$T(z) = T_0(1+z)\ . \tag{8.1}$$

At sufficiently high temperatures several particle species exists. Moreover, if these particles can have reactions with photons at a high rate, they are in thermal equilibrium. As the Universe expands, the temperature drops, and we reach a state where the reaction rates decrease significantly: the particle is then no more in equilibrium, but its distribution retains some "memory" of the previous status. For example, electrons and positrons are in equilibrium with the CMB at $T \gtrsim 1$ MeV through the reaction

$$e^+ + e^- \leftrightarrow \gamma + \gamma\,. \tag{8.2}$$

Similarly, neutrinos are kept in equilibrium through

$$\nu + e \rightarrow \nu + e\ , \qquad\qquad e^+ + e^- \leftrightarrow \nu + \bar{\nu}\ , \tag{8.3}$$

and nucleons throgh

$$p + e^- \leftrightarrow n + \nu\ , \qquad\qquad p + \bar{\nu} \leftrightarrow n + e^+\ . \tag{8.4}$$

At high temperature, say $T \sim 10$ MeV, all these species are in thermal equilibrium with the CMB photons; at even higher temperatures, $T \sim 200$ MeV, other particles (for example μ^+, μ^-, π^+, π^-, and π^0) are also present and in equilibrium.

8.1 Evolution of the momentum

In the rest of this chapter, for simplicity, we set $k_{\rm B} = c = 1$.
Consider two observers separated by a physical distance δr, and let $u_{1,2}$ and $p_{1,2}$ the velocity and momentum of a particle as measured by the two observers. Observer 2 has a relative velocity

$$\delta v = \frac{\dot{a}}{a}\delta r \tag{8.5}$$

with respect to observer 1. Because of the Lorentz transformations, the momentum p_2 of the particle is given by

$$p_2 = \gamma[p_1 - \delta v E_1] = \gamma\left[p_1 - \frac{\dot{a}}{a}\delta t\, u_1 E_1\right] , \tag{8.6}$$

where $E_{1,2} = \sqrt{m^2 + p_{1,2}^2}$ is the energy of the particle as measured in the two reference systems, and where we wrote $\delta r = \delta t\, u_1$ in the last step. We then find

$$p_2 = p_1 - \frac{\dot{a}}{a} p_1 \delta t + \mathscr{O}(\delta t^2) , \tag{8.7}$$

and thus

$$\delta p = p_2 - p_1 = -\frac{\dot{a}}{a} p \delta t , \tag{8.8}$$

$$\frac{\dot{p}}{p} = -\frac{\dot{a}}{a} \quad \Rightarrow \quad pa = \text{const} . \tag{8.9}$$

Therefor, a freely propagating particle has momentum that scales as $p \propto a^{-1}$. For example, for photons we have $p = h\nu/c$ and thus $\nu \propto a^{-1} = 1+z$.

8.2 Thermal distribution function

We first note that, at early epochs, the thermal equilibrium among the various particles is guaranteed to exist at any time. Indeed, although the dynamical time scales as $t \propto (G\rho)^{-1/2}$ [cf. for example Friedmann's Eq. (2.23)], and thus becomes very short at high densities, the two-body reaction rates scale as ρ^2. In addition, since many particles are weakly interacting (hot plasma), it is possible to use the **perfect gas approximation**.

In order to derive the distribution function of particles, we note that if we consider a box of volume $V = L^3$ and assume the usual periodic conditions, then we have allowed wavenumbers $k = n \times 2\pi/L$ in each direction. As a result the number of allowed states in the k space is

$$\mathrm{d}N = g \frac{V}{(2\pi)^3} \mathrm{d}^3 k , \tag{8.10}$$

and thus the number density $n = N/V$ is (as expected) independent of V. In addition, the occupation number of particles follows the usual law

$$f \propto \left[\mathrm{e}^{(\varepsilon - \mu)/kT} \pm 1 \right]^{-1} , \tag{8.11}$$

where ε is the quantum state energy, μ the chemical potential, and the sign $+$ (respectively, $-$), is valid for fermions (bosons). Calling $E = \sqrt{p^2 + m^2}$ the energy of the particles, we find the following general expression for the particle distribution function (Fermi-Dirac or Bose-Einstein distributions)

$$\boxed{f(\boldsymbol{p}, t)\, \mathrm{d}^3 p = \frac{g}{(2\pi\hbar)^3} \frac{\mathrm{d}^3 p}{\mathrm{e}^{(E-\mu)/T} \pm 1} ,} \tag{8.12}$$

where g is the statistical weight (for example, 2 for e^+, e^-, and γ and 1 for ν and $\bar{\nu}$). From this expression, we can find the particle density n, the energy density ρ, and the pressure P:

$$n = \int f(\boldsymbol{p})\, \mathrm{d}^3 p = \frac{g}{2\pi^2\hbar^3} \int_m^\infty \frac{\sqrt{E^2 - m^2} E\, \mathrm{d}E}{\mathrm{e}^{(E-\mu)/T} \pm 1} , \tag{8.13}$$

$$\rho = \int f(\boldsymbol{p}) E\, \mathrm{d}^3 p = \frac{g}{2\pi^2\hbar^3} \int_m^\infty \frac{\sqrt{E^2 - m^2} E^2\, \mathrm{d}E}{\mathrm{e}^{(E-\mu)/T} \pm 1} , \tag{8.14}$$

$$P = \int f(\boldsymbol{p}) \frac{|\boldsymbol{p}|^2}{3E}\, \mathrm{d}^3 p = \frac{g}{6\pi^2\hbar^3} \int_m^\infty \frac{\left(E^2 - m^2\right)^{3/2} \mathrm{d}E}{\mathrm{e}^{(E-\mu)/T} \pm 1} . \tag{8.15}$$

Note that in the limit $T \gg m$ we have $P = \rho/3$, as expected.

8.3 The entropy density

The first principle of thermodynamics can be written as

$$\mathrm{d}E = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N . \tag{8.16}$$

or equivalently, using rules like $\partial S/\partial V = S/V$ for extensive quantities

$$\left(\frac{E}{V}\mathrm{d}V + \frac{\partial E}{\partial T}\mathrm{d}T \right) = \left(T\frac{S}{V}\mathrm{d}V + T\frac{\partial S}{\partial T}\mathrm{d}T \right) - P\mathrm{d}V + \left(\mu\frac{N}{V}\mathrm{d}V + \mu\frac{\partial N}{\partial T}\mathrm{d}T\right) . \tag{8.17}$$

By equating the dV terms, we find a general relation for the entropy S:

$$S = \frac{E + PV - \mu N}{T} \,, \tag{8.18}$$

or, in terms of densities

$$s = \frac{\rho + P - \mu n}{T} \,. \tag{8.19}$$

If we consider the quantity sa^3, we have thus

$$\mathrm{d}(sa^3) = \mathrm{d}\left[\frac{a^3}{T}(\rho + P - n\mu)\right] = \frac{\mu}{T}\mathrm{d}(na^3) \,, \tag{8.20}$$

where for the last step we used the relation $\mathrm{d}(\rho a^3) = -P\mathrm{d}(a^3)$ and Eqs. (8.13–8.15). Hence, the quantity sa^3 is conserved if *either* $\mu/T \ll 1$ or $na^3 =$ const. Note also that, in case of the CMB radiation, the entropy density is extremely large. We have indeed $s = (\rho + P)/T = {}^4\!/_3\rho/T$, and since $\rho_{\mathrm{CMV}} = 4.5 \times 10^{-34}$ g cm^{-3} [cf. Eq. (6.22)], we have the following entropy per unit density in terms of the cosmological parameters:

$$\frac{S}{M} = 1.09 \times 10^{12} \bigl(\Omega h^2\bigr)^{-1} \ \mathrm{J\,K^{-1}\,kg^{-1}} \,. \tag{8.21}$$

By comparison, the same quantity for water at 300 K is only 14 J K^{-1} kg^{-1}. This calculation shows why we can consider the expansion of the Universe as an adiabatic reversible process: although, on small scales, the entropy can increases, these increases are negligible with respect to the total entropy of the CMB.

8.4 Ultra-relitivistic particles

In case $T \gg m$ and $T \gg \mu$, we have

$$\rho = \frac{g}{2\pi^2\hbar^3}\int_0^\infty \frac{E^3\,\mathrm{d}E}{\mathrm{e}^{E/T} \pm 1} = \frac{\pi^2 T^4}{30\hbar^3}\begin{cases} g_B & \text{for bosons} \\ 7g_F/8 & \text{for fermions} \end{cases} . \tag{8.22}$$

Hence, the total energy density of all relativistic species is

$$\boxed{\rho_{\rm tot} = g_{\rm tot}\frac{\pi^2 T^4}{30\hbar^3}\ ,} \tag{8.23}$$

where

$$g_{\rm tot} = \sum_{\rm bosons} g_i \left(\frac{T_i}{T}\right)^4 + \sum_{\rm fermions} \frac{7}{8} g_i \left(\frac{T_i}{T}\right)^4 . \tag{8.24}$$

In this equation T is the temperature of the CMB at the epoch considered. Regarding the entropy density, we find

$$\boxed{s \simeq \frac{\rho + P}{T} = \frac{2\pi^2}{45\hbar^3} q_{\rm tot} T^3\ ,} \tag{8.25}$$

where

$$q_{\rm tot} = \sum_{\rm bosons} g_i \left(\frac{T_i}{T}\right)^3 + \sum_{\rm fermions} \frac{7}{8} g_i \left(\frac{T_i}{T}\right)^3 . \tag{8.26}$$

The particle density n is given by

$$n = \frac{g}{2\pi^2\hbar^3}\int_0^\infty \frac{E^2\,\mathrm{d}E}{\mathrm{e}^{E/T}\pm 1} = \frac{\zeta(3)T^3}{\pi^2\hbar^3}\begin{cases} g_B & \text{for bosons} \\ 3g_F/4 & \text{for fermions} \end{cases}, \tag{8.27}$$

where $\zeta(3) \simeq 1.202$ is the zeta function.

Going back to the energy density, we note that during the radiation-dominated phase we have $a \propto t^{1/2}$. This implies that

$$\left(\frac{\dot a}{a}\right)^2 = H^2 = \frac{1}{4t^2} = \frac{8\pi G}{3}\rho = \frac{8\pi G}{3} g_{\rm tot}\frac{\pi^2 T^4}{30\hbar^3} . \tag{8.28}$$

From this equation we can find a time-temperature relation:

$$\boxed{\frac{t}{1\ \mathrm{s}} \simeq \left(\frac{T}{1\ \mathrm{MeV}}\right)^{-2} g_{\rm tot}^{-1/2} .} \tag{8.29}$$

Note that $g_{\rm tot}$ is the number of degrees of freedoms that are still relativistic at the temperature T. In particular, for $T \in [1, 100]$ MeV, this number is

$$g_{\rm tot} = g_B + \frac{7}{8} g_F = 2 + \frac{7}{8}(2+2+2\cdot 3) = 10.75\ , \tag{8.30}$$

where @@@

8.5 Freeze-out

Let us consider the Bolzmann equation:

$$\frac{\partial f}{t} + \dot{\boldsymbol{x}} \cdot \frac{\partial f}{\partial \boldsymbol{x}} + \dot{\boldsymbol{p}} \cdot \frac{\partial f}{\partial \boldsymbol{p}} = \dot{f}_{\mathrm{c}} \,, \tag{8.31}$$

where the r.h.s. term accounts for possible interactions. For a homogeneous distribution the second term on the l.h.s. vanishes. In addition, if we use $\dot{\boldsymbol{p}} = -(\dot{a}/a)\boldsymbol{p} = -H\boldsymbol{p}$ and if we neglect collisions, then

$$\frac{\partial f}{\partial t} - H\boldsymbol{p} \cdot \frac{\partial f}{\partial \boldsymbol{p}} = 0 \,. \tag{8.32}$$

The general solution of this transport equation is

$$f(\boldsymbol{p},t) = f_{\mathrm{D}}\left(\boldsymbol{p}\frac{a(t)}{a(t_{\mathrm{D}})}, t_{\mathrm{D}}\right) \,. \tag{8.33}$$

In other words, each particle is shifted in the $\boldsymbol{p}$ space along lines with $p(t)a(t) = \text{const}$. In particular, if we consider particles relativistic at the time t_{D} and in thermal equilibrium, then

$$f_D(\boldsymbol{p},t_{\mathrm{D}}) = \frac{g}{(2\pi\hbar)^3}\frac{1}{\mathrm{e}^{p/T(t_{\mathrm{D}})} \pm 1} \,, \tag{8.34}$$

where we have assumed that the chemical potential $\mu \ll T_{\mathrm{D}}$. At later times, then

$$f(\boldsymbol{p},t) = \frac{g}{(2\pi\hbar)^3}\frac{1}{\mathrm{e}^{pa(t)/T(t_{\mathrm{D}})a(t_{\mathrm{D}})} \pm 1} \,. \tag{8.35}$$

In other words, the particles evolution can be understood as a change of the temperature of the distribution

$$T(t) = T(t_{\mathrm{D}})\frac{a(t_{\mathrm{D}})}{a(t)} \,. \tag{8.36}$$

We now consider again Bolzmann equation, but with a collision term:

$$\frac{\partial f}{\partial t} - H\boldsymbol{p} \cdot \frac{\partial f}{\partial \boldsymbol{p}} = \dot{f}_{\mathrm{c}} \,. \tag{8.37}$$

The collision term $\dot{f}_c$ is usually dominated by particle-antiparticle reactions. As a specific example, think of $\nu\bar{\nu} \leftrightarrow \mathrm{e}^+\mathrm{e}^-$. Assuming that the particle-antiparticle density are identical, so that there is no significant asymmetry, the collision term can be written as

$$f_c = -\int \langle \sigma v \rangle f \bar{f} \, \mathrm{d}^3 p \,, \tag{8.38}$$

where $\langle \sigma v \rangle$ is the velocity-average product of the cross-section and velocity. This term is usually slowly varying with p, and in general can be taken outside the integral provided that it is calculated at a suitable momentum. We then integrate the Bolzmann equation over the momentum to obtain

$$\frac{\mathrm{d}n}{\mathrm{d}t} + 3Hn = -\langle \sigma v \rangle n^2 + S \,, \tag{8.39}$$

where S is a source term which has been added to take into account the particle production from thermal processes (pair creation). In order to find the form of this term, we use the following argument. Consider a non-expanding universe ($H = 0$) at temperature T: in this case, n will be constant at the equilibrium value for the temperature T, n_T, and thus

$$S = \langle \sigma v \rangle n_T^2 \,. \tag{8.40}$$

If we define the *comoving number density* $N \equiv a^3 n$, our result takes a simple form:

$$\frac{\mathrm{d} \ln N}{\mathrm{d} \ln a} = -\frac{\Gamma}{H} \left(1 - \frac{N_{\mathrm{eq}}^2}{N^2} \right) \,, \tag{8.41}$$

where $\Gamma = n \langle \sigma v \rangle$ is the interaction rate.

The qualitative behaviour of the solutions of Eq. (8.41) can be easily understood. Suppose initially that we deal with a population of relativistic particles in thermal equilibrium, so that $N = N_{\mathrm{eq}}$ initially. Then, it is easy to see that $N = N_{\mathrm{eq}}$ at future times as well, since for relativistic particles $n_T \propto T^3$ and the temperature scales as $T \propto a^{-1}$. Hence, for these particles it is possible to keep $N = N_T$ at any time, even if Γ/H decreases significantly (this is for example the case of the CMB). For nonrelativistic particles, instead, we have $N_T \propto T^{-3/2} \exp(-mc^2/kT)$, and thus N_T

in general can change significantly during the expansion of the Universe. When N_T changes fast with a, the equilibrium is then maintained only when the prefactor Γ/H is large enough. In practice, there will be a time when the reaction rates drops, and the equilibrium cannot be maintained anymore; this happens approximately when $\Gamma/H = 1$. At later times, the particles density N will thus be frozen to the original value at the froze-out, i.e.

$$N(a \to \infty) \simeq (\Gamma/H = 1) \,. \tag{8.42}$$

This approximate solution is indeed very close to the results shown by more precise numerical calculations.

8.6 Neutrino decoupling

In the early Universe neutrinos are kept in equilibrium by several reactions, including

$$\nu + \bar{\nu} \leftrightarrow e^+ + e^- \,, \qquad \nu + e \leftrightarrow \nu + e \,. \tag{8.43}$$

The cross section for these interactions is approximately given by

$$\sigma \simeq G_F^2 E^2 \simeq G_F^2 T^2 \,, \tag{8.44}$$

where $G_F = 1.17 \times 1.17 \times 10^{-5}\ \mathrm{GeV}^{-2}$ is the Fermi constant. Since the number density of neutrinos is given by [cf. Eq. (8.27)]

$$n \simeq \frac{\zeta(3) g}{\pi^2 \hbar^3} T^3 \,, \tag{8.45}$$

for ultra-relativistic neutrinos we find

$$\frac{\Gamma}{H} \simeq \left(\frac{T}{1.4\ \mathrm{MeV}} \right)^3 \simeq \left(\frac{T}{1.6 \times 10^{10}\ \mathrm{K}} \right) \,. \tag{8.46}$$

As a result, neutrinos decouple from photons and electrons when $T = T_D \simeq 1$ MeV, and after that they form a background radiation field with $T_\nu = T_D a(t_D)/a(t)$.

To understand better the relationship between the neutrino background and the photon background (CMB), let us consider the process

$$\gamma + \gamma \rightarrow e^+ + e^- \ , \tag{8.47}$$

i.e. the production of a electron-positron pair. For $T < 0.5$ MeV this process is suppressed, since photons are not longer energetic enough; the opposite process, annihilation, however proceeds. During the annihilation process, the entropy is transferred from the electron-positron pair into the photons. Since the entropy is conserved for particles in thermal equilibrium, we can study the evolution of the temperature during the annihilation process. Before the annihilation, we have 3 species in thermal equilibrium, γ, e^-, and e^+, and since $T = T_\gamma$ for all of them, we find $q_{\text{tot}} = g_{\text{tot}}$. After the annihilation, we only have photons in thermal equilibrium. Using Eq. (8.25) and imposing the entropy conservation we find then

$$S = q_{\text{tot}}(Ta)^3 = \text{const.} \ , \tag{8.48}$$

and thus

$$\frac{(aT_\gamma)^3_{\text{after}}}{(aT_\gamma)^3_{\text{before}}} = \frac{q_{\text{before}}}{q_{\text{after}}} = \frac{11}{4} \ . \tag{8.49}$$

In other words

$$T_{\gamma,\text{after}} = \left(\frac{11}{4}\right)^{1/3} T_{\gamma,\text{before}} = 1.4 T_{\gamma,\text{before}} = 1.4 T_\nu \ . \tag{8.50}$$

Hence, even today, the temperature of neutrinos is 1.4 times smaller than the temperature of CMB photons, or $T_\nu = 1.9$ K.

8.7 Brief thermal history of the Universe

In this section we briefly summarize our current knowledge on the thermal history of the Universe.

- Early phase (symmetry breaking). At $t < 10^{-2}$ sec ($T > 10$ MeV) several processes are suppose to break original symmetries present in unification theories. The results are asymmetries that we observe today in the Universe, such as the matter-antimatter asymmetry or the lack of magnetic monopoles.

- Nucleosyntesis. From $t \sim 10^{-2}$ sec to $t \sim 10^{2}$ sec the temperature of the Universe is low enough to allow for the production of simple elements (see next chapter). In particular, a consistent quantity of Helium is produced, together with small quantities of Lithium and Deuterium.
- Matter domination. At $t \sim 10^{11}$ sec the matter density becomes equal to the radiation density. This is a crucial moment in the story of the Universe, since it triggers the formation of structures.
- Recombination. At $t \sim 10^{13}$ sec the electrons combine to form atoms, and the matter and radiation decouple. This ends the long epoch of near thermal equilibrium that characterized the early Universe.

9 Big bang nucleosynthesis

In this chapter, for simplicity, we will use *natural* units where $k_B = c = \hbar = 1$.

9.1 Baryon-to-photon ratio

A parameter that plays a fundamental role in nucleosynthesis is the baryon-to-photon ratio $\eta \equiv n_N/n_\gamma$, where n_N is the density of nucleons. To understand better this quantity, we first note that the entropy per comoving volume is conserved [cf. Eq. (8.20)], and as a result the inverse of entropy density s can be taken as a measurement of comoving volume. As shown in Sect. 8.3, the entropy is dominated by relativistic particle, so that [cf. Eq. (8.29)] $q_{\text{tot}}T^3a^3$ remains constant as the Universe expands. Using Eq. (8.27), we can also express the entropy density s in terms of the photon density n_γ, obtaining

$$\boxed{s \simeq 1.80 q_{\text{tot}} n_\gamma\,.} \tag{9.1}$$

Hence, as long as q_{tot} is constant, we can use interchangeably the entropy density s and the photon density n_γ. The baryon number in a comoving volume,

$$B \equiv \frac{n_B}{s} \equiv \frac{n_b - n_{\bar{b}}}{s}\,, \tag{9.2}$$

is conserved as long as baryon number non-conserving interactions are not present. All evidence we have today indicates that there are very few antibaryons in the Universe, and that the only baryons present are nucleons, so that de-facto

$$n_B = n_N = 1.12\times 10^{-5}\,\Omega_B h^2\ \text{cm}^{-3}\,. \tag{9.3}$$

$$B \equiv \frac{n_B}{s} = 3.80\times 10^{-9}\,\Omega_B h^2\,. \tag{9.4}$$

In addition, during the era of e^+e^- annihilations the number of photons per comoving volume $n_\gamma a^3$ increased by a factor of $11/4$ (cf. Sect. 8.6),

and thus η decreased by the same factor. Since then, however, $q_{\rm tot}$ remained constant, and thus η was conserved (note that today we have $s = 7.04 n_\gamma$).
From Eq. (9.4) we can also deduce that the baryon-to-photon ratio is

$$\eta \simeq 7B \simeq 2.74 \times 10^{-8}\,\Omega_B h^2 \,. \tag{9.5}$$

The fact that $\eta \ll 1$ means that the Universe is “hot”, i.e. that there is a very high entropy per baryon. This, as we will see below, plays a crucial role for the nucleosyntesis.

9.2 Nuclear statistical equilibrium

In order to understand the production of light elements in the early universe, let us go back to the equations governing the number density, energy density, and pressure of particles, Eqs. (8.13–8.15). In the non-relativistic limit $m \gg T$ we find for both bosons and fermions

$$n = g\left(\frac{mT}{2\pi}\right)^{3/2} \exp\left(-\frac{m-\mu}{T}\right) \,, \tag{9.6}$$

$$\rho = mn \,, \tag{9.7}$$

$$P = nT \ll \rho \,. \tag{9.8}$$

Let us consider now an atom with mass number A and charge Z. If kinetic equilibrium holds, its number density is given by Eq. (9.6), where of course g, m, and μ refer to the appropriate quantities for this species. If the nuclear reactions that produce this atom out of Z protons and $A-Z$ neutrons occurs rapidly, then the chemical equilibrium implies

$$\mu_A = Z\mu_p + (A-Z)\mu_n \,, \tag{9.9}$$

where μ_p (respectively, μ_n) are the chemical potentials of protons and neutrons. In addition, since Eq. (9.6) applies also to protons and neutrons, we can write

$$\begin{aligned}\exp(\mu_A/T) &= \exp\left[(Z\mu_p+(A-Z)\mu_n)/T\right] \\ &= n_p^Z n_n^{A-Z}\left(\frac{2\pi}{m_N T}\right)^{3A/2} 2^{-A}\exp\left[(Zm_p+(A-Z)m_n)/T\right] .\end{aligned} \tag{9.10}$$

Note that the factor 2^{-A} here arises from the statistical weights of protons and neutrons $g_n = g_p = 2$.

The process of nucleosynthesis is typically associated to the release of energy because of the atom *binding energy*. This quantity is defined as the mass difference between the nucleons composing the atom and the atom itself:

$$B_A \equiv Zm_p + (A-Z)m_n - m_A \, . \tag{9.11}$$

Using this definition in Eq. (9.6) and substituting there Eq. (9.10) we find

$$n_A = g_A A^{3/2} 2^{-A}\left(\frac{2\pi}{m_N T}\right)^{3(A-1)/2} n_p^Z n_n^{A-Z}\exp(B_A/T) \, . \tag{9.12}$$

The binding energies of the simplest elements, deuterium, tritium, and helium, are of the order of a few MeV:

$$\mathrm{D}: 2.22\ \mathrm{MeV} \quad {}^3\mathrm{H}: 6.92\ \mathrm{MeV} \quad {}^3\mathrm{He}: 7.72\ \mathrm{MeV} \quad {}^4\mathrm{He}: 28.3\ \mathrm{MeV} \, . \tag{9.13}$$

Going back to Eq. (9.12), we note that the number densities of particles decreases as a^{-3} (assuming a constant number of comoving volume). It is useful then to define the mass fraction contributed by nuclear species $A(Z)$ as

$$X_A \equiv \frac{n_A A}{n_N} \, . \tag{9.14}$$

In addition, we can express the baryon number density as

$$n_N = n_n + n_p + \sum(A n_A) \, , \tag{9.15}$$

so that $\sum X_A = 1$. Using these definitions we find

$$X_A = g_A \left[\zeta(3)^{A-1} \pi^{(1-A)/2} 2^{(3A-5)/2}\right] A^{5/2} (T/m_N)^{3(A-1)/2} \\ \times \eta^{A-1} X_p^Z X_n^{A-Z} \exp(B_A/T) \,, \tag{9.16}$$

where we have expressed here the baryon density n_N as $n_N = \eta n_\gamma$ and used Eq. (8.27) to write n_γ in terms of T. This equation controls the abundances of elements in nuclear statistical equilibrium, and will be applied in this chapter to determine the elements created by in the few first minutes of the history of the Universe. To this purpose, however, we need first to understand the initial conditions for the nucleosynthesis process.

9.3 Initial conditions

For $t \ll 1$ sec, the temperature of the Universe is $T \gg 1$ MeV, which is much higher than typical binding energies [see above Eq. (9.13)] and thus only protons and neutrons are present. The ratio of neutrons to protons is particularly important here, since essentially all neutrons will form ^{4}He. This ratio can be calculated by taking into account the weak interactions that are responsible for the balance between neutrons and protons:

$$n \leftrightarrow p + e^- + \bar{\nu} \,, \tag{9.17}$$

$$\nu + n \leftrightarrow p + e^- \,, \tag{9.18}$$

$$e^+ + n \leftrightarrow p + \bar{\nu} \,. \tag{9.19}$$

When these reactions proceed at a high rate compared to the expansion rate of the Universe, the chemical equilibrium holds and we have thus

$$\mu_n + \mu_\nu = \mu_p + \mu_e \,, \tag{9.20}$$

which implies

$$\frac{n_n}{n_p} = \frac{X_n}{X_p} = \exp\left(-\frac{Q - \mu_e + \mu_\nu}{T}\right) \,, \tag{9.21}$$

where $Q \equiv m_n - m_p = 1.293$ MeV is the energy difference between neutrons and protons. In order to proceed further, we need first to estimate the chemical potentials entering Eq. (9.21)

9.4 Chemical potential

The chemical potential of the electron μ_e can be estimated by first noting that the chemical potentials of particle and antiparticles are one the opposite of the other. In fact, from the reaction $\gamma + e \leftrightarrow \gamma + \gamma + e$ we can deduce that the chemical potential of the photon vanishes; moreover, from the annihilation process $e^+ + e^- \leftrightarrow \gamma + \gamma$ we have that $\mu_{e^+} = -\mu_{e^-}$. Clearly, the same argument applies to any particle and antiparticle.
Using the latter relation in Eq. (8.13), we then find for the difference of number densities of particle and antiparticle

$$\begin{aligned} n - \bar{n} &= \frac{g}{2\pi^2} \int_m^\infty E\sqrt{E^2 - m^2} \\ &\quad \times \left[\frac{1}{1 + \exp\left[(E - \mu)/T\right]} - \frac{1}{1 + \exp\left[(E = \mu)/T\right]} \right] \mathrm{d}E \\ &\simeq \frac{gT^3}{6\pi^2} \left[\pi^2 \left(\frac{\mu}{T}\right) + \left(\frac{\mu}{T}\right)^3 \right] , \end{aligned} \tag{9.22}$$

where the last relation holds in the limit $T \gg m$. From this equation, under the hypothesis of charge neutrality of the Universe, we can deduce the order of magnitude of the ratio μ_e/T which appears in Eq. (9.21):

$$\frac{\mu_e}{T} \sim \frac{n_{e^-} - n_{e^+}}{n_\gamma} = \frac{n_p}{n_\gamma} \simeq \eta \simeq 10^{-10} . \tag{9.23}$$

Hence, for our purposes we can completely neglect the chemical potential of the electron.
In principle a similar argument applies to neutrinos, but since the relic neutrino background has not been detected, we do not know the neutrino lepton numbers. If we assume that, like the baryon number B, these are small, than we can also conclude that $|\mu_\nu|/T \ll 1$.
In these conditions, Eq. (9.21) simplifies to

$$\frac{n_n}{n_p} = \frac{X_n}{X_p} = \exp\left(-\frac{Q}{T}\right) \simeq \exp\left(-\frac{1.5 \times 10^{10}\ \mathrm{K}}{T}\right) . \tag{9.24}$$

Clearly, the fact that neutrons exists today means that neutrons experienced a freeze-out well before the temperature of the Universe dropped significantly below the value of Eq. (9.24).

9.5 Neutron freeze-out

As shown in Sect. 8.5, the reaction rate Γ plays a key role for the freeze-out of species. In case of neutrons, we need to consider the rates corresponding the reactions of Eqs. (9.17–9.21). These quantities can be found by integrating the square of the matrix element for a given process, weighted by the available phase-space densities of particles (other than the initial nucleon), while enforcing the four-momentum conservation. For example, for $p+e^- \rightarrow \nu+n$ we have

$$\Gamma_{pe\rightarrow\nu n} = \int f_e(E_e)\left[1-f_\nu(E_\nu)\right]|\mathscr{M}|^2_{pe\rightarrow\nu n}(2\pi)^{-5}\delta^{(4)}(p+e-\nu-n) \times \frac{\mathrm{d}^3p_e}{2E_e}\frac{\mathrm{d}^3p_\nu}{2E_\nu}\frac{\mathrm{d}^3p_n}{2E_n}\,. \tag{9.25}$$

The matrix element $\mathscr{M}$ can be calculated using the Fermi theory of weak interactions, and is proportional to $|\mathscr{M}|^2 \propto G_F^2(1+3g_A^2)$, where G_F is the Fermi constant and $G_A \sim 1.26$ is the axial-vector coupling of the nucleon. Since the Fermi Lagrangian can be understood in terms of creation and annihilation operators, the various matrix elements $\mathscr{M}$ corresponding to the different reactions are essentially the same. This allows us to express them in terms of the neutron mean lifetime τ_n. In particular

$$\tau_n^{-1} = \Gamma_{n\rightarrow pe\nu} = \frac{G_F^2}{2\pi^3}(1+3g_A^2)m_e^5\lambda_0\,, \tag{9.26}$$

where λ_0 is a numerical factor arising from the phase space integration:

$$\lambda_0 \equiv \int_1^q \varepsilon(\varepsilon-q)^2(\varepsilon^2-1)\,\mathrm{d}\varepsilon \simeq 1.636\,. \tag{9.27}$$

In this equation we used as integration variable $\varepsilon = E_e/m_2$; the factor q is a dimensionless quantity $q \equiv Q/m_e \simeq 2.53$. If we use τ_n to express the rate $\Gamma_{pe\to\nu n}$ we obtain

$$\Gamma_{pe\to\nu n} = (\tau_n\lambda_0)^{-1}\int_q^\infty \frac{\varepsilon(\varepsilon-q)^2(\varepsilon^2-1)^{1/2}}{\left[1+\exp(\varepsilon z)\right]\left[1+\exp((q-\varepsilon)z_\nu\right]}\,\mathrm{d}\varepsilon\ , \tag{9.28}$$

where $z \equiv m_e/T$ and $z_\nu \equiv m_e/T_\nu$ are two dimensionless quantities. In the conditions considered ($T \gg 1$ MeV), the reaction rates takes the form

$$\Gamma_{pe\to\nu n} \simeq 13.893\,\tau_n^{-1}T^5/m_e^5\ . \tag{9.29}$$

Interesting, an identical result is obtained in the same limit for the other reaction rates involves. The reason is that (1) all integrals have similar forms and (2) since $m_e \sim Q$, at high temperatures the expansions of the denominators provide similar expressions.
In the radiation-dominated era the Hubble parameter drops as $H \propto T^{-2}$, and as a result the ratio Γ/H behaves as

$$\Gamma/H \simeq (T/0.8\ \mathrm{MeV})^3\ . \tag{9.30}$$

Hence, because of the T^3 dependence of this quantity, we deduce that the neutrons go from a chemical equilibrium to a freeze-out in a rather short timescale. Moreover, at temperatures much larger than 0.8 MeV we expect the chemical equilibrium to take place, and thus the neutron-to-proton ratio to closely approach 1.
At temperatures $T \gg 1$ MeV, not only the weak interactions happen at high rate (so that $n_n/n_p \simeq 1$), but also the rates for the nuclear reactions that build up the light elements are fast. However, a detailed study of the chemical equilibrium abundances for the various elements shows that, although for temperatures less than a few MeV nuclei are favoured on energetic grounds, their mass fraction does not approach unity until approximately 0.3 MeV. The reason for that is the small value of η, i.e. the large entropy of the Universe. Indeed, Eq. (9.16) shows that a nucleus with weight mass A is disfavoured by a factor η^{A-1}, which is extremely small for $A > 1$.

9.6 Production of light elements

In this section we will consider more closely the processes that concur to the production of the simple elements in the Universe. For simplicity, we will divide the discussion into three steps

- $t = 10^{-2}$ sec, $T = 10$ MeV. At this epoch, radiation dominates and we have $g_{\rm tot} = 10.75$ (e^+, e^-, γ, and the three neutrino species). All weak interaction rates are much larger than the expansion rate H, and thus $n_n/n_p \simeq 1$ and $T_\nu = T$. The light elements are in thermal equilibrium, but their abundances are very small because of the value of η (for example, for $\eta = 10^{-9}$ we find $X_2 \simeq 6 \times 10^{-12}$ and $X_4 \simeq 2 \times 10^{-34}$).

- $t \simeq 1$ sec, $T \simeq 1$ MeV. Shortly before this epoch the 3 neutrino species decouple from plasma (see Sect. 8.6), and a little later the e^+ and e^- pair annihilates, transferring their entropy to the photons alone and thus increasing their temperature by a factor $(11/4)^{1/3}$ (Sect. 8.6). In addition, at this temperature the weak interaction between neutrons and protons freeze-out, leaving

  $$\frac{n_n}{n_p} = \exp(-Q/T) \simeq \frac{1}{6}\,. \tag{9.31}$$

  In reality, because of the spontaneous decay of neutrons, this ratio does not remains strictly constant, but decreases slowly with time. At this time the nuclear species are still in equilibrium, with very small abundances ($X_2 \simeq 10^{-12}$, $X_4 \simeq 10^{-23}$).

- $t = 1$–3 minutes, $T = 0.3$–0.1 MeV. At this time $g_{\rm tot}$ has reached its current value, 3.36, because $e^\pm$ pairs have completely annihilated. In addition, thermal reactions build the heavier elements. At $T = 0.3$ MeV the ^{4}He mass fraction quickly approaches unity. Shortly before, ^{4}He is actually below its equilibrium value because the reaction rates Γ that control the production of Helium are not high enough (this happens because there is not enough "fuel" for

He, Deuterium, and because of Coulomb-barrier suppression, that starts to be significant at these relatively low temperatures; this is the so-called “light-element bottleneck”). At $T \simeq 0.1$ MeV all neutrons produces essentially are bound within He. The mass fraction of He is thus easily given by

$$X_4 \simeq \frac{4n_4}{n_N} = \frac{4(n_n/2)}{n_n+n_p} = \frac{2(n_n/n_p)}{1+(n_n/n_p)} \simeq 0.25 \; . \tag{9.32}$$

Heavier elements, in principle, could also form abundantly at epoch, because their binding energy is larger than ^{4}He. However, the lack of stable isotopes with $A = 5$ and $A = 8$ and the presence of a significant Coulomb barrier prevents the formation of significant atoms beyond ^{4}He. Some ^{7}Li is synthesized ($X_7 \simeq 10^{-9}$), and as we will see below this element plays an important role for astrophysical applications of primordial nucleosynthesis.

9.7 Observations

Exact abundances for the various elements can be predicted by nuclear reaction networks. Currently, these calculations are rather reliable, and the main sources of uncertainties have disappeared. In the 80’s, indeed, fundamental quantities like τ_n were known with surprisingly large uncertainties (errors of the order of a few percent), and that lead to related uncertainties on the calculations.

The abundances predicted depends non-trivially from η. For example, if η increases, the production of ^{4}He starts earlier, when the ratio n_n/n_p is larger, and thus a slightly larger abundances of ^{4}He is expected. However, other elements can have a significantly different behaviour: for example, D significantly decreases as η increases.

Clearly, if we can measure the abundances of the *primordial* ^{4}He or D we could estimate the value of η and thus of the baryon density Ω_b. The problem clearly that it is not obvious how to obtain an estimate of the “primordial” Helium. Surprisingly, till the 60’s it was not generally accepted that the Helium present in the Universe could be mostly produced by primordial nucleosynthesis, and the situation changed only in the 70's, when it was shown that no astrophysical process could produce the observed fraction of D (D is very easily destroyed in astrophysicall processes, but not created). In particular, in the early 70's the COPERTNICUS UV satellite measured deuterium in the interstellar medium (ISM) of nearby molecular clouds. These observations provided the first evidence that the Universe has a small baryon content.

The abundance of ^{4}He is usually inferred by observing atmospheres of metal-poor stars, and by extrapolating the measured Helium at zero metallicity. Measurements of this species are considered now quite robust, and thus although the dependence of the predicted ^{4}He with η is relatively weak, still the Helium provides a very robust upper limit on η.

Lithium has been traditionally difficult to measure because of its low abundance. Its non-trivial dependence on η, due to two different channels for its production, has provided however relatively strict bounds on η. Today, the combination of estimates ^{4}He and especially of $D+{}^3$He is thought to give more stringent limits. The best estimate, converted into Ω_b, in particular is

$$\Omega_b h^2 \simeq 0.022\,. \tag{9.33}$$

9.8 Neutrino families

In the past, the nucleosynthesis observations have also provided some stringent limits on the number of relativistic neutrino families. Note that the quantity $g_{\rm tot}$ is related to the expansion rate of the Universe by [cf. Eq. (8.28)]

$$H \propto g_{\rm tot}^{1/2} T^2\,. \tag{9.34}$$

As a result, an increase of $g_{\rm tot}$ (for example, as a consequence of a new neutrino species) would increase H and thus would imply an earlier freeze-out of neutrons. This, in turns, trigger a larger production of ^{4}He (because of the larger n_n/n_p ratio at higher temperatures), which might not be in agreement with observations. As a result, one can limit the allowed families of neutrinos by observations of primordial Helium (for example, a new neutrino type would increase $g_{\rm tot}$ from 10.75 to 12.5). Today stronger limits on the neutrino species are put by Z^0 experiments at CERN.

10 Recombination

In this chapter we will describe the process of (re)combination, i.e. the decoupling of matter and radiation. The main source of coupling between photons and matter is provided by (Compton) scattering of the electrons (protons are much more massive and have thus a much smaller cross section). The coupling, hence, is efficient as long as there is a significant fraction of plasma in the Universe.

10.1 Fraction of ionized hydrogen

The binding energy of hydrogen is given by

$$B = m_p + m_e - m_{\mathrm{H}} = 13.6 \text{ eV} . \tag{10.1}$$

When $T \lesssim B$, one would expect that most atoms are neutrals, since H is energetically favored. In reality, since $\eta \equiv n_b/n_\gamma \sim 10^{-10}$ is extremely small, there is always a substantial number of high-energy photons in the Wien tail of the Plank distribution to ionize atoms. As we will see below, then, one needs to reach temperatures T significantly smaller than B before the (almost totality) of H atoms is neutral.
The key reaction we are considering here is

$$p + e \leftrightarrow \mathrm{H} + \gamma , \tag{10.2}$$

and thus we have for the chemical potentials

$$\mu_p + \mu_e = \mu_{\mathrm{H}} . \tag{10.3}$$

Using Eq. (9.6), we can write for the three massive species considered (H, p, and e) the nuclear statistical equilibrium as

$$n_a = g_a \left(\frac{m_a T}{2\pi} \right)^{3/2} \exp\left(-\frac{m_a - \mu_a}{T} \right) . \tag{10.4}$$

From Eq. (10.3) we can then write

$$n_{\mathrm{H}} = \frac{g_{\mathrm{H}}}{g_p g_e} n_p n_e \left(\frac{m_e T}{2\pi} \right)^{-3/2} \mathrm{e}^{B/T} , \tag{10.5}$$

where we also used Eq. (8.27) to express the density of photons in terms of the temperature T.

In the rest of this chapter we will ignore for simplicity the helium, i.e. we will assume that the Universe is composed by hydrogen only. Then, calling as usual $n_b = n_p + n_{\mathrm{H}}$ the number density of barions, we have for the ratios of ionized atoms

$$x_p \equiv \frac{n_p}{n_b} = \frac{n_e}{n_b} \equiv x_e \equiv x \,, \tag{10.6}$$

where the charge neutrality of the Universe has been used. Since the ionization ratio x plays a central role in our discussion, we can obtain an exact solution for it:

$$\begin{aligned} n_{\mathrm{H}} &= (1-x)n_b \,, && (10.7) \\ \frac{1-x}{x^2} &= \frac{n_{\mathrm{H}}}{n_p n_e} n_b = \frac{n_{\mathrm{H}}}{n_p n_e} \eta n_\gamma = \frac{4\sqrt{2}\zeta(3)}{\pi} \nu \left(\frac{T}{m_e}\right)^{3/2} \mathrm{e}^{B/T} \\ &\simeq 3.84\eta \left(\frac{T}{m_e}\right)^{3/2} \mathrm{e}^{B/T} \,. && (10.8) \end{aligned}$$

A sensible threshold to consider is the one when $x \simeq 0.1$: in this case, only 10% of the atoms are still ionized, and the Universe is close to the complete recombination. The temperature T_{atom} at this phase can in principle be estimated by Eq. (10.8); in practice, this equation needs to be solved numerically because of the combination $T^{3/2}\mathrm{e}^{T}$. An approximate solution can nevertheless be obtained by defining $\tau \equiv T/B$, so that

$$\frac{90}{3.84} = \eta \left(\tau \frac{B}{m_e}\right)^{3/2} \mathrm{e}^{1/\tau} \,, \tag{10.9}$$

or

$$\log\left(\frac{90}{3.84}\right) = \log\eta + \frac{3}{2}\log\left(B/m_e\right) + \frac{3}{2}\log\tau + \frac{1}{\tau} \,. \tag{10.10}$$

We obtain then a series of solutions, whose first terms are given by

$$\begin{aligned} \frac{1}{\tau_1} &\simeq \log\left(\frac{90}{3.84}\right) - \log\eta - \frac{3}{2}\log\left(\frac{B}{m_e}\right) \,, && (10.11) \\ \frac{1}{\tau_1} &\simeq \frac{1}{\tau_1} - \frac{3}{2}\log\tau_1 \,, && (10.12) \end{aligned}$$

so that

$$\frac{T_{\rm atom}}{1\ {\rm eV}} \simeq \frac{1}{3.084 - 0.0735\log(\Omega_{\rm b}h^2)}\,, \tag{10.13}$$

$$1+z \simeq \frac{1367}{1-0.024\log(\Omega_{\rm b}h^2)}\,. \tag{10.14}$$

The reaction rate $p+e \leftrightarrow H+\gamma$ is fast enough to maintain the equilibrium for $T > T_{\rm atom}$. However, as the atoms get neutral, there are not enough free electrons to keep a high Compton scattering rate. As discussed above, when

$$\Gamma_{\gamma e}/H \simeq 1\,, \qquad T_{\rm dec} \simeq 0.26\ {\rm eV}\,, \qquad z_{\rm dec} \simeq 1100\,, \tag{10.15}$$

radiation and matter decouple. Note that matter is thus neutral from redshift $z \sim 1000$. However, there is a strong indication that in the local Universe matter is highly ionized: if that would not be the case, we would see a strong decrease on the detected radiation to the blue of the Lyα in high-redshift objects (this is the so-called Gunn-Peterson effect).

10.2 Two-photon emission

In reality, the recombination process shows some peculiarities that needs to be understood in more detail. One problem with the simple picture discussed above is that each recombination to the hydrogen ground state generates a high-energy photon that is capable of ionizing a nearby neutral hydrogen atom. Hence, this way we have no net effect, because each recombination is rapidly followed by a new ionization. Interestingly, since at the epoch considered the photon Plank distribution is characterized by a relatively low temperature (much smaller than the hydrogen binding energy), the photons generated through the process just described constitute a significant deviation from the equilibrium distribution at high frequencies.

There are two possible solution to this problem:

1. Because of the Universe expansion, re-emitted photons are redshited and thus are not capable anymore of ionizing new atoms.

In reality this mechanism is not efficient, because the mean free path of photons is still too short (if a photon finds easily a atom to ionized there has not been enough time to redshift it).

2. Some atoms reach the ground state through a two-photon emission $2s \rightarrow 1s + 2\gamma$. This process is strictly prohibited to first order (with a single photon we cannot have energy and momentum conservation), but is possible with two photons. Interestingly, the inverse process is a three-body reaction, which is rare and thus unimportant: hence we have a mechanism to get rid of energy from the ionized atom in form of non-ionizing photons.

In order to understand the two-photon emission, we consider a simple model for the hydrogen atom composed of two levels only: the ionized state, and the $2s$ state. We have then the following relations for the number density of electrons n_e and of $2s$ hydrogen atoms n_2:

$$\frac{\mathrm{d}n_e}{\mathrm{d}t} = -Rn_e n_p + n_2 \Lambda_u \,, \tag{10.16}$$

$$\frac{\mathrm{d}n_2}{\mathrm{d}t} = Rn_e n_p - n_2(\Lambda_u + \Lambda_{2\gamma}) \,, \tag{10.17}$$

where $R = R(T)$ is the recombination coefficient, $\Lambda_u = \Lambda_u(T)$ is the rate for the $2s + \gamma \rightarrow e + p$ reaction, and $\Lambda_{2\gamma} = \Lambda_{2\gamma}(T)$ is the rate for the $2s \rightarrow 1s + 2\gamma$ reaction. Assuming that n_2 is constant, than we find

$$n_2 = \frac{Rn_e n_p}{\Lambda_u + \Lambda_{2\gamma}} \,, \tag{10.18}$$

and thus

$$\frac{\mathrm{d}n_e}{\mathrm{d}t} = -Rn_e^2 \left(1 - \frac{\Lambda_u}{\Lambda_u + \Lambda_{2\gamma}}\right) = -Rn_e^2 \frac{\Lambda_{2\gamma}}{\Lambda_u + \Lambda_{2\gamma}} \,. \tag{10.19}$$

In this equation the quantities Λ_u, $\Lambda_{2\gamma}$, and R are function of the temperature T and do not directly depend on the cosmological parameters. In order to show the dependence of the solution from the cosmological parameters, note that $n_e = nx \propto \Omega_B h^2$. By rewriting $\mathrm{d}/\mathrm{d}t = (\mathrm{d}z/\mathrm{d}a)(\mathrm{d}a/\mathrm{d}t)\mathrm{d}/z$, and by using the relevant expressions in a matter dominated Universe for the various differentials, we can write the expected dependence of the solution $x(z)$ on the cosmological parameters:

$$x(z) \propto \frac{(\Omega h^2)^{1/2}}{\Omega_B h^2} \,. \tag{10.20}$$

The real solution can be found by direct, numerical integration of Eq. (10.19). Because of the simple factorization of the cosmological parameters expressed in Eq. (10.20), effectively one can solve Eq. (10.19) for a particular model, and rescale the solution from Eq. (10.20). The final result obtained is well approximated by

$$x(z) \simeq 2.4 \times 10^{-3} \frac{(\Omega h^2)^{1/2}}{\Omega_B h^2} \left(\frac{z}{1000}\right)^{12.75} \,, \tag{10.21}$$

where the redshift z is taken to be $z \simeq 1000$.
The optical depth for Thomson scattering is given by

$$\tau = \int n_e x \sigma_T \, \mathrm{d}w \, m \tag{10.22}$$

where σ_T is the Thomson cross section, and where the integral is carried out in comoving coordinates, in the desired range. Using $n_e \propto \Omega_B$ and $\mathrm{d}w = c\,\mathrm{d}t/a$, and making use of Friedmann's equation in the matter dominated case, $\dot{a}/a = H_0 \Omega_m^{1/2} a^{-3/2}$, we find

$$\tau \simeq 0.37 \left(\frac{z}{1000}\right)^{14.25} \,, \tag{10.23}$$

which surprisingly does not depend on the cosmological parameters. The probability of a photon travel from 0 to w of *not* being scattered is thus just $P(w) = \mathrm{e}^{-\tau}$. We can related this probability to the integral of the probability of last scattering, $p(z)$, using

$$1 - P(z) = 1 - \mathrm{e}^{-\tau} = \int_0^z p(z)\,\mathrm{d}z \,, \qquad p(z) = \mathrm{e}^{-\tau} \frac{\mathrm{d}\tau}{\mathrm{d}z} \,. \tag{10.24}$$

Since $\tau(z)$ has a strong dependence on z (cf. above the exponent 14.25), the probability of last scattering $p(z)$ will be very peaked around $z \sim$

1000. More precisely, a numerical fit shows that we can approximate very well $p(z)$ with a Gaussian of mean $z = 1065$ and standard deviation 80. In other words, at redshifts significantly larger than 1065 the Universe was opaque to the electromagnetic radiation, and thus behaved as a black body. At $z \sim 1065$ the Universe became optically thin. Note that, in principle, the epoch of freeze-out of photons, provided by Eq. (10.14), is slightly larger than the epoch of recombination, discussed here.

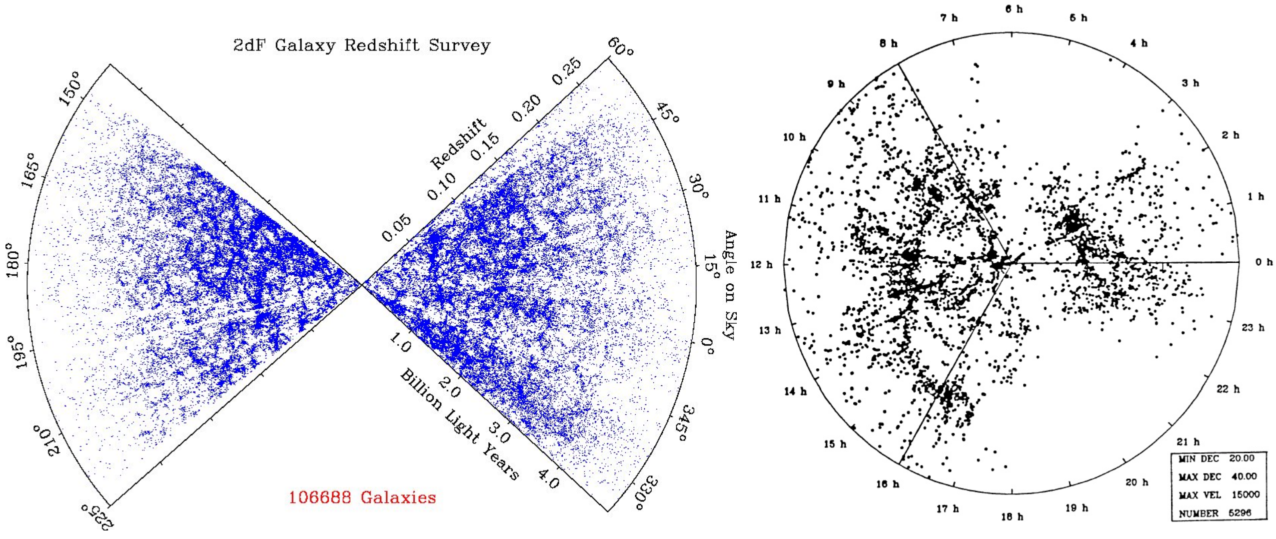

Figure 6: Left: The 2dF galaxy survey. Right: The Great Wall, an example of large-scale filament.

11 Linear cosmological perturbation theory

So far we considered a perfectly homogeneous Universe. In reality, the true Universe shows structures on various scales up to $\sim 200h^{-1}$Mpc.

11.1 Zero-pressure spherical perturbations

In a matter dominated background cosmology the simplest way to construct a perturbation is to excise a sphere of matter and replace it with a smaller sphere of the same gravitational mass: this generates a "top-hat" perturbation ("Swiss-Cheese" model).
We will consider here two main perturbation: one, where the inside sphere has a *delayed* expansion (i.e., expands like the Universe, but with a constant delay); another, where the average *energy* inside the sphere is smaller than the one in the Universe, so that effectively the sphere expands like in a modified Friedmann-Lemaître model.
In both cases, because of the matter conservation, we must have $\rho(t)r^3(t) = \tilde{\rho}(t)\tilde{r}^3(t)$, where $r(t)$ is the radius of a sphere as a function of the cosmic time t, and variables with the tilde indicates perturbed quantities. Calling

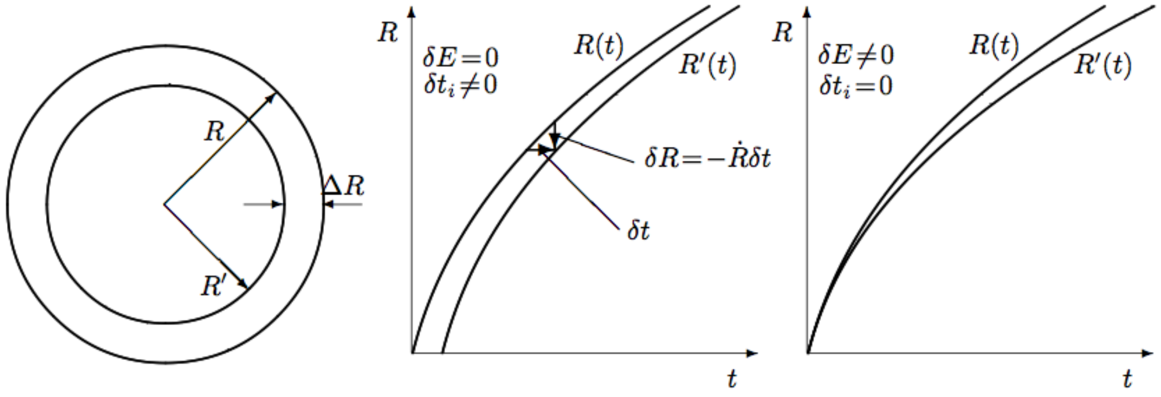

Figure 7: Example of spherical perturbations charcterized by a different "big-bang time" or different energy.

$\Delta\rho \equiv \tilde{\rho} - \rho$ and analogously for r we find then

$$\frac{\Delta\rho}{\rho} = -3\frac{\Delta r}{r}\,. \tag{11.1}$$

If we call $\Delta t_r \equiv \tilde{t}(r) - t(r)$ the time delay with which the perturbed sphere arrives at a given radius r, then $\Delta r = -\dot{r}\,\Delta t_r$, and thus

$$\boxed{\frac{\Delta\rho}{\rho} = 3\frac{\dot{r}}{r}\Delta t_r = 3H\Delta t_r\,.} \tag{11.2}$$

If we then make the first kind of perturbation by delaying the Big Bang time within the sphere, but keeping the energy unaltered, the time delay Δt_r, so this produces a density perturbation with $\Delta\rho/\rho \propto H$, and the density perturbation will deacy as $1/t$: for perturbations generated in the early universe, these decaying mode density perturbations will be negligible at late times.

A more interesting way to perturb the universe is to keep the bang time fixed, but to perturb the energy of the explosion, so that the interior will again resemble part of another Friedmann-Lemaître Universe, but with a lower energy $\tilde{K} = K + \Delta K$. The equation for the perturbed radius is

$$\left(\frac{\mathrm{d}\tilde{r}}{\mathrm{d}t}\right)^2 = \frac{2GM}{\tilde{r}} + 2Kc^2 + 2\Delta Kc^2\,. \tag{11.3}$$

We can integrate this equation as usual, obtaining

$$\begin{aligned}\tilde{t}(r) &= \int_0^r \frac{\mathrm{d}\tilde{r}}{\sqrt{\frac{2GM}{\tilde{r}} + 2Kc^2 + 2\Delta Kc^2}} \\ &\simeq t(r) - c^2 \Delta K \int_0^r \mathrm{d}\tilde{r} \left(\frac{2GM}{\tilde{r}} + 2Kc^2 \right)^{-3/2} . \end{aligned} \tag{11.4}$$

Inserting this equation into Eq. (11.2) we obtain

$$\frac{\Delta\rho}{\rho} = 3c^2 \Delta K \frac{\dot{r}}{r} \int_0^r \mathrm{d}\tilde{r} \left(\frac{2GM}{\tilde{r}} + 2Kc^2 \right)^{-3/2} . \tag{11.5}$$

We have two interesting limiting cases

- In an EdS model, or at early times in all models, K is negligible, $r \propto a \propto t^{2/3} \Rightarrow \dot{r}/r = 2/3t$, and the integral gives a term $r^{5/2} \propto t^{5/3}$: hence, we have in this case

  $$\boxed{\frac{\Delta\rho}{\rho} \propto \Delta K t^{2/3} \propto \Delta K a ,} \tag{11.6}$$

  i.e. growing perturbations.

- At later times in a low density Universe ($\Omega_\mathrm{m} \ll 1$), the energy term dominates, $\dot{r} \to$ const and the perturbartion is "frozen"

  $$\boxed{\frac{\Delta\rho}{\rho} = \mathrm{const} .} \tag{11.7}$$

Note that the transition between these two extreme cases appens when $\Omega_\mathrm{m}(a) \sim 1$, which is at redshift $1 + z \sim \Omega_\mathrm{m}^{-1}$ (this because $\Omega_\mathrm{m}(a) \propto 1/a$ for a low density Universe).

11.2 Zero-pressure general perturbations

We consider now more general linear ($\Delta\rho \ll \rho$) perturbations in a Universe with negligible pressure (and thus $z \ll z_\mathrm{eq}$). We will use the fluid equations

$$\frac{\partial \rho}{\partial t} + \nabla_{\boldsymbol{r}}(\rho \boldsymbol{u}) = 0\,, \tag{11.8}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla_{\boldsymbol{r}})\boldsymbol{u} = -\nabla_{\boldsymbol{r}}\phi\ , \tag{11.9}$$

$$\nabla_{\boldsymbol{r}}^2 \phi = 4\pi G\rho - \Lambda\,, \tag{11.10}$$

where all spatial derivatives are taken with respect to the physical coordinate $\boldsymbol{r}$. This system has the homogeneous solution

$$\rho(\boldsymbol{r},t) = \bar{\rho}\ , \tag{11.11}$$

$$\boldsymbol{u}(\boldsymbol{r},t) = \frac{\dot{a}}{a}\boldsymbol{r}\ , \tag{11.12}$$

$$\phi(\boldsymbol{r},t) = \frac{1}{6}(4\pi G\bar{\rho} - \Lambda)|\boldsymbol{r}|^2\ . \tag{11.13}$$

which, when substituted back into the original equations, gives

$$\dot{\bar{\rho}} + \frac{3\dot{a}}{a}\bar{\rho} = 0 \quad \Rightarrow \quad \bar{\rho} = \rho_0 a^{-3}\ , \tag{11.14}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi}{3}G\bar{\rho} + \frac{\Lambda}{3} \quad \text{(Friedmann's equation)} \tag{11.15}$$

In order to isolate the perturbations from the Hubble flow, it is convenient to rewrite these equations in terms of comoving coordinates. Define $\boldsymbol{x}$, $\hat{\rho}$, and $\boldsymbol{v}$ implicitly from

$$\boldsymbol{r} = a(t)\boldsymbol{x}\ , \tag{11.16}$$

$$\rho(\boldsymbol{r},t) = \hat{\rho}(\boldsymbol{x},t) = \hat{\rho}\left(a^{-1}(t)\boldsymbol{r},t\right)\ , \tag{11.17}$$

$$\boldsymbol{u}(\boldsymbol{r},t) = \dot{a}(t)\boldsymbol{x} + \boldsymbol{v}(\boldsymbol{x},t) = \dot{a}(t)a^{-1}(t)\boldsymbol{r} + \boldsymbol{v}\left(a^{-1}(t)\boldsymbol{r},t\right)\ . \tag{11.18}$$

Then, from the continuity equation

$$\begin{aligned} 0 &= \frac{\partial \rho}{\partial t} + \nabla_{\boldsymbol{r}}(\rho\boldsymbol{u}) \\ &= \frac{\partial \hat{\rho}}{\partial t} - \frac{\dot{a}}{a^2}\boldsymbol{r}\cdot\nabla_{\boldsymbol{x}}\hat{\rho} + \frac{1}{a}\Big[(\dot{a}\boldsymbol{x} + \boldsymbol{v})\cdot\nabla_{\boldsymbol{x}}\hat{\rho} + \hat{\rho}(3\dot{a} + \nabla_{\boldsymbol{x}}\cdot\boldsymbol{v})\Big] \\ &= \boxed{\frac{\partial \hat{\rho}}{\partial t} + \frac{3\dot{a}}{a}\hat{\rho} + \frac{1}{a}\nabla_{\boldsymbol{x}}(\hat{\rho}\boldsymbol{v}) = 0\ .} \end{aligned} \tag{11.19}$$

In a similar manner

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial r_j} = \frac{\ddot{a}}{a} r_i - \frac{\dot{a}^2}{a^2} r_i + \frac{\partial v_i}{\partial t} - \frac{\partial v_i}{\partial x_j} \frac{\dot{a} \partial r_j}{a^2} + \frac{1}{a}(\dot{a} x_j + v_j) \left(\dot{a} \delta_{ij} + \frac{\partial v_j}{\partial x_j} \right)$$

$$= \boxed{\frac{\partial v_i}{\partial t} + \ddot{a} x_i + \frac{\dot{a}}{a} v_i + \frac{1}{a} (\boldsymbol{v} \cdot \nabla_{\boldsymbol{x}}) v_i = -\frac{\partial \phi}{\partial r_i}\,.} \tag{11.20}$$

Next, let us define the comoving gravitational potential

$$\Phi(\boldsymbol{x}, t) \equiv \phi(a\boldsymbol{x}, t) + \frac{\ddot{a} a}{2} |\boldsymbol{x}|^2\,, \tag{11.21}$$

which satisfies the Poisson equation

$$\nabla_{\boldsymbol{x}}^2 \Phi = a^2 \left(\nabla_{\boldsymbol{r}}^2 \phi + \frac{3\ddot{a}}{a} \right) = 4\pi G a^2 \left[\hat{\rho}(\boldsymbol{x}, t) - \bar{\rho}(t) \right]\,. \tag{11.22}$$

Note that in the last step we used Friedmann's and Poisson's equations. With this definition Euler's equations becomes

$$\boxed{\frac{\partial \boldsymbol{v}}{\partial t} + \frac{\dot{a}}{a} \boldsymbol{v} + \frac{1}{a} (\boldsymbol{v} \cdot \nabla_{\boldsymbol{x}}) \boldsymbol{v} = -\frac{1}{a} \nabla_{\boldsymbol{x}} \Phi\,.} \tag{11.23}$$

By defining the density contrast

$$\delta(\boldsymbol{x}, t) \equiv \frac{\hat{\rho}(\boldsymbol{x}, t) - \bar{\rho}(t)}{\bar{\rho}(t)} \tag{11.24}$$

then the continuity equation yields

$$\boxed{\frac{\partial \delta}{\partial t} + \frac{1}{a} \nabla_{\boldsymbol{x}} \left[(1 + \delta) \boldsymbol{v} \right] = 0\,,} \tag{11.25}$$

and the Poisson equation can be rewritten as

$$\boxed{\nabla_{\boldsymbol{x}}^2 \Phi = \frac{3 H_0^2 \Omega_{\mathrm{m}}}{2a} \delta\,.} \tag{11.26}$$

Equations (11.23), (11.25), and (11.26) form a complete set of equations in the density contranst δ, peculiar velocirty $\boldsymbol{v}$, and comoving potential Φ. A solution is $\delta = 0$, $\boldsymbol{v} = \boldsymbol{0}$, and $\Phi = 0$.

11.3 Zero-pressure linear perturbations

We now specialize the previous set of equations to linear perturbations, i.e. consider only terms linear in δ and $\boldsymbol{v}$:

$$\boxed{\frac{\partial \delta}{\partial t} + \frac{1}{a}\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v} = 0\,,} \tag{11.27}$$

$$\boxed{\frac{\partial \boldsymbol{v}}{\partial t} + \frac{\dot{a}}{a}\boldsymbol{v} = -\frac{1}{a}\nabla_{\boldsymbol{x}}\Phi\,.} \tag{11.28}$$

The solution of these equations is better found in the Fourier space. Let us use a standard decomposition:

$$\delta(\boldsymbol{x},t) = \int_{\mathbb{R}^3} \frac{\mathrm{d}^3 k}{(2\pi)^3}\hat{\delta}(\boldsymbol{k},t)\mathrm{e}^{i\boldsymbol{k}\cdot\boldsymbol{x}}\,, \tag{11.29}$$

and similarly for $\hat{\boldsymbol{v}}(\boldsymbol{k},t)$ and $\hat{\Phi}(\boldsymbol{k},t)$. We have then

$$\frac{\mathrm{d}\hat{\delta}}{\mathrm{d}t} + \frac{i}{a}\boldsymbol{k}\cdot\hat{\boldsymbol{v}} = 0\,, \tag{11.30}$$

$$\frac{\mathrm{d}\hat{\boldsymbol{v}}}{\mathrm{d}t} + \frac{\dot{a}}{a}\hat{\boldsymbol{v}} = -\frac{i}{a}\boldsymbol{k}\hat{\Phi}\,, \tag{11.31}$$

$$-|\boldsymbol{k}|^2\hat{\Phi} = \frac{3H_0^2\Omega_{\mathrm{m}}}{2a}\hat{\delta}\,. \tag{11.32}$$

Let us consider first solutions such that $\boldsymbol{k}\cdot\hat{\boldsymbol{v}} = 0$ (rotational modes). A simple inspection of Eqs. (11.30–11.32) shows that in this case $\hat{\delta} = 0$, i.e. the perturbations are not coupled to density fluctuactions; moreover $\hat{\boldsymbol{v}} \propto 1/a$, i.e. these modes die quickly. We will thus ignore them.

Taking the time derivative of the continuity equation and the divergence of Euler's equation, and combining them together with Poisson's equation we obtain

$$\boxed{\frac{\mathrm{d}^2\hat{\delta}}{\mathrm{d}t^2} + \frac{2\dot{a}}{a}\frac{\mathrm{d}\hat{\delta}}{\mathrm{d}t} - \frac{3H_0^2\Omega_{\mathrm{m}}}{2a^3}\hat{\delta} = 0\,.} \tag{11.33}$$

Since there is no explicit spatiatial dependence anymore, the same equation holds in the real space; moreover, the solution can be factorized and written as

$$\delta(\boldsymbol{x},t) = D_+(t)\Delta_+(\boldsymbol{x}) + D_-(t)\Delta_-(\boldsymbol{x})\,, \tag{11.34}$$

where $\Delta_\pm(\boldsymbol{x})$ are arbitrary function (set by the initial conditions of the perturbation), and $D_\pm(t)$ are two linearly independent solutions of the differential equation

$$\ddot{D}+\frac{2\dot{a}}{a}\dot{D}-\frac{3H_0^2\Omega_{\rm m}}{2a^3}D=0\,. \tag{11.35}$$

The solutions of this equation clearly depend on the cosmological model considered.

Exercise 11.1 *Show that in an EdS Universe, the equation above reduces to*

$$\ddot{D}+\frac{4}{3t}\dot{D}-\frac{2}{3t^2}=0\,, \tag{11.36}$$

and find two explicit solutions of this equation. How do these solutions compare with the ones found in Sect. 11.1?

In a general Universe, there will be a growing and a decaying solutions, but finding them will be in general non-trivial.

Exercise 11.2 *Show that the Hubble parameter, $H(t)$, is a decaying solution of Eq.* (11.35).

Interestingly, once a solution of a homogeneous linear second-order differential equation is found, the other can be constructed using Sturm-Liuville theory:

$$D_+(t)\propto H(t)H_0^2\int_0^t\frac{{\rm d}t'}{a^2(t')H^2(t')}\,, \tag{11.37}$$

which can be rewritten as

$$\boxed{D_+(t)\propto\frac{H(t)}{H_0}\int_0^a\frac{{\rm d}a'}{\left[\Omega_{\rm m}a'^{-1}+\Omega_\Lambda a'^2+\Omega_{\rm c}\right]^{3/2}}\,.} \tag{11.38}$$

The constant of proportionality should be chosen by requiring that $D_+(t_0)=1$.

11.4 Pressure perturbations – the Jeans equation

We now consider the same set of linear equations used in Sect. 11.3 with the inclusion of a pressure term. Let us first write Euler's equation as

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla_{\boldsymbol{r}})\boldsymbol{u} = -\nabla_{\boldsymbol{r}}\phi - \frac{1}{\rho}\nabla_{\boldsymbol{r}} p \, . \tag{11.39}$$

In presence of a constant pressure $p = \bar{p}$, this equation, together with Eqs. (11.8) and (11.10), admit homogeneous solutions of the form (11.11–11.13). If we now allow for pressure perturbations δp and retain only terms linear in δ, $\boldsymbol{v}$, and δp, we obtain

$$\boxed{\frac{\partial \boldsymbol{v}}{\partial t} + \frac{\dot{a}}{a}\boldsymbol{v} = -\frac{1}{a}\nabla_{\boldsymbol{x}}\Phi - \frac{1}{\bar{\rho}}\nabla_{\boldsymbol{x}}\delta p \, .} \tag{11.40}$$

Similarly to what done before, we take the time derivative of the continuity equation and the divergence of the above equation, and consider a single Fourier mode $\delta(\boldsymbol{x},t) = \delta(t)\mathrm{e}^{i\boldsymbol{k}\cdot\boldsymbol{x}}$. Then, the inclusion of the pressure term modifies Eq. (11.33) into

$$\boxed{\frac{\mathrm{d}^2\hat{\delta}}{\mathrm{d}t^2} + \frac{2\dot{a}}{a}\frac{\mathrm{d}\hat{\delta}}{\mathrm{d}t} = \left(4\pi G\bar{\rho} - \frac{c_\mathrm{s}^2 k^2}{a^2}\right)\hat{\delta} \, ,} \tag{11.41}$$

where c_s is the “sound speed”

$$c_s^2 \equiv \frac{\partial p}{\partial \rho} \, . \tag{11.42}$$

Hence, the fate of perturbations in presence of pressure will depend on the sign of the r.h.s. of Eq. (11.41):

if r.h.s. $>$ 0: perturbations can grow (small k);

if r.h.s. $<$ 0: oscillatory solutions are expected (large k).

In terms of *physical* wavelength, the turning point for the two cases is given by the *Jeans wavelength*:

$$\boxed{\lambda_J \equiv a\frac{2\pi}{k_J} = \frac{2\pi c_s}{\sqrt{4\pi G\rho}} \, .} \tag{11.43}$$

To order of magnitude, the Jeans length is just the distance a sound wave can propagate in the age of the universe ($\lambda_J \sim c_s t$) which is just the physical sound horizon. Perturbations of wavelength $\lambda \ll \lambda_J$ can oscillate many times, but the equation that these perturbations obey is not a simple harmonic oscillator, as it contains a damping term $2H\dot{\delta}$ and also the frequency of the oscillations $\omega \simeq c_s k/a$ is not constant. To determine the secular evolution of the perturbations, let us write the density contrast as $\delta(\boldsymbol{r},t) = \chi(\boldsymbol{r})t^{\alpha}$, with α constant. Let us now consider for simplicity an EdS universe, where $a \propto t^{2/3}$: then a simple calculation shows that if we take $\alpha = -2/3$ then the equation for χ is

$$\ddot{\chi} + \left(\frac{c_s^2 k^2}{a^2} - H^2\right)\chi = 0\,. \tag{11.44}$$

This is an un-damped oscillator $\ddot{\chi} + \Omega^2\chi = 0$ with time varying frequency $\Omega^2 = c_s^2 k^2/a^2 - H^2$. We are here mostly interested in waves with $\lambda \ll c_s t$ since longer wavelengths would not have had time to oscillate, but this condition is $k = 2\pi a/\lambda \gg Ha/c_s$. To a good approximation we can neglect the term H^2 in the frequency, so $\Omega \simeq c_s k/a$. The sound speed is $c_s \sim c\sqrt{\rho_{\mathrm{r}}/\rho_{\mathrm{m}}}$, which is proportional to $a^{-1/2}$, so $\Omega \propto a^{-3/2}$.
In the limit that the time-scale for variation of the frequency is much greater than the period of oscillation – which is just the condition $k \gg Ha/c_s$ again – we can apply the principle of adiabatic invariance (see, e.g., @@@), which tells us that the amplitude of the χ-field fluctuations should scale as $\chi \propto 1/\sqrt{\Omega} \propto a^{3/4}$. Hence, the actual density fluctuation $\delta = \chi t^{\alpha} = \chi t^{-2/3} \propto \chi/a$, with the net result that the amplitude of acoustic fluctuations damp adiabatically as

$$\boxed{\delta \propto a^{-1/4} \propto (1+z)^{1/4}\,.} \tag{11.45}$$

Although this result has been obtain for the EdS case, the use of the adiabatic invariance makes it largely independent of the details of the expansion.

11.5 Diffusive damping and free-streaming

The Jeans analysis assumed a tightly coupled baryon-photon plasma. There are two situations where this is inappropriate. At early times the plasma is very optically thick, but as the Universe expands the mean-free path for photons increases and the photons tend to leak out of the sound waves and consequently they damp out. This was first analyzed by Joe Silk and is called Silk damping. Around $z \simeq 1000$, the ions and electrons (re)combine and become neutral and the photons are no longer locked to the baryons. The consequence of this is that sufficiently small scale perturbations (interestingly close to the mass of galaxies as it happens) are damped out. Nowadays the emphasis is on dark matter models where the damping of acoustic fluctuations is not so critical.

Another interesting (and possibly very relevant) situation is that of hot-dark-matter (see below) such as a neutrino with a mass on the order of 10 eV. Such particles have large thermal velocities, but are not locked to the other components by scattering. One can still define a neutrino Jeans length which is the distance a neutrino can travel in one expansion time (cf. above in Sect. 11.4 where $\lambda_J \sim c_s t$). This starts off small (in comoving terms), grows and peaks around the time the neutrinos go non-relativistic. Once the neutrinos become non-relativistic their velocities decay adiabatically as $v \propto 1/a$ (this can be seen from Eq. (11.31) with $\hat{\Phi} = 0$ which shows that in comoving coordinates there appears to be a *cosmic dragging*) and the comoving Jeans length decreases. As with acoustic waves, perturbations bigger than the Jeans length grow, but smaller perturbations damp out very rapidly as the neutrinos stream out of the perturbation.

11.6 Matter perturbations in radiation dominated background

In this section we consider the evolution of matter perturbations at early times, in the radiation dominated era. We will take the radiation field uniform, so that no pressure gradients are acting upon the matter perturbations. In these conditions Eq. (11.33) still describes the evolution of matter perturbations

$$\frac{\mathrm{d}^2\hat{\delta}}{\mathrm{d}t^2} + \frac{2\dot{a}}{a}\frac{\mathrm{d}\hat{\delta}}{\mathrm{d}t} = 4\pi G\rho_{\mathrm{m}}\hat{\delta}\,. \tag{11.46}$$

Hence, the only difference with respect to the perturbations considered in Sect. 11.3 is that now the evolution of the Universe (and in particular of the scale parameter a) is mainly driven by radiation rather than matter. To this purpose, we define y as the ratio between matter and radiation density

$$y \equiv \frac{\rho_{\mathrm{m}}}{\rho_{\mathrm{r}}} = \frac{a}{a_{\mathrm{eq}}}\,. \tag{11.47}$$

We then rewrite Friedmann's equation as

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 = H^2 &= \frac{8\pi}{3}G(\rho_{\mathrm{m}}+\rho_{\mathrm{r}}) \\ &= \frac{8\pi}{3}G\rho_{0m}\frac{1}{a_{\mathrm{eq}}^3 y^3}\left(1+\frac{1}{y}\right)\,,\end{aligned} \tag{11.48}$$

where in the last step we used the a^{-3} dependence of the matter density. Since Friedmann's equation can be written completely in terms of y (instead of a), it is convenient to use y as "time" variable. Let us start from the relations

$$\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}t}\frac{\mathrm{d}}{\mathrm{d}y}\,, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\dot{a}}{a_{\mathrm{eq}}} = Hy\,, \tag{11.49}$$

so that

$$\dot{\delta} = Hy\frac{\mathrm{d}\delta}{\mathrm{d}y}\,, \tag{11.50}$$

$$\begin{aligned}\ddot{\delta} &= \frac{\mathrm{d}y}{\mathrm{d}t}\frac{\mathrm{d}}{\mathrm{d}y}\left(Hy\frac{\mathrm{d}\delta}{\mathrm{d}y}\right) \\ &= Hy\left(\frac{\mathrm{d}H}{\mathrm{d}y}y\frac{\mathrm{d}\delta}{\mathrm{d}y} + H\frac{\mathrm{d}\delta}{\mathrm{d}y} + Hy\frac{\mathrm{d}^2\delta}{\mathrm{d}y^2}\right)\,.\end{aligned} \tag{11.51}$$

Inserting here Eq. (11.46) we finally have

$$\boxed{\delta'' + \frac{2+3y}{2y(1+y)}\delta' - \frac{3}{2y(1+y)}\delta = 0\,.} \tag{11.52}$$

The growing solution for this differential equation are of the form

$$\boxed{\delta \propto y + \frac{2}{3} = \frac{a}{a_{\rm eq}} + \frac{2}{3}\,.} \tag{11.53}$$

In other words, *in the radiation dominated phase matter perturbations are approximately constant (Meszaros effect).* Note that, in contrast, in the matter dominated phase perturbations grow as $\delta \propto a$ (cf. Sect. 11.3).

11.7 Superhorizont perturbations

The perturbation theory developed so far only applies to perturbations on scales smaller than the Universe horizon.

In order to study larger perturbations (without developing a General Relativity perturbation theory), we consider spherical homogeneous perturbations: these, essentially, will follow the same expansion law of the Universe, but with different "local" cosmological parameters.

Let us consider Friedmann's equation without the cosmological constant (which is unimportant at the early times considered):

$$H^2 = \left(\frac{\dot a}{a}\right)^2 = \frac{8\pi}{3}G\rho - \frac{Kc^2}{a^2}\,. \tag{11.54}$$

With the usual technique we can integrate this equation and get

$$t = \int_0^a \frac{\mathrm{d}a'}{a'H(a')} \simeq \int_0^a \frac{\mathrm{d}a'}{\sqrt{8\pi G\rho a'^2/3}}\left(1 + \frac{3Kc^2}{16\pi G\rho a'^2}\right)\,. \tag{11.55}$$

Let us now consider first the *radiation dominated phase*. In this case, since $\rho = \Omega_{\rm r}\rho_{\rm cr}a^{-4}$, the term inside square root in Eq. (11.55) becomes $8\pi G\rho a^2/3 = \Omega_{\rm r}H_0^2 a^{-2}$. Hence, we simply have

$$t \simeq \frac{1}{\sqrt{\Omega_{\rm r}}H_0}\int_0^a \mathrm{d}a'\,a'\left(1 + Za'^2\right)\,, \qquad \text{where} \qquad Z \equiv \frac{Kc^2}{2H_0^2\Omega_{\rm r}}\,. \tag{11.56}$$

Consider now the evolution of a Universe described by Ω_r and $Z = Z_0$, for which

$$tH_0 \simeq \frac{1}{\sqrt{\Omega_\mathrm{r}}}\left(\frac{a_0^2}{2} + \frac{Z_0}{4}a_0^4\right) . \tag{11.57}$$

A perturbation with density parameter $\Omega_\mathrm{r} + \omega_\mathrm{r}$ and $Z = Z_1 = Z_0 + \Delta Z$ will instead follow the evolution

$$tH_0 \simeq \frac{1}{\sqrt{\Omega_\mathrm{r} + \omega_\mathrm{r}}}\left(\frac{a_1^2}{2} + \frac{Z_1}{4}a_1^4\right) . \tag{11.58}$$

If we compare these two expressions at equal times we obtain

$$\frac{a_1^2}{2} + \frac{Z_1}{4}a_1^4 = \sqrt{1 + \frac{\omega_\mathrm{r}}{\Omega_\mathrm{r}}}\left(\frac{a_0^2}{2} + \frac{Z_0}{4}a_0^4\right) \tag{11.59}$$

$$\begin{aligned} \sqrt{1 + \frac{\omega_\mathrm{r}}{\Omega_\mathrm{r}}\frac{a_0^2}{a_1^2}} &= 1 + \left(\frac{1}{2}Z_1 a_1^2 - \sqrt{1 + \frac{\omega_\mathrm{r}}{\Omega_\mathrm{r}}\frac{Z_0 a_0^4}{2a_1^2}}\right) \\ &\simeq 1 + Xa^2 . \end{aligned} \tag{11.60}$$

In the last step we used the fact that $a_0 \simeq a_1$. The density contrast for the perturbation then can be finally evaluated:

$$\delta = \frac{\Delta\rho}{\rho_0} = \frac{(\Omega_\mathrm{r} + \omega_\mathrm{r})a_1^{-4} - \Omega_\mathrm{r}a_0^{-4}}{\Omega_\mathrm{r}a_0^{-4}} = \left(1 + \frac{\omega_\mathrm{r}}{\Omega_\mathrm{r}}\right)\frac{a_0^4}{a_1^4} - 1 \simeq 2Xa^2 . \tag{11.61}$$

In summary, *in a radiation dominated Universe superhorizon perturbations grow as* $\delta \propto a^2$.

Exercise 11.3 *Using a similar technique, repeat the calculation for a matter dominated Universe and show that there superhorizon perturbations grow like* $\delta \propto a$.

11.8 Perturbations in radiation dominated era

In order to complete our picture of perturbation growing, we need to add a few ingredients to out study. We did not study in details perturbations in the radiation dominated era except for *isothermal* matter perturbations (Sect. 11.6). However, using calculations similar to the one done in this chapter, one case show that sub-horizon radiation perturbations have an oscillatory behaviour for $a \ll a_{\rm eq}$, with constant amplitude.

12 The power spectrum of perturbations

Density perturbations in the Universe clearly cannot be modeled in a deterministic way, i.e. we cannot expect cosmology to predict the function $\delta(\boldsymbol{x},t)$. However, the *statistical properties* of the perturbations can be addressed if we take the density contrast to be a stochastic random field.

12.1 Random fields

A random field is the generalization of a random variable to functions. If we neglect for the moment the time dependence of the density contrast, $\delta(\boldsymbol{x})$ can be taken to be a random field if for any possible realization of a density perturbation $\delta(\boldsymbol{x})$ we are able to evaluate a "probability." Since the definition of a probability for a continuous field has a few subtleties which cannot be addressed here, we will assume that for our purposes $\delta(\boldsymbol{x})$ can be described sufficiently well by its values at a regular grid. If we number the points of the grid as $\{\boldsymbol{x}_i\}$ and call $\delta_i = \delta(\boldsymbol{x}_i)$, then we can consider the probability distribution

$$p(\delta_1, \delta_2, \dots)\, \mathrm{d}\delta_1\, \mathrm{d}\delta_2 \dots \tag{12.1}$$

that $\bigl\{\delta(\boldsymbol{x}_i)\bigr\}$ lies within $\{\mathrm{d}\delta_i\}$ from $\{\delta_i\}$. In this notation, *the description of a random field reduces to a joint probability distribution of (possibly infinite) random variables.*

Since the Universe is taken to be homogeneous and isotropic, the random field $\delta(\boldsymbol{x})$ will have some related properties. In particular, the probability distribution written in Eq. (12.1) will not change if we either (i) translate all points $\boldsymbol{x}_i \mapsto \boldsymbol{x} + \boldsymbol{d}$ or (ii) rotate all points $\boldsymbol{x}_i \mapsto \mathscr{R}\boldsymbol{x}_i$, where $\mathscr{R}$ is an orthogonal matrix. Furthermore, we will assume that the density contrast random field will have vanishing average, i.e. that

$$\bigl\langle \delta(\boldsymbol{x}) \bigr\rangle = \int \mathrm{d}\delta\, p\bigl(\delta(\boldsymbol{x})\bigr)\, \delta(\boldsymbol{x}) = 0\,. \tag{12.2}$$

This relation is a consequence of replacing the *ensemble average* of δ with the *spatial average*, and of using the definition of density contrast.

12.2 Correlation function and power spectrum

Let us consider the two-point correlation function of the density contrast

$$C_{\delta\delta}(\boldsymbol{x},\boldsymbol{y}) = \left\langle \delta(\boldsymbol{x})\delta^*(\boldsymbol{y})\right\rangle , \tag{12.3}$$

where, although the density contrast is a real field, in order to keep the discussion at a general level we have used the complex conjugate for $\delta(\boldsymbol{y})$. The homogeneity and isotropy of δ severely limit the form of the correlation function:

- Because of the homogeneity, the correlation function can only depend on the difference $\boldsymbol{x}-\boldsymbol{y}$;
- If we now further consider the isotropy, then $C_{\delta\delta}$ can only be a function of $|\boldsymbol{x}-\boldsymbol{y}|$.

We now define the Fourier transform of $\delta(\boldsymbol{x})$ as

$$\hat{\delta}(\boldsymbol{k}) = \int_{\mathbb{R}^3} \mathrm{d}^3x\,\delta(\boldsymbol{x})\mathrm{e}^{-i\boldsymbol{x}\cdot\boldsymbol{k}} , \qquad \delta(\boldsymbol{x}) = \int_{\mathbb{R}^3} \frac{\mathrm{d}^3k}{(2\pi)^3}\,\hat{\delta}(\boldsymbol{k})\mathrm{e}^{i\boldsymbol{x}\cdot\boldsymbol{k}} . \tag{12.4}$$

Let us now evaluate the equivalent of the two-point correlation function in Fourier space, i.e.

$$\left\langle \hat{\delta}(\boldsymbol{k})\hat{\delta}^*(\boldsymbol{k}')\right\rangle = \int_{\mathbb{R}^3} \mathrm{d}^3x\mathrm{e}^{-i\boldsymbol{x}\cdot\boldsymbol{k}} \int_{\mathbb{R}^3} \mathrm{d}^3x'\,\mathrm{e}^{i\boldsymbol{x}'\cdot\boldsymbol{k}'} \left\langle \delta(\boldsymbol{x})\delta(\boldsymbol{x}')\right\rangle . \tag{12.5}$$

Inserting here Eq. (12.3) and using the properties of the two-point correlation function we have

$$\begin{aligned}\left\langle \hat{\delta}(\boldsymbol{k})\hat{\delta}^*(\boldsymbol{k}')\right\rangle &= \int_{\mathbb{R}^3} \mathrm{d}^3x\mathrm{e}^{-i\boldsymbol{x}\cdot\boldsymbol{k}} \int_{\mathbb{R}^3} \mathrm{d}^3y\,\mathrm{e}^{i(x+y)\cdot\boldsymbol{k}'} C_{\delta\delta}\left(|\boldsymbol{y}|\right)\\ &= (2\pi)^3\delta^{(3)}(\boldsymbol{k}-\boldsymbol{k}') \int_{\mathbb{R}^3} \mathrm{d}^3y\,\mathrm{e}^{-iy\cdot\boldsymbol{k}'} C_{\delta\delta}\left(|\boldsymbol{y}|\right)\\ &\equiv (2\pi)^3\delta^{(3)}(\boldsymbol{k}-\boldsymbol{k}')P_{\delta}\left(|\boldsymbol{k}|\right) . \end{aligned} \tag{12.6}$$

The final line defines the *power spectrum of the density contrast* which, as a consequence of the homogeneity and isotropy of δ, only depends on the modulus of $\boldsymbol{k}$. Hence, the power spectrum is the Fourier transform of the two-point correlation function:

$$\boxed{P_\delta(|\boldsymbol{k}|) = \int_{\mathbb{R}^3} \mathrm{d}^3 y \, \mathrm{e}^{-i\boldsymbol{y}\cdot\boldsymbol{k}'} C_{\delta\delta}(|\boldsymbol{y}|) \, .} \tag{12.7}$$

12.3 Gaussian random fields

A random field $\delta(\boldsymbol{x})$ can be completely described by its Fourier transform $\hat{\delta}(\boldsymbol{k})$; moreover, similarly to the real space, we can imagine to use a discretized grid in the Fourier space.

A random field $\delta(\boldsymbol{x})$ is a *Gaussian random field* if

1. its Fourier components $\hat{\delta}(\boldsymbol{k})$ are mutually statistically *independent* and
2. the probability density for each $\hat{\delta}(\boldsymbol{k})$ is *normal.*

Note that clearly the various components $\hat{\delta}(\boldsymbol{k})$ will in general have different variances: indeed, it is easily realized that the variance $\sigma^2(\boldsymbol{k})$ for a component is proportional to the power spectrum $P_\delta(\boldsymbol{k})$. Hence, we can deduce that

A Gaussian random field is fully specified by its power spectrum.

A Gaussian random field can be defined from an alternative property:

The probability distribution of any linear combination of the random variables $g(\boldsymbol{x}_i)$ is normal and more generally various linear combination of the random variables are a multivariate Gaussian. Hence, the quantity

$$X_i = \sum_{j=1}^{N} U_{ij} \delta(\boldsymbol{x}_j) \, , \tag{12.8}$$

where U_{ij} are simple coefficients, is distributed as n-dimensional multivariate normal.

Finally, we note that for a Gaussian random field any correlator $\langle \hat{\delta}(\boldsymbol{k}_1) \ldots \hat{\delta}(\boldsymbol{k}_N) \rangle$ either (i) vanish if N is odd or (ii) can be expressed in terms of the power spectrum if N is even.

12.4 The initial power spectrum

Consider the simple Poisson's equation for ϕ/c^2 and write it in Fourier space:

$$\nabla^2 \frac{\phi}{c^2} = \frac{4\pi G \bar{\rho}}{c^2} \delta \,, \qquad -k^2 \frac{\hat{\phi}}{c^2} = \frac{4\pi G \bar{\rho}}{c^2} \hat{\delta} \,. \tag{12.9}$$

These equation, in terms of power spectrum for ϕ/c^2 and δ, imply

$$k^3 P_{\phi/c^2}(k) \propto k^{-1} P_\delta(k) \,. \tag{12.10}$$

We now note that ϕ/c^2 is dimensionless, and that thus the l.h.s. of this equation is also dimensionless. At very early time the Universe horizon is much smaller than any other fluctuation scale, and thus it is reasonable to expect that $k^3 P_{\phi/c^2}(k)$ is constant and

$$\boxed{P_\delta(k) \propto k \,.} \tag{12.11}$$

This is the *scale-invariant spectrum of Harrison-Zel'dovich.*

12.5 Normalization of the power spectrum

Observations show that the relative fluctuations on the number of galaxies within spheres of $8\,h^{-1}$Mpc radius are of the order of unity:

$$\frac{\left\langle (N - \langle N \rangle)^2 \right\rangle}{\langle N \rangle^2} \simeq 1 \qquad \text{at } R = 8\,h^{-1}\,\text{Mpc} \,. \tag{12.12}$$

Under the hypothesis that galaxies trace the matter distribution, this information can be used to fix the normalization of the power spectrum, as shown below.

Let us define first a *smoothed* density field

$$\delta_W(\boldsymbol{x}) = \int_{\mathbb{R}^3} \mathrm{d}^3 y\, \delta(\boldsymbol{y}) W\big(|\boldsymbol{x}-\boldsymbol{y}|\big) , \tag{12.13}$$

where W is a filter function, taken to be normalized:

$$\int_{\mathbb{R}^3} \mathrm{d}^3 x\, W\big(|\boldsymbol{x}|\big) = 1 . \tag{12.14}$$

From the convolution theorem in the Fourier space we find $\hat{\delta}_W = \hat{\delta}\hat{W}$, and thus for the power spectrum

$$P_W(k) = \big|\hat{W}(k)\big|^2 P(k) . \tag{12.15}$$

The dispersion of the smoother density field can be then written as

$$\sigma_W^2 = \big\langle \delta_W^2(\boldsymbol{x}) \big\rangle = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} P_W(k) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \big|\hat{W}(k)\big|^2 P(k) . \tag{12.16}$$

Hence, the observational result of Eq. (12.12) can be rewritten in terms of σ_W^2, where W is a top-hat filter of radius $R = 8\,h$ Mpc:

$$W\big(|\boldsymbol{x}|\big) = \frac{3}{4\pi R^3} H\big(R - |\boldsymbol{x}|\big) . \tag{12.17}$$

Conventionally the square root of σ^2 with this weight function is written as σ_8; its measurement allows us to fix the normalization of the power spectrum using Eq. (12.16).

So far we assumed that galaxies trace the mass, but they might be *biased* with respect to the matter fluctuations. One can thus introduce a *bias factor b* defined as

$$\boxed{\delta_{\mathrm{g}} = \frac{\Delta N}{N} = b \frac{\Delta \rho}{\rho} = b\delta .} \tag{12.18}$$

12.6 The peculiar velocity field

In linear perturbation theory the velocity field is non-rotational, and thus can be written as

$$\boldsymbol{v}(\boldsymbol{x},t) = \nabla_x \psi , \tag{12.19}$$

i.e. derived from a potential. Since typically observations only allow us to measure the *radial* peculiar velocity $\boldsymbol{v}\cdot\boldsymbol{x}/|\boldsymbol{x}|$, the velocity field is usually derived from a measurement of the ψ potential (POTENT method). Using the continuity equation

$$\frac{\partial\delta}{\partial t}+\frac{1}{a}\nabla_x\cdot\boldsymbol{v}=0\,, \tag{12.20}$$

and the law for growing perturbations $\delta(\boldsymbol{x},t)=D_+(t)\Delta_+(\boldsymbol{x})$, we find

$$-\nabla^2\psi=a\frac{\dot{D}_+}{D_+}\delta=\frac{a\dot{a}}{D_+}\frac{\mathrm{d}D_+}{\mathrm{d}a}\delta=aH\frac{\mathrm{d}\ln D_+}{\mathrm{d}\ln a}\delta\,. \tag{12.21}$$

Peebles (1980) found that the logarithmic derivative above, f, is very well approximated in all cosmological models by

$$f\simeq\Omega_m^{0.6}+\frac{\Omega_\Lambda}{70}\left(1+\frac{\Omega_\mathrm{m}}{2}\right)\,. \tag{12.22}$$

Since the Ω_Λ correction is so small, we will neglect it and use simply $f\simeq\Omega_\mathrm{m}^{0.6}$. We then have

$$\nabla^2\psi=-aH\Omega_\mathrm{m}^{0.6}\delta\,, \tag{12.23}$$

which can be inverted into

$$\boldsymbol{v}(\boldsymbol{x},a)=\frac{\Omega_\mathrm{m}^{0.6}}{4\pi}aH\int\mathrm{d}^3y\,\frac{\delta(\boldsymbol{y},a)(\boldsymbol{x}-\boldsymbol{y})}{|\boldsymbol{x}-\boldsymbol{y}|^3}\,. \tag{12.24}$$

In other words, if you know density contrast δ you can predict the peculiar velocity field $\boldsymbol{v}$. The density contrast δ can be inferred from the galaxy distribution. To the purpose, one defines $\delta_\mathrm{g}\equiv(\rho_\mathrm{g}-\bar{\rho}_\mathrm{g})/\bar{\rho}_\mathrm{g}$ with the averages evaluated on some large scales, and one assume linear biasing $\delta_\mathrm{g}=b\delta$. Measurements show that the combination $\beta=\Omega_\mathrm{m}^{0.6}/b\simeq0.5$, but apparently the exact value depends on the galaxy type (and hence, different galaxies have different biasing factors b).

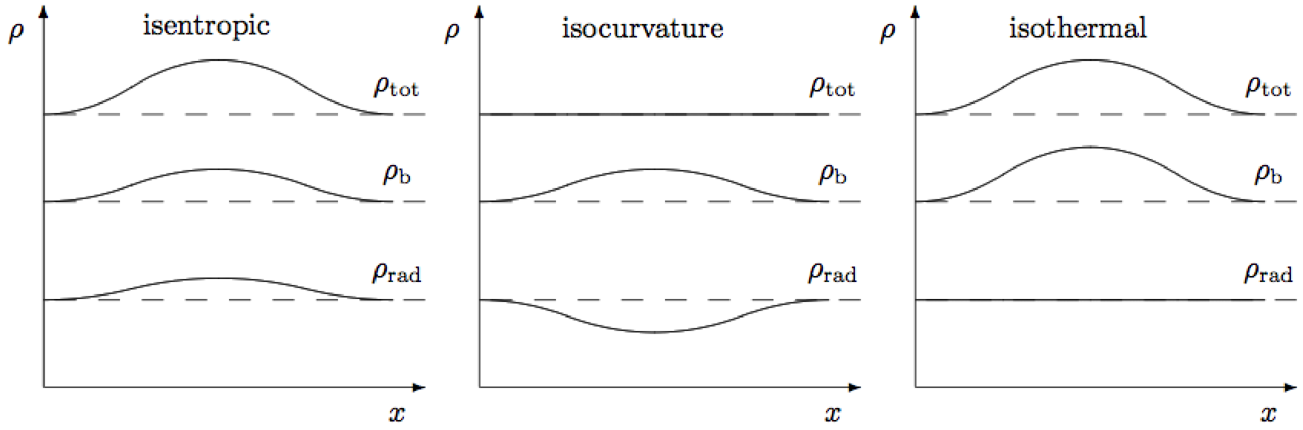

Figure 8: Most perturbations considered so far are called “adiabatic” or “isentropic:” one compress in some region both the density and the radiation field (left). We have then $\delta_{\rm rad} = (4/3)\delta_{\rm b}$. Alternatively, one can consider an “isothermal” perturbation, where the radiation field is left unchanged (right; cf. Sect. 11.6). Finally, there is a third natural way to to generate a perturbation in which the initial perturbation in the baryon density is cancelled by a corresponding under-density in the radiation. Such perturbations have, initially, no net density perturbation and therefore no associated curvature perturbation, and are called “isocurvature” (middle).

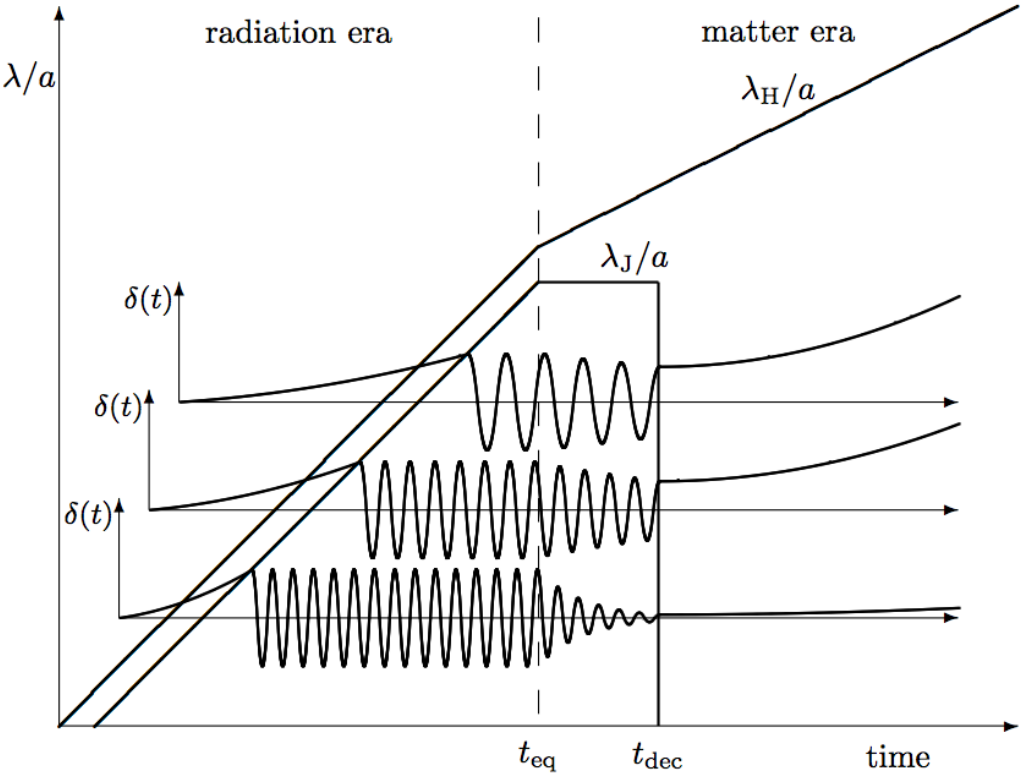
radiation era
matter era
λ/a
λ_H/a
λ_J/a
$\delta(t)$
$\delta(t)$
$\delta(t)$
t_{eq}
t_{dec}
time

Figure 9: Schematic evolution of initially adiabatic (or isentropic) perturbations for perturbations of three different wavelengths. The perturbation passes through three phases. First, when outside the horizon, the perturbation amplitude grows as $\delta \propto a^2$ in the radiation era and as $\delta \propto a$ in the matter dominated era. Perturbations which enter the horizon before t_{eq} oscillate at constant amplitude until t_{eq}. For $t_{eq} < t < t_{dec}$ the amplitude decays adiabatically as $\delta \propto 1/a^{1/4}$. Short wavelength perturbations are, in addition, subject to diffusive damping, and are strongly attenuated. Perturbations which persist to t_{dec} then couple to growing and decaying perturbations in the now pressure-free neutral gas.

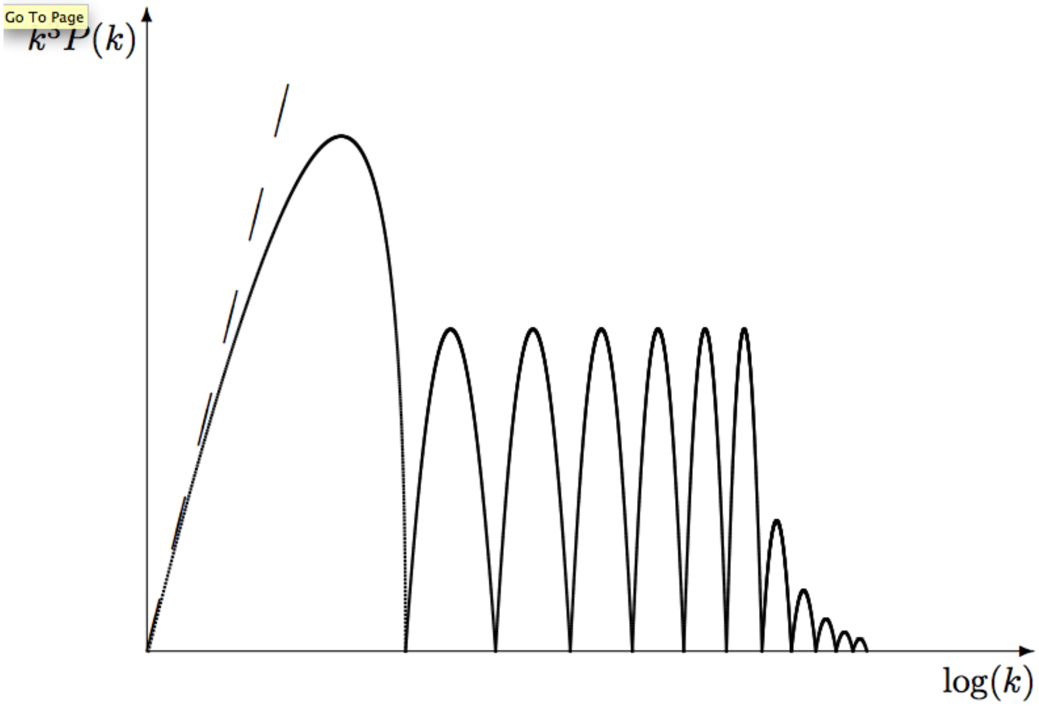

Figure 10: Schematic plot of the power spectrum of adiabatic model. The dashed line shows the original power spectrum (a power law).

12.7 The adiabatic-baryonic model

The first scenario to be worked out in any detail was the adiabatic baryonic model (Peebles and Yu). In this calculation one assumes some initial *adiabatic* density fluctuations imposed at some very early epoch, and one assumes that the universe is dominated by baryons and radiation. At sufficiently early times all the modes are superhorizon and the perturbations grow as $\delta \propto a^2$ (see Sect. 11.7). Eventually, short wavelength perturbations enter the horizon. For galaxy scale perturbations ($\lambda_0 \simeq 1$ Mpc) this happens when the Universe is radiation dominated. These perturbations then oscillate as acoustic waves with constant amplitude (Sect. 11.8). As time goes on progressively longer wavelengths enter the horizon and start oscillating. Eventually we reach the epoch $z_{\rm eq}$ and then something rather interesting happens. Before $z_{\rm eq}$, $P \simeq c^2\rho/3$, $c_s \simeq c/\sqrt{3}$, and the "sound horizon" c_st/a grows like $a \propto t^{1/2}$. After $z_{\rm eq}$, $P/\rho \propto 1/a$, the sound speed falls like $1/\sqrt{a}$, $t \propto a^{3/2}$ and consequently the comoving sound horizon (or comoving Jeans length) is approximately constant. At

$z \sim 1000 \sim z_{\mathrm{dec}} \sim z_{\mathrm{eq}}/10$, the plasma recombines and the pressure support from the radiation is lost, the Jeans length falls, and all perturbations grow according to the usual $\delta \propto a$ law.

The result of this calculation can be expressed in terms of a transfer function $T(k)$ expressing the final amplitude of a perturbation of (comoving) wavenumber k relative to it's initial amplitude. For a power-law initial power spectrum $P(k) \propto k^n$ for instance, the final power spectrum is $P(k) \propto k^n T^2(k)$. This linear output spectrum can then be given as initial conditions to a numerical simulator who can evolve the density fluctuations into the non-linear regime and compare the results with observational data.

The transfer function for the adiabatic baryonic model has some rather interesting features. For wavelengths longer than the maximum Jeans length (roughly the horizon size at z_{eq}) $T(k)$ is constant, but for shorter wavelengths there are oscillations. If one models the decoupling process as instantaneous, it is easy to show that the amplitude of the growing mode perturbation is then an oscillatory function of k. This results in quasi-periodic bumps in the power spectrum. These oscillations extend to about a factor 10 in wavelength, below which they are damped out by photon diffusion. There is a large bump in $T(k)$ at around the maximum Jeans length. This comes about because of the plateau in the Jeans length: a mode with wavelength just greater than $\lambda_{J_{\max}}$ will grow uninterrupted while one of slightly shorter wavelength will oscillate with actually slowly decreasing amplitude between z_{eq} and z_{rec}.

This calculation really marked a turning point in structure formation. While the creation of the initial fluctuations was still a subject of speculation, it seemed reasonable to assume a power law initial state, and one then obtained a quantitative prediction for the post-z_{eq} power spectrum. This model has only two free parameters: the initial amplitude of the power spectrum and the initial spectral index n. Moreover, this spectrum had two prominent features, the bump and the damping cut-off. The present horizon size is $\sim c/H_0 \sim 3000$ Mpc, and since the comoving horizon is proportional to $t/a \propto a^{1/2} \propto z^{-1/2}$, the power spectrum has a

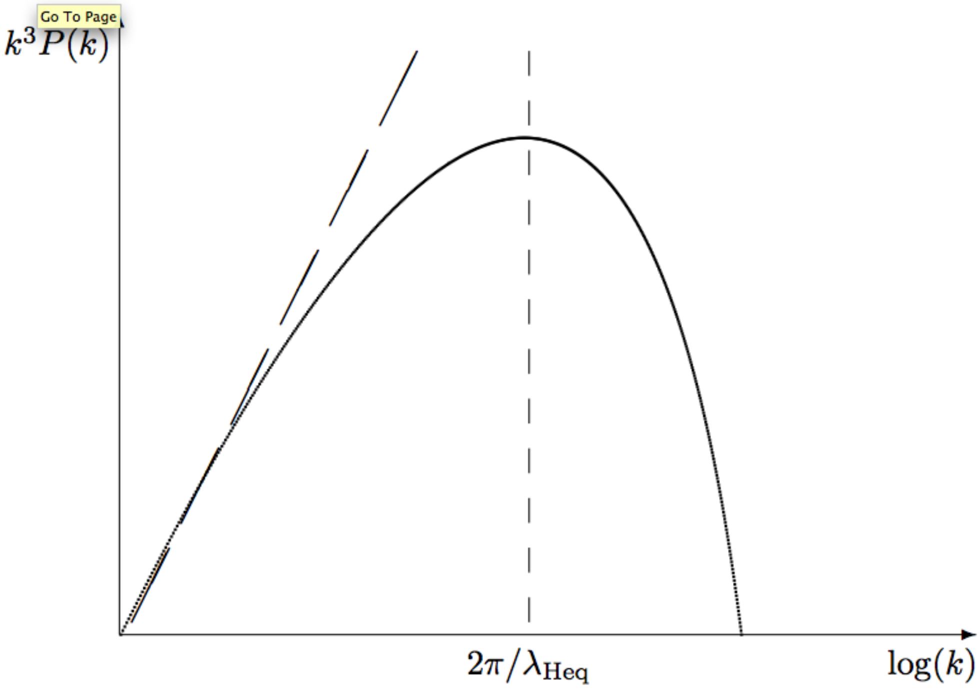

Figure 11: Power spectrum in the Hot Dark Matter (HDM) model (schematic). The long-dashed line indicates the initial power spectrum, while the vertical dashed line the horizon size at the time the neutrinos become non-relativistic at $z \simeq z_{\rm eq}$.

maximum at $\lambda \sim 30$ Mpc which is the scale of superclusters, the largest prominent structures we observe in the universe; a remarkable fact indeed! The damping scale sets a minimum scale for the first structures to form, and it is obviously attractive to identify this scale with galaxies in this theory.

12.8 Hot Dark Matter (HDM)

The first non-baryonic model to be seriously studied was the hot dark matter model (HDM), with the candidate particle being a massive neutrino. Neutrinos are produced in thermal abundance at high temperatures, but the weak interactions freeze-out at $T \sim$ MeV and the neutrino are thereafter effectively uncoupled from the rest of matter save through their gravitational influence.

In the HDM model, fluctuations on small scales are damped out by free-streaming. The characteristic smoothing length in this theory is just the comoving distance traveled by a neutrino. At early times, $z > z_{\rm eq}$, the neutrinos are relativistic and the comoving distance traveled is just the horizon size; after they go non-relativistic the velocity of the neutrinos redshifts as $v \propto 1/a$, so the comoving distance traveled per expansion time is $\propto vt/a \propto t/a^2 \propto a^{-1/2}$ which decreases with time. Thus, the total distance traveled is on the order of the horizon size at $z_{\rm eq}$, which, as we have seen above, is roughly the scale of superclusters today. As there is essentially no power remaining on small scales, the first structures to go non-linear are superclusters and galaxies must then form by fragmentation; this is called a *top down* scenario , as opposed to a *bottom up* scenario in which structures form first on smallscales and then cluster hierarchically into progressively larger entities.

12.9 Cold Dark Matter (CDM)

The cold dark matter model (CDM) also assumes that the bulk of the matter is non-baryonic, but that this material is not in thermal equilibrium at early times. At early times, the superhorizon perturbation grows as usual. The perturbation enters the horizon in the radiation dominated era; the baryon-photon plasma then oscillates acoustically and the growth of the sub-dominant CDM component stagnate (Meszaros effect, Sect. 11.6). Since the dark matter is decoupled from the radion, however, fluctuations are not erased as they are in the HDM model, so at $z_{\rm eq}$ the fluctuations can start to grow. On scales exceeding the horizon size at $z_{\rm eq}$ there is uninterrupted growth. The end result is a suppression of small scale fluctuations by a factor $\sim (\lambda/\lambda_{\rm Heq})^{-2}$ in amplitude. This, for a primordial linear power spectrum $P_0(k) \propto k$ implies that $k^3P(k)$ is asymptotically constant, i.e. the fluctuations becomes asymptotically constant. Numerical simulations show that the progression from the large-scale to small scale asymptotes is very gradual, and consequently one has in effect a hierarchical or *bottom-up* scenario.

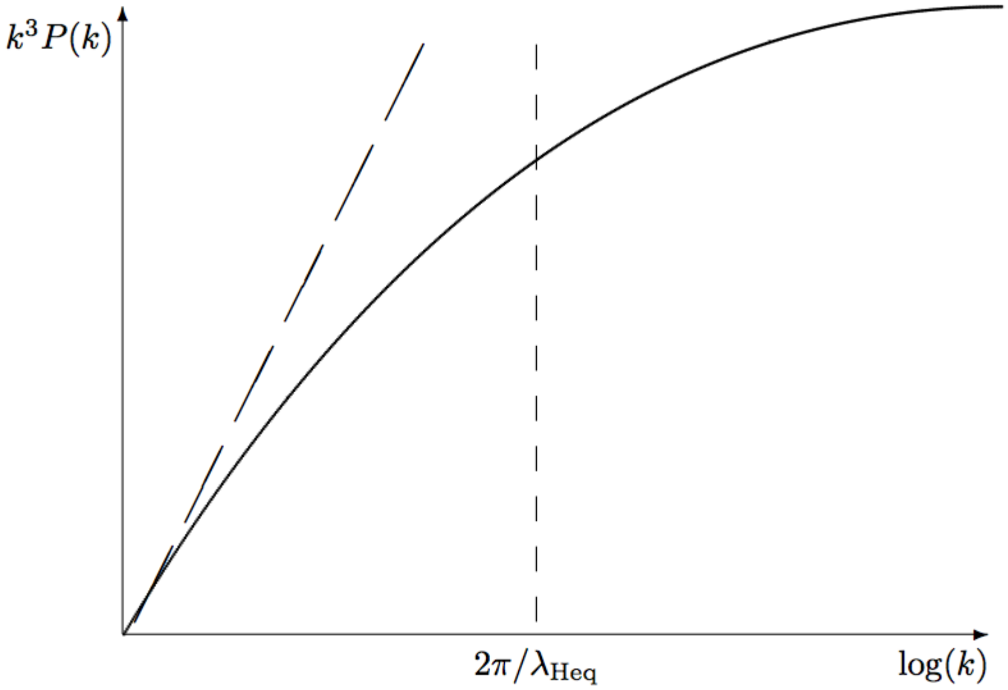

Figure 12: Power spectrum in the Cold Dark Matter (CDM) model (schematic). The long-dashed line indicates the initial power spectrum, while the vertical dashed line the horizon size at the time the neutrinos become non-relativistic at $z \simeq z_{\mathrm{eq}}$.

13 Non-linear cosmological structure

13.1 The Press & Schechter mass function

The Press-Schechter approximation is designed for hierarchical type initial fluctuation fields. It provides one with a useful approximation for the *differential* mass function $n(M) = \mathrm{d}N(>M)/\mathrm{d}M$, where the cumulative mass function $N(>M)$ is the comoving number density of bound structures with mass larger than M.

The density contrast at some early time t_i is a Gaussian random field. If we filter δ with a top-hat window function of comoving radius R, the filtered contrast $\delta_W \equiv \delta_R$ is still a Gaussian random field. Hence, if we consider this function at a fixed point $\boldsymbol{x}$, its distribution will be normal

$$P(\delta_R, t_i, R) = \frac{1}{\sqrt{2\pi}\sigma(R,t_i)} \exp\left(-\frac{\delta_R^2}{2\sigma^2(R,t_i)}\right) , \tag{13.1}$$

$$\sigma^2(R,t_i) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \left|\hat{W}_R(k)\right|^2 P(k,t_i) . \tag{13.2}$$

Note that for the last term of the second equation we have $P(k,t_i) = D_+^2(t_i) P_0(k)$. The fraction *of volume* where the filtered functions have a contrast larger than a given critical value δ_c is then the cumulative probability of Eq. (13.1), i.e.

$$\begin{aligned} P(\delta_c, t_i, R) &= \int_{\delta_c}^{\infty} \frac{\mathrm{d}\delta'}{\sqrt{2\pi}\sigma(R,t_i)} \exp\left(-\frac{\delta_R'^2}{2\sigma^2(R,t_i)}\right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{\delta_c/\sigma}^{\infty} \mathrm{d}x\, \mathrm{e}^{-x^2/2} . \end{aligned} \tag{13.3}$$

The mass of the collapsed object is given simply by

$$M = \frac{4\pi}{3} R^3 \rho_0 = \frac{4\pi}{3} R^3 \Omega_{\mathrm{m}} \rho_{\mathrm{cr}} , \tag{13.4}$$

where R is taken to be measured in comoving coordinates. Since $M \propto R^3$, it is reasonable to assume that the fraction in volume of Eq. (13.3) can be identified with the fraction of mass in objects more massive than M:

$$P_M(>M) = \int_M^\infty Mn(M,t)\,\mathrm{d}M\,. \tag{13.5}$$

If we differenciate both Eq. (13.3) and Eq. (13.5) with respect to M we obtain

$$\boxed{n(M) = \frac{\delta_c}{\sqrt{2\pi}}\frac{1}{M\sigma^2(M)}\frac{\mathrm{d}\sigma(M)}{\mathrm{d}M}\exp\left(-\frac{\delta_c^2}{2\sigma^2(M)}\right)\,.} \tag{13.6}$$

While hard to justify rigorously, the idea obviously contains an element of truth, and moreover seems to give predictions which agree with the results of N-body experiments.

13.2 Redshift space distortions

If the density field is statistically homogeneous and isotropic, and galaxies are tracers of the underlaying mass distribution, then the galaxy power spectrum $P_{\mathrm{g}}(\boldsymbol{k})$ depends only on $|\boldsymbol{k}|$, and the galaxy two-point correlation function $\xi(\boldsymbol{r})$ depends only on $|\boldsymbol{r}|$.

However, we do not observe the real 3D position of a galaxy, but only its angular position and redshift. The measured redshift constists of the cosmic expansion and of the peculiar velocity v:

$$z \simeq \frac{DH_0 + v}{c}\,. \tag{13.7}$$

One therefore can define a 3D "redshift space" $\boldsymbol{s}$ related to the real 3D space $\boldsymbol{r}$ and to the angular position of the galaxy θ by

$$s_1 = r_1 = \frac{zc}{H_0}\theta_1\,, \qquad s_2 = r_2 = \frac{zc}{H_0}\theta_2\,, \qquad s_3 = r_3 + \frac{v_3}{H_0}\,, \tag{13.8}$$

where the third axis is taken to be aligned along the line of sight. Hence, as measured in the redshift space the correlation function is no longer isotropic since the (line os sight component of the) peculiar velocity field is related to the density constrast.

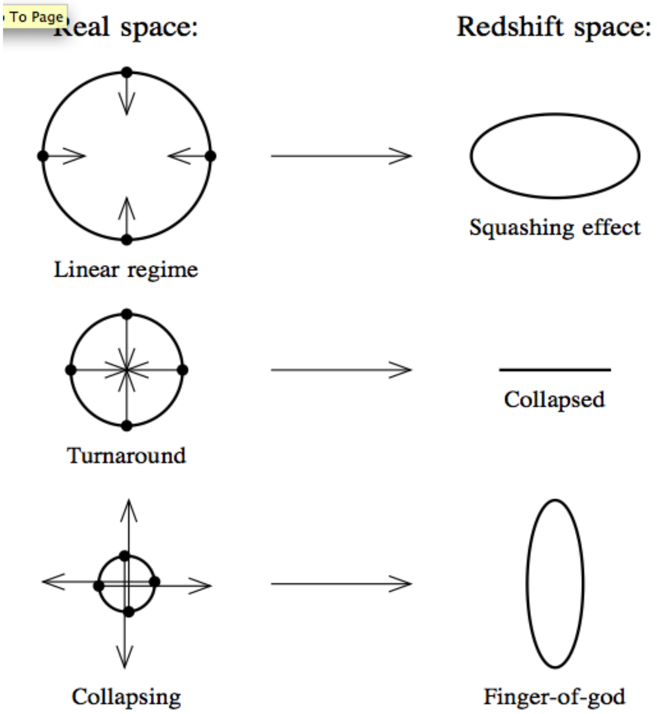
To Page
Real space:
Redshift space:
Linear regime
Squashing effect
Turnaround
Collapsed
Collapsing
Finger-of-god

Figure 13: Effects of a correlation between pecular velocities and spatial distribution of galaxies.

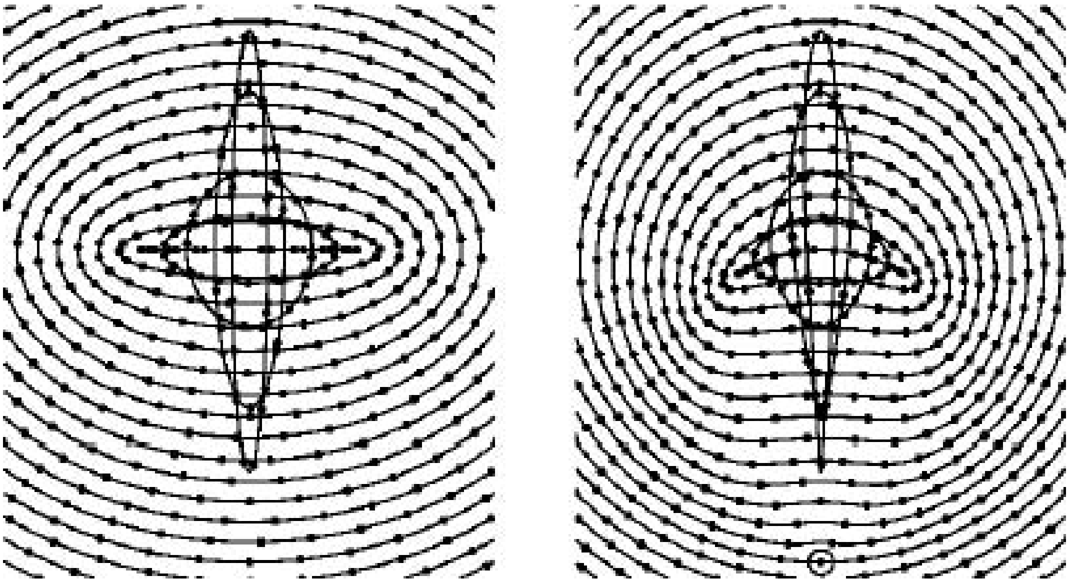

Figure 14: Effects of a correlation between pecular velocities and spatial distribution of galaxies.

Let $n(\boldsymbol{r})$ and $n(\boldsymbol{s})$ be the number density of galaxies in real and redshift space respectively, and write them as

$$n(\boldsymbol{r}) = \bar{n}(\boldsymbol{r})\left[1+\delta_{\mathrm{g}}(\boldsymbol{r})\right] , \qquad n(\boldsymbol{s}) = \bar{n}(\boldsymbol{s})\left[1+\delta_{\mathrm{g}}^{s}(\boldsymbol{s})\right] , \tag{13.9}$$

there $\bar{n}(\boldsymbol{r})$ and $\bar{n}(\boldsymbol{s})$ denote the mean galaxy densities. Since both these quantities are taken to be slowly varying, we will assume that they are equal.

Since the number of galaxies is conserved, we have

$$\bar{n}(\boldsymbol{r})\left[1+\delta_{\mathrm{g}}(\boldsymbol{r})\right] \mathrm{d}^3 r = \bar{n}(\boldsymbol{s})\left[1+\delta_{\mathrm{g}}^{s}(\boldsymbol{s})\right] \mathrm{d}^3 s , \tag{13.10}$$

which leads to

$$1+\delta_{\mathrm{g}}^{s}(\boldsymbol{s}) = \frac{\mathrm{d}r_3}{\mathrm{d}s_3}\left[1+\delta_{\mathrm{g}}(\boldsymbol{r})\right] . \tag{13.11}$$

To first degree the Jacobian can be evaluated as

$$\frac{\mathrm{d}r_3}{\mathrm{d}s_3} \simeq 1 - \frac{1}{H_0}\frac{\mathrm{d}v_3}{\mathrm{d}r_3} , \tag{13.12}$$

so that we finally have

$$\boxed{\delta_{\mathrm{g}}^{s}(\boldsymbol{r}) = \delta_{\mathrm{g}}(\boldsymbol{r}) - \frac{1}{H_0}\frac{\mathrm{d}v_3}{\mathrm{d}r_3} .} \tag{13.13}$$

Note that here we neglected the difference between $\boldsymbol{s}$ and $\boldsymbol{r}$ in the argument of δ_{g}^{s}, since this would lead to a second-order term.

As before, we write the peculiar velocity as the gradient of a potential

$$\boldsymbol{v}(\boldsymbol{r}) = \nabla_{\boldsymbol{r}}\psi(\boldsymbol{r}) , \tag{13.14}$$

now expressed in proper coordinates. For ψ we found previously the Poisson equation

$$\nabla_{\boldsymbol{x}}^2\psi = -aH(a)\frac{\mathrm{d}\ln D_+}{\mathrm{d}\ln a}\delta , \tag{13.15}$$

so that

$$\nabla_{\boldsymbol{r}}\psi(\boldsymbol{r}) = \frac{1}{a^2}\nabla_{\boldsymbol{x}}^2\psi = -\frac{H(a)}{a}\frac{\mathrm{d}\ln D_+}{\mathrm{d}\ln a}\delta \simeq -H_0 f\delta , \tag{13.16}$$

where the last equality holds at the present time. In the Fourier space this equation is

$$\hat{\psi}(\boldsymbol{k}) = \frac{H_0 f}{k^2}\hat{\delta}\ , \tag{13.17}$$

so that

$$\hat{v}_3(\boldsymbol{k}) = -ik_3\hat{\psi}(\boldsymbol{k}) = -\frac{ik_3}{k^2}H_0 f\hat{\delta}\ . \tag{13.18}$$

If we take the Fourier transform of the Eq. (13.13) we find

$$\hat{\delta}_{\rm g}^s(\boldsymbol{k}) = \hat{\delta}_{\rm g}(\boldsymbol{k}) + \frac{ik_3}{H_0}\hat{v}_3(\boldsymbol{k})\ , \tag{13.19}$$

so that we finally have

$$\boxed{\hat{\delta}_{\rm g}^s(\boldsymbol{k}) = \left[1+\beta\mu^2(\boldsymbol{k})\right]\hat{\delta}_{\rm g}(\boldsymbol{k})\ ,} \tag{13.20}$$

where we defined the direction cosione

$$\mu(\boldsymbol{k}) \equiv \frac{k_3}{|k|}\ , \tag{13.21}$$

and as before

$$\beta \equiv f/b \simeq \Omega_{\rm m}^{0.6}/b\ . \tag{13.22}$$

Analogously, the power spectrum of galaxies in redshift space is related to the one in real space by

$$\boxed{\hat{P}_{\rm g}^s(\boldsymbol{k}) = \left[1+\beta\mu^2(\boldsymbol{k})\right]^2 P_{\rm g}(\boldsymbol{k})} \tag{13.23}$$

is anisotropic in general!
Note that

- The galaxy power spectrum is amplified with respect to the matter power spectrum.
- The ratio between the isotropic and anisotropic part of $P_{\rm g}^s(\boldsymbol{k})$, and similarly of the isotropic and anisotropic part of $\xi(\boldsymbol{x})$, only depends on β. Thus if one of these quantities can be measured, β can be determined.

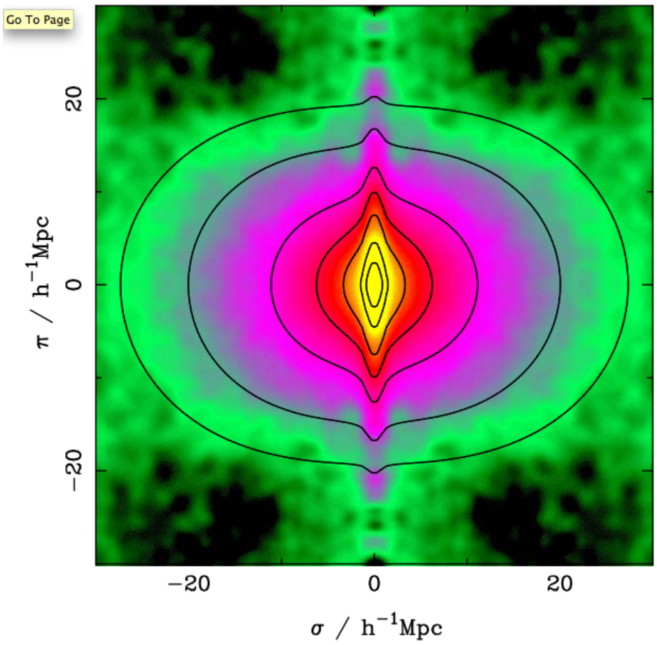

Figure 15: Redshift space two-point correlation function from the 2dFGRS; contours show a model with $\beta = 0.4$ and $\sigma_\mathrm{p} = 400$ km s^{-1}. The quadrupole anisotropy prevails at large scales, while the "finger-of-god" at small scales. The notation used is $\sigma \equiv \sqrt{s_1^2 + s_2^2}$ and $\pi \equiv s_3$.

- Galaxies do not follow precisely the peculiar velocity field obtained from the linear analysis performed here, but have an additional velocity component due to the small-scale non-linear density field. This is usually modeled as a random velocity component with a normal distribution with variance σ_p.

13.3 Clustering of galaxies

From the current surveys, such as the 2dFGRS, we can measure the two-point correlation function in redshift space $\xi^s(\sigma, \pi)$. This, taken as a function of σ, is a power law only for a very limited range. Let us consider however the integral of ξ^s over π:

$$\Xi(\sigma) \equiv \int_{-\infty}^{\infty} \mathrm{d}\pi \, \xi^r(\sigma\pi) \,, \tag{13.24}$$

which from Eq. (13.23) can be rewritten as

$$\Xi(\sigma) = 2\int_{\sigma}^{\infty} \frac{\mathrm{d}r\, r}{\sqrt{r^2-\sigma^2}}\xi(r)\ . \tag{13.25}$$

If we now assume that

$$\boxed{\xi(r) = \left(\frac{r}{r_0}\right)^{-\gamma}\ ,} \tag{13.26}$$

then we find

$$\Xi(\sigma) = A(\gamma)\left(\frac{\sigma}{r_0}\right)^{-\gamma+1}\ , \tag{13.27}$$

where the amplityde $A(\gamma)$ is a numerical factor. Since Ξ is well approximated with a power law, then we can deduce that $\xi(r)$ is also well described by Eq. (13.26).
Observationally, for galaxies one finds $\gamma \simeq 1.7$ over a wide range of distances, with very little variation for the galaxy type; the parameter $r_0 \sim 5\,h^{-1}$ Mpc ofr optical galaxies. Surprisingly, the law (13.26) describes well also the two-point correlation function of galaxy clusters, with a similar exponent!

14 Gravitational lensing

A mass inhomogeneity perturbs the simple and symmetric form of the Robertson-Walker metric and, as a result, geodesics are modified and do not appear as "straight lines" anymore (cf. Gravity as a that acts at distance in Newton's theory with Einstein's metric perturbation). Similarly to massive bodies, light is also influenced by gravity in several respects. In this chapter we will in particular consider the *bending* of light by massive bodies, called Gravitational Lensing.

14.1 A bit of history

- 1804: Soldner gave first written account of deflection of light by Gravity: calculated expected deflection at Solar limb;
- 1915: General Relativity from Einstein
- 1919: Eddington measured light deflection near Solar limb with $\sim 20\%$ accuracy;
- 1924: First discussion of "lens effect": multiple images, rings from Chwolson;
- 1936: Einstein's paper on point-mass (stellar) lensing;
- 1937: Zwicky's visions: "extragalactic nebulæ" as lenses, probability for observing this "a certainty"; use magnification effect as "natural telescope";
- Klimov (1963), Liebes (1964), Refsdal (1964): Revival of gravitational lensing, cosmological applications (including H_0);
- 1979: First lens, 1957+561 discovered by Walsh, Carswell, & Weymann;
- 1986: First giant luminous arcs discovered by Lynds & Petrosian;

- 1987: First Einstein ring (Hewitt et al.);
- 1990: Weal lensing in clusters of galaxies (Tyson et al.);
- 1993: Microlensing towards LMC (MACHO searches) and Galactic bulge;
- 1997: First solid measurement of H_0;
- 2000: Cosmic shear detections.

14.2 Deflection of light

Consider a metric $g_{\alpha\beta}$ that differs little from the flat, Minkowskian metric $\eta_{\alpha\beta} = \mathrm{diag}(1, -1, -1, -1)$:

$$g_{\alpha\beta} = \left(1 - \tfrac{1}{2}h\right)\eta_{\alpha\beta} + h_{\alpha\beta} \; , \tag{14.1}$$

$$h \equiv \eta^{\alpha\beta} h_{\alpha\beta} \; , \tag{14.2}$$

where $h_{\alpha\beta}$ describes the perturbation to the flat metric (the definition of the scalar is introduced to simplify some of equations of this chapter). Einstein's field equations, linearized in $H_{\alpha\beta}$ are then

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) h^{\alpha\beta} = \frac{16\pi G}{c^4} T^{\alpha\beta} \; . \tag{14.3}$$

Note that this equation is reminiscent of the equation for gravitational waves, with the important different that here we have a non-vanishing r.h.s. In any case, the solution for an isolated source can be written in terms of the *delayed* energy-momentum tensor:

$$h^{\alpha\beta}(t, \boldsymbol{x}) = -\frac{4G}{c^4} \int \frac{T^{\alpha\beta}(t - |\boldsymbol{y}|/c, \boldsymbol{x} + \boldsymbol{y})}{|\boldsymbol{y}|} \, \mathrm{d}^3 y \; . \tag{14.4}$$

In the following, for simplicity, we will neglect the pressure and assume $p \ll \rho c^2$, and assume also that the matter perturbation is stationary. Under these hypotheses, the stress-energy tensor is dominated by the "dust" term $T^{00} \simeq \rho c^2$ and the final metric can be written as

$$\boxed{\mathrm{d}s^2 = \left(1 + \frac{2U}{c^2}\right) c^2\,\mathrm{d}t^2 - \left(1 - \frac{2U}{c^2}\right) \mathrm{d}x^2\,,} \tag{14.5}$$

where U is the usual Gravitational potential:

$$U(t,\boldsymbol{x}) = -G \int \frac{\rho(t,\boldsymbol{x}+\boldsymbol{y})}{|\boldsymbol{y}|}\,\mathrm{d}^3 y\,. \tag{14.6}$$

Exercise 14.1 *Show that the path of the light in a gravitational field can be described in terms of an effective refraction index given by*

$$n = 1 - \frac{2U}{c^2}\,. \tag{14.7}$$

Exercise 14.2 *A version of Fermat's principle holds also in General Relativity. Using Eq.* (14.7) *together with the variational principle (deduced from Fermat's principle)*

$$\delta \int n\,\mathrm{d}\ell = 0\,, \tag{14.8}$$

show that the associated Euler-Lagrange equations can be written as

$$\frac{\mathrm{d}\boldsymbol{e}}{\mathrm{d}l} = -\frac{2}{c^2}\nabla_\perp U\,, \tag{14.9}$$

where $\boldsymbol{e}$ *is the unit tangent vector of the ray and* $\nabla_\perp U \equiv \nabla U - \boldsymbol{e}(\boldsymbol{e}\cdot\nabla U)$ *denotes the projection of* ∇U *onto the plane orthogonal to the direction* $\boldsymbol{e}$ *of the ray.*

We now evaluate the final deflection angle using a perturbative method. In principle, the bending angle is given by the integral of the differential equation (14.9),

$$\hat{\alpha} = \frac{2}{c^2}\int \nabla_\perp U\,\mathrm{d}\ell\,. \tag{14.10}$$

The integral of this equation can be obtained by using an iterative approach:

- Consider first the *unperturbed* light path, $\boldsymbol{x}^{(0)}(t) = \boldsymbol{x}_0 + \boldsymbol{e}(t - t_0)$, which is a straight line;
- Evaluate the deflection along the previous path by using it in the integral of Eq. (14.10);
- Calculate the *perturbed* path $\boldsymbol{x}^{(1)}(t)$ by adding-up all deflections.
- Repeat the whole process with the perturbed path and calculate $x^{(2)}(t)$.

Although, in principle, the scheme above can be applied any number of times, in practice it is usually enough to stop the calculation to the first perturbed path $x^{(1)}$.

Exercise 14.3 *Consider a point mass M and show that in this case the deflection angle obtained from the unperturbed path is given by*

$$\hat{\alpha}(\xi) = \frac{4GM}{c^2} \frac{\xi}{|\xi|^2}\,, \tag{14.11}$$

where ξ is the impact parameter of the light ray. Note that this deflection angle is twice the Newtonian value.

In most situation of astrophysical interest, the extent of the lens is much smaller than the observer-lens and lens-source distances ("thin lens" approximation). In this case, it is possible to sum all deflections from the various part of the lens and write [cf. Eq. (14.11)]

$$\boxed{\hat{\alpha}(\xi) = \frac{4G}{c^2} \int_{\mathbb{R}^2} \frac{(\xi - \xi')\Sigma(\xi')}{|\xi - \xi'|^2} \mathrm{d}^2\xi'\,,} \tag{14.12}$$

where $\Sigma(\xi)$ is the lens *projected surface mass density*, and the integration is performed over the lens plane.
Note: Eqs. (14.10) and (14.12) are reminiscent of Newton's gravitational force, except for the different exponent in the denominator; in reality, the different exponent is the "correct one" to have a "Poisson equation" in 2D.

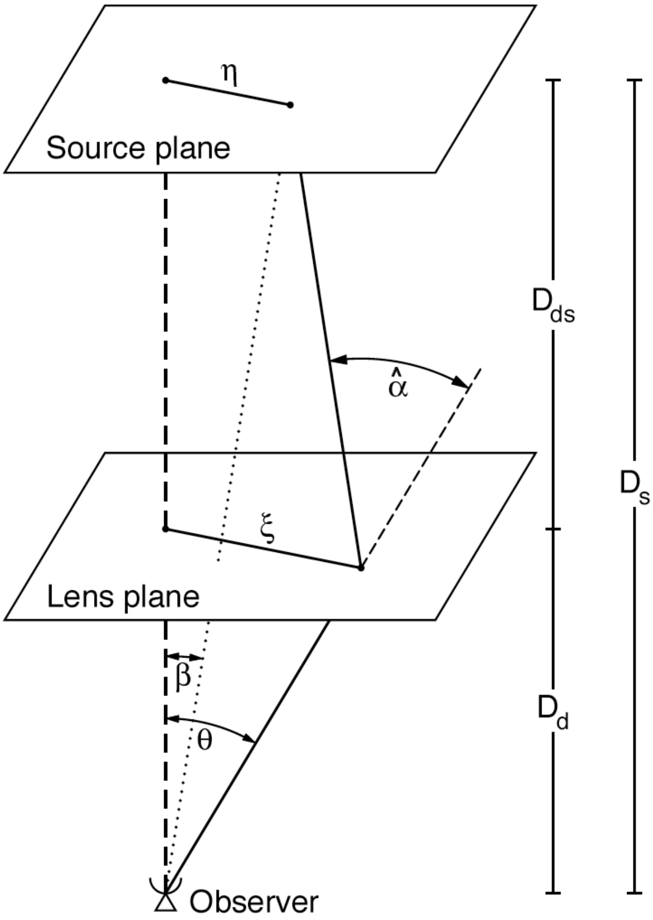
η
Source plane
D_{ds}
$\hat{\alpha}$
D_s
ξ
Lens plane
β
θ
D_d
Observer

Figure 16: Lensing configuration.

14.3 The lens equation

From the simple geometry of Fig.16, we can see that

$$\boxed{\eta = \frac{D_\mathrm{s}}{D_\mathrm{d}}\xi - D_\mathrm{ds}\hat{\alpha}(\xi)\ .} \tag{14.13}$$

Introducing the angular coordinates $\eta = D_\mathrm{s}\beta$ and $\xi = D_\mathrm{d}\theta$, the lens equation becomes

$$\boxed{\beta = \theta - \frac{D_\mathrm{ds}}{D_\mathrm{s}}\hat{\alpha}(D_\mathrm{d}\theta) \equiv \theta - \alpha(\theta)\ ,} \tag{14.14}$$

where $\alpha(\theta)$ is the scaled deflection angle. Equation (14.14) is also called the *ray-tracing equation*.
A few observations:

- Note that for any direction θ there is one and only one β: on other words, to an *observed* direction on the sky corresponds a single position on the source plane.

- For a source position β one can have in principle several *observed* directions θ; in other words, the ray-tracing equation is not (always) *injective* (multiple images).

- It is possible to show that the ray-tracing is however *subjective*, i.e. to a source position β corresponds at least one observed image ξ.

Example: lensing for an axially symmetrical lens. In this case, since the Gauss theorem holds for Eq. (14.12), we find

$$\beta = \theta - \frac{D_{\mathrm{ds}}}{D_{\mathrm{s}} D_{\mathrm{d}}} \frac{4GM(\theta)}{c^2 \theta} , \tag{14.15}$$

where $M(\theta)$ is the mass enclosed within a radius θ. If the source lies on the optic axis, i.e. $\beta = \boldsymbol{0}$, then something peculiar occurs: there is no preferential direction for light deflection, so an Einstein ring is produced. Its radius is

$$\theta_{\mathrm{E}} = \sqrt{\frac{D_{\mathrm{ds}}}{D_{\mathrm{s}} D_{\mathrm{d}}} \frac{4GM(\theta_{\mathrm{E}})}{c^2}} . \tag{14.16}$$

For a typical galaxy lens system, $\theta_{\mathrm{E}} \sim 1$ arcsec, while $\theta_{\mathrm{E}} \sim 20$ arcsec for a typical cluster lens system.

We define the *convergence* or *dimensionless surface mass density* $\kappa(\theta)$ as

$$\kappa(\theta) \equiv \frac{\Sigma(D_{\mathrm{d}} \theta)}{\Sigma_{\mathrm{c}}} , \tag{14.17}$$

where Σ_{c} is the *critical surface mass density*

$$\Sigma_{\mathrm{c}} \equiv \frac{c^2}{4\pi G} \frac{D_{\mathrm{s}}}{D_{\mathrm{d}} D\mathrm{ds}} . \tag{14.18}$$

Note that:

- The critical density sets the typical scale for the projected mass density: in other words, if $\Sigma \ll \Sigma_{\mathrm{c}}$ then $\kappa \ll 1$ and the lensing effect is very small.

- It can be shown (see below) that if there is a direction where $\Sigma(\theta) > \Sigma_c$, i.e. if $\kappa(\theta) > 1$ somewhere, then the lens will produce multiple images if there is a suitably located source (clearly no multiple images are created if the source is not well aligned with the lens).

- The critical density is a purely *geometrical* quantity, defined only in terms of physical constants and of the relative distances between the observer, the source, and the lens.

Using the convergence, can rewrite the deflection as

$$\alpha(\theta) = \frac{1}{\pi} \int_{\mathbb{R}^2} \mathrm{d}^2\boldsymbol{\theta}\, \kappa(\theta) \frac{\theta - \theta'}{|\theta - \theta'|} \,. \tag{14.19}$$

From the identity $\nabla \ln|\theta| = \theta/|\theta|^2$ we see that

$$\boxed{\alpha = \nabla \psi \,,} \tag{14.20}$$

where the deflection potential is

$$\boxed{\psi(\theta) = \frac{1}{\pi} \int_{\mathbb{R}^2} \mathrm{d}^2\boldsymbol{\theta}\, \kappa(\theta') \ln|\theta - \theta'| \,.} \tag{14.21}$$

Furthermore, using $\nabla^2 \ln|\theta| = 2\pi\delta(\theta)$, we find

$$\boxed{\nabla^2 \psi = 2\kappa \,,} \tag{14.22}$$

which is Poisson's equation in 2D.

14.4 Image distortion and magnification

We now consider the effect of lensing on a small patch of the sky, i.e. show how the ray-tracing affects the images of small sources. On small scales, we can Taylor expand the lens mapping (14.14). In this expansion, the first interesting term is the linear one

$$\mathscr{A}_{ij}(\theta) \equiv \frac{\partial \beta_i}{\partial \theta_j} = \delta_{ij} - \frac{\partial^2 \psi(\theta)}{\partial \theta_i \partial \theta_j} \,. \tag{14.23}$$

Because of Eq. (14.22), the trace of the matrix $\mathscr{A}$ is fixed and simply related to the convergence κ; the other elements are called *shear* and usually indicated as γ_1 and γ_2:

$$\mathscr{A} = \begin{bmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{bmatrix} . \tag{14.24}$$

Note that:

- κ leads to an isotropic focusing and hence *magnification* of the images;
- γ_1 and γ_2, usually written as a complex quantity $\gamma = \gamma_1 + \mathrm{i}\gamma_2$, introduces anisotropy, and squeeze round images into ellipses.
- Lensing has the fundamental property of preserving the *surface brightness* of sources: if $I(\theta)$ is the observed surface brightness of a lensed source, and $I^{\mathrm{s}}(\beta)$ is its real (unlensed) surface brightness, then

  $$\boxed{I(\theta) = I^{\mathrm{s}}(\beta) = I^{\mathrm{s}}\big(\theta - \alpha(\theta)\big) \ .} \tag{14.25}$$

  As a result, the magnification of small images is given by $|\mu|$, where

  $$\boxed{\mu = \frac{1}{\det \mathscr{A}} = \frac{1}{(1-\kappa)^2 - |\gamma|^2} \ .} \tag{14.26}$$

14.5 Caustics and critical curves

With the term *critical lines* ones indicate the curves on the lens plane where the lensing equation is not locally invertible, i.e. where $\delta\mathscr{A} = 0$. Note that since $\mathscr{A}$ depends on the source distance through the critical density Σ_{c}, the location of the critical lines also changes with the redshift of the source. Typically, critical lines for high-redshift sources are larger than critical lines for low-redshift ones.

The mapping of the critical lines into the source plane with the lens equation defines the *caustics*. A source located on a caustic will (in principle) have an infinite magnification; in practice the magnification is very large but always finite (for various reasons, but mostly because real sources are extended and thus their true amplification is an average of μ over the source profile).

When a source crosses a caustic, a pair of images on the lens plane are created or destroyed. This way, the number of images that a gravitational lens produces out of a pointlike source is always odd (because, for example, when a source is angularly distant from the lens only a single, weakly distorted image is produced).

Although caustic lines are smooth and typically quite round, caustics are not and are usually characterized by *cusps*; moreover, the number of cusps typically increases when the lens has a large amount of *substructures* (i.e., is composed by many different clumps of matter). An source close to a cusps usually produces a triplet in the image plane.

14.6 Fermat's principle

As anticipated in Sect.14.2, a version of the Fermat's principle holds in General Relativity. Its simple formulation is

> In any static metric ($g_{0i} = g_{\alpha\beta,0} = 0$) consider all null curves between two points in the space, $x^i = a^i$ and $x^i = b^i$. Each such null curve requires a particular coordinate time Δt to get from a^i to b^i. The curves of extremal time Δt are null geodesics of the spacetime.

In gravitational lensing, one can show that the coordinate time Δt is proportional to the so-called Fermat potential, defined as

$$\boxed{\phi(\theta,\beta) = \frac{1}{2}(\theta-\beta)^2 - \psi(\theta)\,.} \tag{14.27}$$

Exercise 14.4 *Show that as a result of Fermat's principle the lens equation can be obtained from*

$$\nabla_\theta \phi(\theta,\beta) = \boldsymbol{0}\,. \tag{14.28}$$

Also, compare this equation with Eq. (14.14).

If we fix the source position β, the surface described by $\phi(\theta,\beta)$ for various θ can be used to characterize the images of the lens. Images are expected on the extremal points of ϕ, i.e. for minima, maxima, and saddle-points of this function. Since $\psi(\theta)$ vanishes for large $\|\theta\|$, and since the Fermat potential has a quadratic term in θ, the surface considered will tend to be a paraboloid in the limit $\|\theta\| \to \infty$. In these conditions, a topological theorem ensure that the number of minima plus the number of maxima of this function is equal to the number of saddle-points plus one: i.e.

$$\#(\text{minima}) + \#(\text{maxima}) - \#(\text{saddle points}) = 1 \; . \qquad (14.29)$$

This again shows that the number of multiple images of a lens is odd. In addition, given the boundary conditions for the Fermat potential, we will have at least a minimum: hence a lens produces at least a single image out of a point source (i.e., a transparent gravitational lens cannot “hide” sources).

14.7 Time delay

If a source observed through a lens has luminosity variability, then the various images produced will show luminosity changes at different times. In particular, one can evaluate the *time delay* between two images from the Fermat potential

$$\delta t = \frac{D_\mathrm{d} D_\mathrm{s}}{c D_\mathrm{ds}} (1 + z_\mathrm{d}) \left[\phi(\theta_1,\beta) - \phi(\theta_2,\beta) \right] \; . \qquad (14.30)$$

The time delay is important because it is a measurable quantity for lenses with variable sources, and because it is the only non-dimensionless quantity that lensing provides (other data such as angles, magnification μ, convergence κ, and shear γ are all dimensionless). In addition, for a given lensing model the time delay is proportional to H_0^{-1}, and thus it can be used to estimate the Hubble constant. Current measurements of H_0 from lensing appear to be consistent with independent methods.

14.8 Simple lens models

Uniform disk

Consider a uniform disk of matter with constant density Σ and radius θ_r. Then for $\theta < \theta_r$ we have [cf. Eq. (14.15)]

$$\beta = \theta - \kappa\theta \, , \tag{14.31}$$

where $\kappa = \Sigma/\Sigma_c$. Hence, if $\kappa = 1$ the lens focuses perfectly the observer with a source located at the center of the disk. The uniform disk thus in gravitational lensing is the equivalent of a perfect converging standard lens.

Singular isothermal sphere

The singular isothermal sphere (SIS) is a simple galaxy model characterized by three-dimensional density

$$\rho(r) = \frac{\sigma_v^2}{2\pi G r^2} \, , \tag{14.32}$$

where σ_v is the 1-dimensional velocity dispersion of the lens. Projecting along the line of sight, we obtain the surface mass density

$$\Sigma(\xi) = \frac{\sigma_v^2}{2G}\xi^{-1} \, , \tag{14.33}$$

and the lensing equation is thus

$$\beta = \theta - \theta_E \frac{\theta}{|\theta|} \, , \tag{14.34}$$

where θ_E is the Einstein radius

$$\theta_E = 4\pi \left(\frac{\sigma_v}{c}\right)^2 \frac{D_{ds}}{D_s} \, . \tag{14.35}$$

Hence, this lens produces two images located at $\beta \pm \theta_E$ for $\beta \equiv |\beta| < \theta_E$, and just a single image for $\beta > \theta_E$.

14.9 Lensing in galaxy and galaxy clusters

A gravitational lens producing multiple images, such as galaxies and galaxy clusters, can be used to study the matter distribution of the deflector. The standard method used is based on an inversion of the data:

- A preliminary model for the mass distribution of the lens is constructed.
- This model is used together with the position of the observed images in the ray-tracing equation to find the position(s) of the relative source(s).
- Typically, since the original model is not realistic, images of the same source will be ray-traced to different positions in the source plane. A single fiducial source position thus has to be chosen by using a sort of "average" of the various positions obtained.
- The ray-tracing equation is then inverted for this source, and the corresponding various images positions are found.
- The lens model and source positions are then finally modified until a good match is obtained between the predicted image positions and the observed ones.

In addition, galaxy clusters often produce elongated images called *arcs*. A normal lens bends light rays so that a relatively small area of the source plane is shown in a large area in the image plane (equivalently, most lenses have $\mu > 1$ in the central region). Since the lens strength is larger in the central part of a the cluster, from a phenomenological point of view a typical lensing cluster acts as a "repulsive" lens: the image plane is stretched so that background galaxies on average are at larger distances among them. In order to "make space" on the central regions of the cluster, a typical lens will squeeze images radially: hence, background galaxies often appear as arcs or arclets in galaxy clusters. These peculiar configurations provides many constraints for lensing models and can also be used for more advanced inversion techniques.

15 Weak lensing

A lens is called *weak* if $\kappa(\theta) = \Sigma(\theta)/\Sigma_\mathrm{c}$ everywhere, or a stronger lens on locations far away from the lens center. In these condition, the lens is not able to produce multiple images out of individual sources, but still produce some measurable distortions on the background images.

15.1 Weak distortions

As discussed in Sect. 14.4, the magnification matrix can be used to describe the ray-tracing map on small regions of the sky (Taylor expansion). Let us now consider a small image characterized by a surface brightness $I(\theta)$, and define its first moments as

$$I \equiv \int I(\theta)\,\mathrm{d}^2\boldsymbol{\theta}\ , \tag{15.1}$$

$$\xi \equiv \frac{1}{I}\int \theta I(\theta)\,\mathrm{d}^2\boldsymbol{\theta}\ , \tag{15.2}$$

$$Q_{ij} \equiv \frac{1}{I}\int \theta_i\theta_j I(\theta)\,\mathrm{d}^2\boldsymbol{\theta}\ . \tag{15.3}$$

The first quantity, I, represents the total flux observed from the source; the second, ξ, the source "center of light"; the third, Q_{ij}, the (symmetric) tensor of quadrupole moments, describing the shape of the source (e.g., a quadrupole proportional to the identity would represent a circular source). Similar definitions hold for the similar quantities of the unlensed source surface brightness $I^\mathrm{s}(\beta)$:

$$I^\mathrm{s} \equiv \int I^\mathrm{s}(\beta)\,\mathrm{d}^2\boldsymbol{\beta}\ , \tag{15.4}$$

$$\xi^\mathrm{s} \equiv \frac{1}{I^\mathrm{s}}\int \beta I(\beta)\,\mathrm{d}^2\boldsymbol{\beta}\ , \tag{15.5}$$

$$Q^\mathrm{s}_{ij} \equiv \frac{1}{I^\mathrm{s}}\int \beta_i\beta_j I(\beta)\,\mathrm{d}^2\boldsymbol{\beta}\ . \tag{15.6}$$

Using the conservation of the surface brightness and the linearized lensing equation we then find two interesting relations:

$$I^{\mathrm{s}} = |\det A| I = \mu^{-1} I \,, \tag{15.7}$$

$$Q^{\mathrm{s}}_{ij} = A_{ii'} A_{jj'} Q_{i'j'} \qquad \Rightarrow \qquad Q^{\mathrm{s}} = A Q A^{\mathrm{T}} \,. \tag{15.8}$$

Hence, the matrix A is responsible for both the magnification and the distortion of an image.

15.2 Weak lensing measurements

For a single source we cannot estimate the matrix A using Eq. (15.8), because we only observe Q_{ij} and Q^{s}_{ij} remains unknown. However, if we observe a relatively large number of galaxies all subject to the same distortion A, then we can obtain an estimate of this matrix. In absence of lensing, there is no preferred direction in the sky, and thus the average quadrupole moment must be proportional to the identity:

$$\left\langle Q^{\mathrm{s}}_{ij} \right\rangle = M \delta_{ij} \,, \tag{15.9}$$

where M is a positive quantity. From Eq. (15.8) we find then

$$A^{\mathrm{T}} A = A A = M Q^{-1} \,, \tag{15.10}$$

where we have used the symmetry of A. Since A is symmetric, we can diagonalize it with a orthogonal transformation $A = U \Lambda U^{\mathrm{T}}$, which gives

$$A^2 = U \Lambda U^{\mathrm{T}} U \Lambda U^{\mathrm{T}} = U \Lambda^2 U = M U \Upsilon^{-1} U = M Q^{-1} \,, \tag{15.11}$$

where the last equation shows that U is diagonalizable on the same base as A. This equation shows that we can obtain the eigenvectors of A (in practice, a direction on the plane), while the eigenvalues are determined modulus their signs. In reality, since galaxies have a large scatter in size, the value of M is usually poorly constrained, and we only measure easily the (absolute value of the) *ratio* of the eigenvalues of A.

We now note that the matrix A can be rewritten as

$$A = (1-\kappa)\begin{pmatrix} 1-g_1 & -g_2 \\ -g_2 & 1+g_1 \end{pmatrix} , \tag{15.12}$$

where $g_i = \gamma_i/(1-\kappa)$ is the so-called *reduced shear*. The eigenvalues of A are $\lambda_1 = 1-\kappa-|\gamma| = (1-\kappa)\bigl(1-|g|\bigr)$ and $\lambda_2 = 1-\kappa+|\gamma| = (1-\kappa)\bigl(1+|g|\bigr)$. A little algebra shows that, in case M is taken to be undetermined, then the method described above provides a measurement for g or for $1/g^*$. This ambiguity, as we will see, is usually resolved by *global* considerations on the behaviour of g.

A more widely used technique to measure the weak lensing signal of a cluster, i.e. to obtain an estimate of the reduced shear g, is based on the *ellipticity* of background galaxies. This quantity is defined in therms of the quadrupole moments as

$$\varepsilon = \frac{Q_{11} - Q_{22} + 2iQ_{12}}{Q_{11} + Q_{22} + 2\sqrt{Q_{11}Q_{22} - Q_{12}^2}} . \tag{15.13}$$

Exercise 15.1 *Show that $|\varepsilon| \le 1$ for any galaxy, and $|\varepsilon| = 0$ if Q_{ij} is proportional to the identity matrix.*

Exercise 15.2 *Show that for a source with elliptical isophotes of axis ratio $r = a/b \le 1$, and of position angle ϕ, the complex ellipticity ε is given by*

$$\varepsilon = \frac{1-r}{1+r}\mathrm{e}^{2i\phi} \tag{15.14}$$

Note in particular the 2ϕ angle in the argument of the ellipticity.

One can define the ellipticity of the unlesed source ε^{s} using the unlensed quadrupole moment Q^{s}_{ij}. From the transformation properties of the quadrupole moment [Eq. (15.8)] we get then, after some algebra

$$\varepsilon^{\mathrm{s}} = \begin{cases} \dfrac{\varepsilon - g}{1 - g^*\varepsilon} & \text{for } |g| \le 1 , \\[2ex] \dfrac{1 - g\varepsilon^*}{\varepsilon^* - g^*} & \text{for } |g| > 1 . \end{cases} \tag{15.15}$$

Note that in case of very weak lensing, $\gamma \ll 1$ and $\kappa \ll 1$, we find $g \simeq \gamma$ and $\varepsilon^{\mathrm{s}} \simeq \varepsilon - g \simeq \varepsilon - \gamma$.

Exercise 15.3 *Show that the inverse of Eq.* (15.15) *is*

$$\varepsilon = \begin{cases} \dfrac{\varepsilon^{\mathrm{s}} + g}{1 + g^* \varepsilon^{\mathrm{s}}} & \textit{for } |g| \leq 1 \, , \\[2ex] \dfrac{1 + g \varepsilon^{\mathrm{s}*}}{\varepsilon^{\mathrm{s}*} + g^*} & \textit{for } |g| > 1 \, , \end{cases} \tag{15.16}$$

i.e. can be obtained by swapping ε with ε^{s} and by changing the sign of the reduced shear g.

Exercise 15.4 *Using the previous exercise, show that in the hypothesis of a isotropic probability distribution for ε^{s}, i.e. of a probability distribution that depends only on the modulus of this quantity and not on its argument, then the expectation value of ε is*

$$\langle \varepsilon \rangle = \begin{cases} g & \textit{if } |g| \leq 1 \, , \\ 1/g^* & \textit{if } |g| > 1 \, . \end{cases} \tag{15.17}$$

Hint: *note that $\big\langle (\varepsilon^{\mathrm{s}})^n \big\rangle = 0$ for any positive integer n, and use the fact that the transformation* (15.16) *involves only ε or only ε^* (and not a mixture of them) and is thus (anti)holomorphic.*